TRIPATHI STUDY ZONE

JEE MAIN | NEET | AIIMS | KVPY | BITSAT

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CHAPTER 1

DEFINITE INTEGRATION & THEIR APPLICATIONS

1. INTRODUCTION

In the previous chapter, we have studied about indefinite integral, which is anti derivative of a function. Now in this chapter we will study definite integral, which is used in various field of Mathematics, Physics and Chemistry. Symbolically $\int_a^b f(x)dx$ is the integration of $f(x)$ w.r.t. x with $x = a$ as lower limit and $x = b$ as upper limit.

2. GEOMETRICAL INTERPRETATION OF DEFINITE INTEGRAL

Let $f(x)$ be a function defined on a closed interval $[a, b]$. Then $\int_a^b f(x)dx$ represents the algebraic sum of the area of the region bounded by the curve $y = f(x)$, x-axis and the lines $x = a$, $x = b$. Here algebraic sum means that area which is above the x-axis will be added in this sum with + sign and area which is below the x-axis will be added in this sum with –sign. So value of the definite integral may be positive, zero or negative.

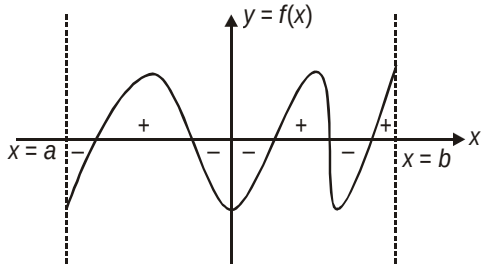
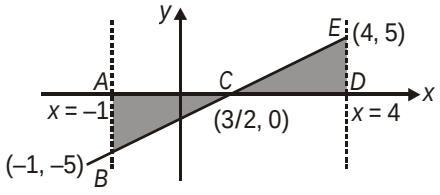


Illustration 1: Evaluate : $\int_{-1}^{4}(2x-3)dx$

Solution : $y = 2x – 3$ is a straight line, which lie below the x-axis in $\left[-1,\frac{3}{2}\right)$ and above in $\left(\frac{3}{2},4\right]$

Now area of $\Delta ABC = \frac{1}{2}\times\frac{5}{2}\times 5 = \frac{25}{4}$

Area of $\Delta CDE = \frac{1}{2}\times\frac{5}{2}\times 5 = \frac{25}{4}$

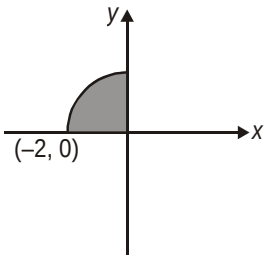


So $\int_{-1}^{4}(2x-3)dx = -\frac{25}{4}+\frac{25}{4} = 0$

***Illustration 2*:** Evaluate : $\int_{-2}^{0}\sqrt{4-x^2}\,dx$

Solution : $y = \sqrt{4-x^2}, x \in [-2, 0]$

Represents a quarter circle in 2nd quadrant, which is above the x-axis radius of circle is 2.



So $\int_{-2}^{0}\left(\sqrt{4-x^2}\right)dx = \frac{1}{4}\left[\pi(2)^2\right] = \pi$ square unit

Student Task

Find $\int_{-2}^{2}|x|dx$ by geometrical interpretation of definite integral?

3. FUNDAMENTAL THEOREM OF INTEGRAL CALCULUS

If $f(x)$ is a continuous function on $[a, b]$, then $\frac{d}{dx}\int_{a}^{x} f(t)dt = f(x)\,(x \in [a,b])$

Now if we take $F'(x) = f(x)$, then by the above theorem

$$\frac{d}{dx}\left[F(x) - \int_{a}^{x} f(t)dt\right] = f(x) - f(x) = 0$$

$\Rightarrow \quad F(x) - \int_{a}^{x} f(t)dt$ = constant = c (say)

$\Rightarrow \quad F(x) = \int_{a}^{x} f(t)dt + c$

Now $F(a) = \int_{a}^{a} f(t)dt + c = 0 + c = c$

So $F(x) = \int_{a}^{x} f(t)dt + F(a)$

$\Rightarrow \quad F(b) = \int_{a}^{b} f(t)dt + F(a)$

$\Rightarrow \quad F(b) - F(a) = \int_{a}^{b} f(t)dt$

Hence if $\int f(x)dx = g(x) + c$, then

$$\int_{a}^{b} f(x)dx = g(x)\Big|_{a}^{b} = g(b) - g(a)$$

***Illustration 3*:** Evaluate : $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$

Solution : $\int \frac{dx}{\sqrt{2-x^2}} = \sin^{-1}\frac{x}{\sqrt{2}} + c$

So $\int_0^1 \frac{dx}{\sqrt{2-x^2}} = \sin^{-1}\frac{x}{\sqrt{2}}\Big|_0^1 = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) - \sin^{-1}(0)$

$= \frac{\pi}{4} - 0 = \frac{\pi}{4}$

Student Task

If $\int f(x)dx = g(x) + c$ and $g(b) = g(a)$, will it always implies $\int_a^b f(x)dx = 0$?

4. GENERAL PROPERTIES OF DEFINITE INTEGRAL

(i) $\boxed{\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(t)dt}$

***Illustration 4*:** Evaluate : $\int_4^9 \frac{dx}{\sqrt{x}(1+\sqrt{x})}dx$

Solution : $I = \int_4^9 \frac{dx}{\sqrt{x}(1+\sqrt{x})}dx$

Put $1 + \sqrt{x} = t$

$\Rightarrow \quad \frac{dx}{2\sqrt{x}} = dt$

$\Rightarrow \quad \frac{dx}{\sqrt{x}} = 2dt$

Now When $x = 4$, $t = 1 + \sqrt{4} = 3$

where $x = 9$, $t = 1 + \sqrt{9} = 4$

So $I = \int_3^4 \frac{2dt}{t} = 2\left|\ln|t|\right|_3^4 = 2(\ln 4 - \ln 3) = \ln\left(\frac{16}{9}\right)$

(ii) $\boxed{\int_a^b f(x)dx \pm \int_a^b g(x)dx = \int_a^b (f(x) \pm g(x))dx}$

***Illustration 5*:** Evaluate : $\int_2^3 \left(\frac{2x^2}{x^4+3x^2+1} \right) dx$

Solution : $I = \int_2^3 \left(\frac{2x^2}{x^4+3x^2+1} \right) dx = \int_2^3 \frac{(x^2+1)}{x^4+3x^2+1} dx + \int_2^3 \frac{(x^2-1)dx}{x^4+3x^2+1}$

$$= \int_2^3 \frac{(1+(1/x^2))dx}{(x-(1/x))^2+5} + \int_2^3 \frac{(1-(1/x^2))dx}{(x+(1/x))^2+1}$$

In 1st put $x - \frac{1}{x} = t$, in 2nd put $x + \frac{1}{x} = y$

$$I = \int_{3/2}^{8/3} \frac{dt}{t^2+5} + \int_{5/2}^{10/3} \frac{dy}{y^2+1}$$

$$= \frac{1}{\sqrt{5}} \left[\tan^{-1}\left(\frac{8}{3\sqrt{5}} \right) - \tan^{-1}\left(\frac{3}{2\sqrt{5}} \right) \right] + \tan^{-1}\left(\frac{10}{3} \right) - \tan^{-1}\left(\frac{5}{2} \right)$$

$$= \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{7\sqrt{5}}{54} \right) + \tan^{-1}\left(\frac{5}{56} \right)$$

(iii) $\boxed{\int_a^b f(x)dx = \int_a^b f(t)dt = \int_a^b f(y)dy}$

i.e. variable of integration in definite integral is a dummy variable.

***Illustration 6*:** If $f(x) = \int_1^x \left(\frac{\ln t}{1+t} \right) dt$ $(x > 0)$, prove that $f(x) + f\left(\frac{1}{x} \right) = \frac{(\ln x)^2}{2}$ **[JEE]**

Solution : $f(x) = \int_1^x \left(\frac{\ln t}{1+t} \right) dt$

$$\Rightarrow \quad f\left(\frac{1}{x} \right) = \int_1^{1/x} \left(\frac{\ln t}{1+t} \right) dt$$

Now to add $f(x)$ and $f\left(\frac{1}{x} \right)$, we have to make both the limits of integration same (by (ii))

Put $\frac{1}{t} = y \Rightarrow \frac{dt}{-t^2} = dy$

$$\Rightarrow \quad dt = -\frac{dy}{y^2}$$

So $\int_1^{1/x} \left(\frac{\ln t}{1+t} \right) dt = \int_1^x \frac{\ln(1/y)}{1+\frac{1}{y}} \left(\frac{-dy}{y^2} \right)$

$$= \int_1^x \frac{\ln y}{y(1+y)} dy = \int_1^x \frac{\ln t}{t(1+t)} dt \text{ (by (iii))}$$

Hence $f(x)+f\left(\frac{1}{x}\right)=\int_1^x\left(\frac{\ln t}{1+t}+\frac{\ln t}{t(1+t)}\right)dt$

$=\int_1^x\frac{\ln t}{1+t}\left(1+\frac{1}{t}\right)dt=\int_1^x\frac{\ln t}{t}dt=\frac{(\ln t)^2}{2}\Big|_1^x=\frac{(\ln x)^2}{2}$

(iv) $\boxed{\int_a^b f(x)dx=-\int_b^a f(x)dx}$

***Illustration 7*:** Evaluate : $\int_2^3\frac{dx}{x\sqrt{4x^2+1}}$

Solution : $I=\int_2^3\frac{dx}{x\sqrt{4x^2+1}}$

Put $x=\frac{1}{t}\Rightarrow dx=-\frac{dt}{t^2}$

So $I=\int_{1/2}^{1/3}\frac{-dt}{t^2\left(\frac{1}{t}\right)\sqrt{\frac{4}{t^2}+1}}=-\int_{1/2}^{1/3}\frac{dt}{\sqrt{4+t^2}}$

$=\int_{1/3}^{1/2}\frac{dt}{\sqrt{4+t^2}}=\ln\left(t+\sqrt{4+t^2}\right)\Big|_{1/3}^{1/2}=\ln\left(\frac{3}{2}\left(\frac{\sqrt{17}+1}{\sqrt{37}+1}\right)\right)$

(v) $\boxed{\int_a^b f(x)dx=\int_a^{c_1} f(x)dx+\int_{c_1}^{c_2} f(x)dx+....+\int_{c_n}^b f(x)dx}$

***Illustration 8*:** Evaluate : $\int_{-2}^3|x^2-1|dx$

Solution : $\int_{-2}^3|x^2-1|dx=\int_{-2}^{-1}|x^2-1|dx+\int_{-1}^1|x^2-1|dx+\int_1^3|x^2-1|dx$

(Here modulus function will change at the points, when $x^2 - 1 = 0$ i.e. at $x = \pm 1$)

So $I=\int_{-2}^{-1}(x^2-1)dx+\int_{-1}^1(1-x^2)dx+\int_1^3(x^2-1)dx$

$=\frac{x^3}{3}-x\Big|_{-2}^{-1}+x-\frac{x^3}{3}\Big|_{-1}^1+\frac{x^3}{3}-x\Big|_1^3$

$=\left(\frac{-1}{3}+1\right)-\left[\frac{-8}{3}+2\right]+\left[1-\frac{1}{3}\right]-\left[-1+\frac{1}{3}\right]+\left[\frac{27}{3}-3\right]-\left[\frac{1}{3}-1\right]$

$=\frac{2}{3}+\frac{2}{3}+\frac{2}{3}+\frac{2}{3}+6+\frac{2}{3}=\frac{28}{3}$

(vi) $\boxed{\int_a^b f(x)dx = \int_a^b f(a+b-x)dx}$

***Illustration 9*:** Evaluate : $\int_2^7 \frac{\sqrt{x}\,dx}{\sqrt{x}+\sqrt{9-x}}$

Solution : $I = \int_2^7 \frac{\sqrt{x}\,dx}{\sqrt{x}+\sqrt{9-x}}$ .... (i)

$$I = \int_2^7 \frac{\sqrt{9-x}}{\sqrt{9-x}+\sqrt{9-(9-x)}}dx$$

$$I = \int_2^7 \frac{\sqrt{9-x}}{\sqrt{9-x}+\sqrt{x}}dx \quad \text{.... (ii)}$$

Adding (i) and (ii), we get

$$2I = \int_2^7 \left(\frac{\sqrt{x}}{\sqrt{x}+\sqrt{9-x}} + \frac{\sqrt{9-x}}{\sqrt{x}+\sqrt{9-x}} \right) dx = \int_2^7 dx = x\Big|_2^7 = 7-2 = 5 \text{ so } I = \frac{5}{2}$$

Student Task

If $f(x)$ is an even function $\int_{-a}^{0} f(x)dx = \int_{-a}^{0} f(a+x)dx$?

(vii) $\boxed{\int_{-a}^{a} f(x)dx = \int_0^a f(x)+f(-x))dx}$

***Illustration 10*:** Evaluate : $\int_{-\sqrt{3}}^{\sqrt{3}} \frac{dx}{(1+e^x)(1+x^2)}$

Solution : $I = \int_{-\sqrt{3}}^{\sqrt{3}} \frac{dx}{(1+e^x)(1+x^2)}$

Here $f(x) = \frac{1}{(1+e^x)(1+x^2)}$

$$\Rightarrow \quad f(-x) = \frac{1}{(1+e^{-x})(1+(-x)^2)} = \frac{e^x}{(1+e^x)(1+x^2)}$$

$$f(x)+f(-x) = \frac{1}{1+x^2}\left[\frac{1}{1+e^x}+\frac{e^x}{1+e^x}\right] = \frac{1}{1+x^2}$$

So $I = \int_0^{\sqrt{3}} \frac{dx}{1+x^2} = \tan^{-1}x\Big|_0^{\sqrt{3}} = \frac{\pi}{3}$

Student Task

If $f(x)$ is an odd function, what will be the value of $\int_{-a}^{a} f(x)dx$?

(viii) $\boxed{\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx}$

***Illustration 11*:** Evaluate : $\int_0^{\pi} \frac{x\,dx}{1+\cos^2 x}$

Solution : $I = \int_0^{\pi} \frac{x\,dx}{1+\cos^2 x}$

$$I = \int_0^{\pi} \frac{(\pi - x)dx}{1+\cos^2(\pi - x)} = \int_0^{\pi} \frac{(\pi - x)dx}{1+\cos^2 x}$$

Adding both, we get

$$2I = \int_0^{\pi} \frac{\pi\,dx}{1+\cos^2 x}$$

$$\Rightarrow \quad I = \frac{\pi}{2}\int_0^{\pi} \frac{dx}{1+\cos^2 x}$$

$$= \frac{\pi}{2}\left[\int_0^{\pi/2} \frac{dx}{1+\cos^2 x} + \int_0^{\pi/2} \frac{dx}{1+\cos^2(\pi - x)}\right]$$

$$= \frac{\pi}{2}\left[2\int_0^{\pi/2} \frac{dx}{1+\cos^2 x}\right] = \pi\int_0^{\pi/2} \frac{dx}{1+\cos^2 x}$$

$$= \pi\int_0^{\pi/2} \frac{\sec^2 x\,dx}{2+\tan^2 x} \text{ put } \tan x = t$$

$$I = \pi\int_0^{\infty} \frac{dt}{t^2+2} = \frac{\pi}{\sqrt{2}}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)\Big|_0^{\infty} = \frac{\pi^2}{2\sqrt{2}}$$

(ix) $\boxed{\int_a^b f(x)dx = (b-a)\int_0^1 f((b-a)x + a)dx}$

***Illustration 12*:** Prove that $\int_{-5}^{-4} e^{(x+4)^2}dx = 3\int_{1/3}^{2/3} e^{9(x-2/3)^2}dx$.

Solution : Let $I = 3\int_{1/3}^{2/3} e^{9(x-2/3)^2}dx$

$$= 3\left[\left(\frac{2}{3}-\frac{1}{3}\right)\right]\int_0^1 e^{9\left(\left(\frac{2}{3}-\frac{1}{3}\right)x+\frac{1}{3}-\frac{3}{3}\right)^2} dx$$

$$= \int_0^1 e^{9\left(\frac{x}{3}-\frac{1}{3}\right)^2} dx = \int_0^1 e^{(x-1)^2} dx$$

$$\text{Also } \int_{-5}^{-4} e^{(x+4)^2} dx = (-4-(-5))\int_0^1 e^{((-4-(-5))x-5+4)^2} dx$$

$$= \int_0^1 e^{(x-1)^2} dx$$

EXERCISE - 01

Q.1 Evaluate : $\int_2^3 \frac{dx}{x^2}$

Q.2 Evaluate : $\int_0^{\pi/2} \sqrt{\sin x}\cos^5 x\, dx$

Q.3 Evaluate : $\int_0^{\pi} \frac{dx}{1+\sin x}$

Q.4 Evaluate : $\int_{-5}^{5} |x+2|\, dx$

Q.5 Evaluate : $\int_0^{\pi/2} \frac{dx}{1+\tan^5 x}$

Q.6 Evaluate : $\int_0^1 \ln x\, dx$

Q.7 Evaluate : $\int_0^1 x(1-x)^n dx$

Q.8 Evaluate : $\int_0^{\pi/2} (2\ln(\sin x)-\ln(\sin 2x))dx$

Q.9 Evaluate : $\int_0^{\pi/2} \frac{x\, dx}{\sin x+\cos x}$

Answer Key

1. 1/6 2. $\frac{64}{231}$ 3. 2 4. 29 5. $\pi/4$ 6. –1

7. $\frac{1}{(n+1)(n+2)}$ 8. $\frac{\pi}{4}\ln\frac{1}{4}$ 9. $\frac{\pi}{2\sqrt{2}}\ln(1+\sqrt{2})$

5. PERIODIC PROPERTIES OF DEFINITE INTEGRAL

(i) If f(x) is a periodic function with period p, then $\int_a^{a+np} f(x)dx = n\int_0^p f(x)dx, n \in I$

***Illustration 13*:** Prove that $\int_0^{n\pi+v} |\sin x|\,dx = (2n+1) - \cos v$, where $n \in N$ and $0 \le v < \pi$.

Solution : $I = \int_0^{n\pi+v} |\sin x|\,dx = \int_0^v |\sin x|\,dx + \int_v^{n\pi+v} |\sin x|\,dx = I_1 + I_2$

$I = \int_0^v |\sin x|\,dx = \int_0^v \sin x\,dx$

(as $0 \le v < \pi$ and $\sin x \ge 0$, when $n \in [0, \pi]$)

$= -\cos x\big|_0^v = -\cos v + 1 = 1 - \cos v$

$I_2 = \int_v^{n\pi+v} |(\sin x)|\,dx = n\int_0^{\pi} |(\sin x)|\,dx$ (as $|\sin x|$ is periodic with period π)

$= n\int_0^n \sin x\,dx = n[-\cos x]_0^{\pi}$

$= -n\,[-1 - 1] = 2n$

So $I = 1 - \cos v + 2n = (2n + 1) - \cos v$

(ii) If f(x) is a periodic function with period p, then $\int_{mp}^{np} f(x)dx = (n-m)\int_0^p f(x)dx, n, m \in I$

***Illustration 14*:** Evaluate : $\int_{-3/2}^{10} \{2x\}dx$, where { } denotes the fractional part of x.

Solution : $f(x) = \{2x\}$ is a periodic function with period $\frac{1}{2}$

Let $I = \int_{-3/2}^{10} \{2x\}dx = \int_{-3(1/2)}^{20(1/2)} \{2x\}dx$

$= (20 - (-3))\int_0^{1/2} \{2x\}dx$

$= 23\int_0^{1/2} 2x\,dx$ (as $\{2x\} = 2x - [2x]$ and where $x \in [0, ½)$, $[2x] = 0$)

$= 23x^2\big|_0^{1/2} = \frac{23}{4}$

(iii) If f(x) is a periodic function with period p, then $\int_{a+np}^{b+np} f(x)dx = \int_{a}^{b} f(x)dx, n \in I$

***Illustration 15*:** Evaluate : $\int_{10\pi+\frac{\pi}{6}}^{10\pi+\frac{\pi}{3}} (\sin x + \cos x)dx$.

Solution : $f(x) = \sin x + \cos x$ is periodic with period 2π.

Let $I = \int_{10\pi+\frac{\pi}{6}}^{10\pi+\frac{\pi}{3}} (\sin x + \cos x)dx = \int_{\pi/6}^{\pi/3} (\sin x + \cos x)dx$

$= (\sin x - \cos x)\Big|_{\pi/6}^{\pi/3}$

$= \left[\frac{\sqrt{3}}{2} - \frac{1}{2}\right] - \left[\frac{1}{2} - \frac{\sqrt{3}}{2}\right] = (\sqrt{3} - 1)$

Student Task

Q. Find $\int_{0}^{10\pi} (|\sin x| + |\cos x|)dx$.

6. DIFFERENTIATION OF DEFINITE INTEGRAL

(i) If $F(x) = \int_{f_1(x)}^{f_2(x)} g(t)dt$ then $F'(x) = g(f_2(x))f'_2(x) - g(f_1(x))f'_1(x)$

***Illustration 16*:** If a, b are variable real numbers such that $a + b = 4$, $a < 2$ and $f'(x) > 0 \ \forall \ x \in R$, then prove that

$\left(\int_{0}^{a} f(x)dx + \int_{0}^{b} f(x)dx\right)$ will increase as $(b - a)$ increases.

Solution : Let $(b - a) = t$

$b + a = 4$

$\Rightarrow \quad b = \frac{4+t}{2}, a = \frac{4-t}{2}$

Let $g(t) = \int_{0}^{\frac{4-t}{2}} f(x)dx + \int_{0}^{\frac{4+t}{2}} f(x)dx$

So, $g'(t) = f\left(\frac{4-t}{2}\right)\left(-\frac{1}{2}\right) + f\left(\frac{4+t}{2}\right)\left(\frac{1}{2}\right)$

$= \frac{1}{2}\left[f\left(\frac{4+t}{2}\right) - f\left(\frac{4-t}{2}\right)\right]$

Now $a < 2$ and $a + b = 4 \Rightarrow a < b$

$\Rightarrow \quad \frac{4-t}{2} < \frac{4+t}{2}$

$\Rightarrow \quad f\left(\frac{4-t}{2}\right) < f\left(\frac{4+t}{2}\right)$ (as $f'(x) > 0 \Rightarrow f(x)$ is increasing)

$\Rightarrow \quad g'(t) > 1 \Rightarrow g(t)$ will increases as t increases

$\Rightarrow \quad \int_0^a f(x)dx + \int_0^b f(x)dx$ will increases as $(b - a)$ increases

Student Task

Q. Find the interval in which $\int_0^x \frac{\ln t}{t}$ is decreasing ?

(ii) If $I(t) = \int_a^b f(x,t)dx$, then $\frac{dI}{dt} = \int_a^b f'(x,t)dx$, where $f'(x, t)$ denotes the derivative of f w.r.t. 't' keeping x constant.

***Illustration 17*:** Evaluate : $\int_0^1 \left(\frac{x^t - 1}{\ln x}\right)dx; \ (t \geq 0)$

Solution : Let $I(t) = \int_0^1 \left(\frac{x^t - 1}{\ln x}\right)dx$

$\Rightarrow \quad \frac{dI}{dt} = \int_0^1 \left(\frac{x^t \ln x}{\ln x}\right)dx$

$= \int_0^1 x^t dx = \frac{x^{t+1}}{t+1}\Big|_0^1 = \frac{1}{t+1}$

$\Rightarrow \quad dI = \frac{dt}{t+1}$

$\Rightarrow \quad \int dI = \int \frac{dt}{t+1}$

$\Rightarrow \quad I = \ln |t + 1| + c$

but at $t = 0$, $I(t) = 0$

So $c = 0$

Hence $I = \ln |t + 1|$

7. APPROXIMATION IN DEFINITE INTEGRAL

(i) If $f_1(x) \le f(x) \le f_2(x) \ \forall \ x \in [a, b]$, then $\boxed{\int_a^b f_1(x) \le \int_a^b f_2(x) \le \int_a^b f_3(x)}$.

***Illustration 18*:** Prove that $\int_0^1 \frac{dx}{\sqrt{4 - x^2 - x^3}} \le \frac{\pi}{4\sqrt{2}}$.

Solution : $0 \le x \le 1$

$\Rightarrow \ 0 \le x^3 \le x^2 \le 1$

$\Rightarrow \ -x^2 \le -x^3 \le 0$

$\Rightarrow \ 4 - x^2 - x^2 \le 4 - x^2 - x^3 \le 4 - x^2$

$\Rightarrow \ \frac{1}{\sqrt{4 - x^2}} \le \frac{1}{\sqrt{4 - x^2 - x^3}} \le \frac{1}{\sqrt{4 - 2x^2}}$

$\Rightarrow \ \int_0^1 \frac{dx}{\sqrt{4 - x^2}} \le \int_0^1 \frac{dx}{\sqrt{4 - x^2 - x^3}} \le \int_0^1 \frac{dx}{\sqrt{4 - 2x^2}}$

$\Rightarrow \ \sin^{-1}\left(\frac{x}{2}\right)\Big|_0^1 \le \int_0^1 \frac{dx}{\sqrt{4 - x^2 - x^3}} \le \frac{1}{\sqrt{2}} \sin^{-1}\left(\frac{x}{\sqrt{2}}\right)\Big|_0^1$

$\Rightarrow \ \frac{\pi}{6} \le \int_0^1 \frac{dx}{\sqrt{4 - x^2 - x^3}} \le \frac{\pi}{4\sqrt{2}}$

(ii) $\boxed{\text{If absolute maximum and minimum value of f(x), when } x \in [a, b] \text{ is M and m respectively, then } m(b - a) \le \int_a^b f(x)dx \le M(b - a)}$

***Illustration 19*:** Prove that $\frac{\pi}{\pi^3 + 10\pi + 5} < \int_0^{\pi} \frac{dx}{x^3 + 10x + 9\sin x + 5} < \frac{\pi}{5}$.

Solution : Let $f(x) = x^3 + 10x + 9 \sin x + 5$

$f'(x) = 3x^2 + 10 + 9 \cos x > 0 \ \forall \ x \in R$

$\Rightarrow$ $f(x)$ is entirely increasing $\Rightarrow \frac{1}{f(x)}$ is decreasing in $(0, \pi)$

$\Rightarrow$ Absolute maximum of $f(x)$ in $[0, \pi]$ is $\frac{1}{5}$ and absolute minimum is $\frac{1}{\pi^3 + 10\pi + 5}$

So $\frac{\pi}{\pi^3 + 10\pi + 5} < \int_0^{\pi} \frac{dx}{x^3 + 10x + 9\sin x + 5} < \frac{\pi}{5}$

8. DEFINITE INTEGRAL OF PIECEWISE CONTINUOUS FUNCTIONS

Suppose we have to evaluate $\int_a^b f(x)dx$, but either $f(x)$ is not continuous at $x = c_1, c_2 \ldots, c_n$ or it is not defined at these points. In both cases we have to break the limit at $c_1, c_2, \ldots, c_n$.

***Illustration 20*:** Evaluate $\int_1^2 [x^3 - 1]dx$ where [.] denotes the greatest integer function.

Solution : $1 \le x \le 2$

$\Rightarrow \quad 1 \le x^3 \le 8$

$\Rightarrow \quad 0 \le x^3 - 1 \le 7$

So $I = \int_1^2 [x^3 - 1]dx$

$$= \int_1^{2^{1/3}} [x^3 - 1]dx + \int_{2^{1/3}}^{3^{1/3}} [x^3 - 1]dx + \ldots\ldots + \int_{7^{1/3}}^{2} [x^3 - 1]dx$$

Now if $x \in \left[1, 2^{\frac{1}{3}}\right)$, then $x^3 \in [1, 2)$ or $[x^3 - 1] = 0$ and so on

therefore $I = \int_1^{2^{1/3}} 0.dx + \int_{2^{1/3}}^{3^{1/3}} 1.dx + \ldots\ldots\ldots \int_{7^{1/3}}^{2} 6.dx$

$= 3^{1/3} - 2^{1/3} + 2[4^{1/3} - 3^{1/3}] + 3[5^{1/3} - 4^{1/3}] + 4[6^{1/6} - 5^{1/3}] + 5[7^{1/3} - 6^{1/3}] + 6[2 - 7^{1/3}]$

$= 12 - [7^{1/3} + 6^{1/3} + 5^{1/3} + 4^{1/3} + 3^{1/3} + 2^{1/3}]$

***Illustration 21*:** Prove that $\int_a^b \frac{|x|}{x}dx = |b| - |a|$.

Solution : We can divide all the possible values of a and b in 3 cases

Case I : $0 \le a < b$

$$I = \int_a^b \frac{|x|}{x}dx = \int_a^b \frac{x}{x}dx = b - a = |b| - |a|$$

Case II : $a < b \le 0$

$$I = \int_a^b \frac{|x|}{x}dx = \int_a^b \frac{-x}{x}dx = a - b = -|a| - (-|b|) = |b| - |a|$$

Case III : $a < 0 < b$

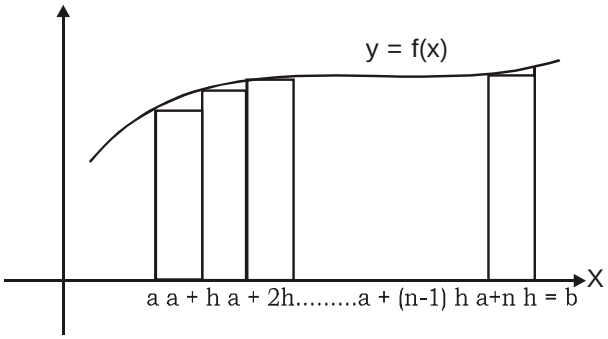
$$I = \int_a^0 (-1)dx + \int_0^b 1\,dx$$

$= a + b = -|a| + |b|$

$= |b| - |a|$

9. DEFINITE INTEGRAL AS THE LIMIT OF A SUM

Let $f(x)$ be a continuous real valued function defined on the closed interval $[a,b]$ which is divided into n parts as shown in figure.



The point of division on x-axis are $a, a+h, a+2h.....a+(n-1)h, a+nh$, where $\frac{b-a}{n}=h$.

Let S_n denotes the area of these n rectangles.

Then, $S_n = hf(a)+hf(a+h)+hf(a+2h)+.....+hf(a+(n-1)h)$

Clearly, S_n is area very close to the area of the region bounded by curve $y=f(x)$, x-axis and the ordinates $x=a, x=b$.

Hence $\int_a^b f(x)dx = \lim_{n\to\infty} S_n \Rightarrow \int_a^b f(x)dx = \lim_{n\to\infty}\sum_{r=0}^{n-1} h\ f(a+rh) = \lim_{n\to\infty}\sum_{r=0}^{n-1}\left(\frac{b-a}{n}\right) f\left(a+\frac{(b-a)r}{n}\right)$

Note :

1. We can also write

$S_n = hf(a+h)+hf(a+2h)+.....+hf(a+nh)$ and $\int_a^b f(x)dx = \lim_{n\to\infty}\sum_{r=1}^{n}\left(\frac{b-a}{n}\right) f\left(a+\left(\frac{b-a}{n}\right)r\right)$

2. If $a=0, b=1 \Rightarrow \int_0^1 f(x)dx = \lim_{n\to\infty}\sum_{r=0}^{n-1}\frac{1}{n} f\left(\frac{r}{n}\right)$

Step to express the limit of sum as definite integral

Step - 1. Replace $\frac{r}{n}$ by $x, \frac{1}{n}$ by dx and $\lim_{n\to\infty}\sum$ by $\int$

Step - 2. Evaluate $\lim_{n\to\infty}\left(\frac{r}{n}\right)$ by putting least and greatest values of r as lower and upper limits respectively.

For example : $\lim_{n\to\infty}\sum_{r=1}^{np}\frac{1}{n} f\left(\frac{r}{n}\right) = \int_0^p f(x)dx \qquad \left(\because \lim_{n\to\infty}\left(\frac{r}{n}\right)\Big|_{r=1} = 0, \lim_{n\to\infty}\left(\frac{r}{n}\right)\Big|_{r=np} = p\right)$

***Illustration 22*:** Evaluate : $\lim_{n\to\infty}\left[\frac{1}{\sqrt{4n^2-1}}+\frac{1}{\sqrt{4n^2-4}}+\frac{1}{\sqrt{4n^2-9}}+.....\frac{1}{\sqrt{3n^2}}\right]$.

Solution : $L = \lim_{n\to\infty}\left[\frac{1}{\sqrt{4n^2-1}}+\frac{1}{\sqrt{4n^2-4}}+\frac{1}{\sqrt{4n^2-9}}+.....\frac{1}{\sqrt{3n^2}}\right]$

$= \lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{\sqrt{4n^2-r^2}}$

$$= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n} \frac{1}{\sqrt{4-(r/n)^2}}$$

$$= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{(1-0)}{n} \frac{1}{\sqrt{4-\left(0+4\left(\frac{1-0}{n}\right)\right)^2}}$$

which of the form

$$\lim_{n\to\infty} \sum_{r=1}^{n} \frac{b-a}{n} f\left(a+r\left(\frac{b-a}{n}\right)\right)$$

Here $b = 1$, $a = 0$ and $f(x) = \frac{1}{\sqrt{4-x^2}}$

So $L = \int_0^1 \frac{dx}{\sqrt{4-x^2}} = \sin^{-1}\frac{x}{2}\Big|_0^1 = \frac{\pi}{6}$

***Illustration 23*:** Evaluate : $\lim_{n\to\infty}\left[\frac{n^2}{(n+1)^3}+\frac{n^2}{(n+2)^3}+.....+\frac{1}{64n}\right]$.

Solution : $L = \lim_{n\to\infty}\left[\frac{n^2}{(n+1)^3}+\frac{n^2}{(n+2)^3}+.....+\frac{1}{64n}\right]$

$$= \lim_{n\to\infty} \sum_{r=1}^{3n} \frac{n^2}{(n+r)^3}$$

Put $3n = m$, we get

$$L = \lim_{m\to\infty} \sum_{r=1}^{m} \frac{m^2/9}{\left(\frac{m}{3}+r\right)^3}$$

$$= \lim_{m\to\infty} \sum_{r=1}^{m} \frac{3m^2}{(m+3r)^3}$$

$$= \lim_{m\to\infty} \sum_{r=1}^{m} \frac{3}{m}\left(\frac{1}{\left(1+\frac{3r}{m}\right)^3}\right)$$

$$= \lim_{m\to\infty} \sum_{r=1}^{m} \frac{(3-0)}{m}\left(\frac{1}{1+\left(0+\left(\frac{3-0}{m}\right)\right)r}\right)^3$$

$$= \int_0^3 \frac{dx}{(1+x)^3} = \frac{-1}{2(1+x)^2}\Big|_0^3 = -\frac{1}{2}\left[\frac{1}{16}-\frac{1}{1}\right] = \frac{1}{2}\left[\frac{15}{16}\right] = \frac{15}{32}$$

***Illustration 24*:** Evaluate : $\int_{-1}^{4}(e^{2x}-3x)dx$ as limit of sums.

Solution : Let $I=\int_{-1}^{4}(e^{2x}-3x)dx=\int_{-1}^{4}f(x)dx$,

where $f(x)=e^{2x}-3x$, $nh=b-a=4+1=5$

$$\therefore \quad \int_{a}^{b}f(x)dx=\lim_{h\to 0}h\{f(-1+h)+f(-1+2h)+f(-1+3h)+.....+f(-1+nh)\}$$

$$\therefore \quad \int_{-1}^{4}f(x)dx=\lim_{h\to 0}h\{f(-1+h)+f(-1+2h)+f(-1+3h)+.....+f(-1+nh)\}$$

$$=\lim_{h\to 0}h\left\{(e^{2(-1+h)}-3(-1+h)+(e^{2(-1+2h)}-3(-1+2h)+(e^{2(-1+3h)}-3(-1+3h)+....+(e^{2(-1+nh)}-3(-1+nh)\right\}$$

$$=\lim_{h\to 0}h\left\{e^{-2}.e^{2h}(1+e^{2h}+e^{4h}+.....\text{to } n \text{ terms})-3h(1+2+3+.....+n)+3n)\right\}$$

$$=\lim_{\substack{h\to 0\\ n\to\infty}}h\left[e^{-2}e^{2h}\left[\frac{(e^{2h})^{n}-1}{e^{2h}-1}\right]-3h\frac{n(n+1)}{2}+3n\right]$$

$$=\lim_{\substack{h\to 0\\ n\to\infty\\ nh=5\\ y_n=0}}\frac{e^{-2}e^{2h}(e^{2nh}-1)}{2\left(\frac{e^{2h}-1}{2h}\right)}-\frac{3}{2}(nh)^2(1+1/n)+3(nh)$$

$$=\frac{e^{-2}(e^{10}-1)}{2}-\frac{3}{2}(5)^2(1)+3(5)\ =\frac{e^{8}-e^{-2}}{2}-\frac{45}{2}$$

10. REDUCTION FORMULA IN DEFINITE INTEGRALS

$$\int_{0}^{\pi/2}\sin^{m}x\cos^{n}x\,dx=\frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{m+n+2}{2}\right)}$$

Where $\Gamma(n)$ is called Gamma function which satisfies the following properties

$$\Gamma(n+1)=n\Gamma(n)=n!,\Gamma(1)=1,\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$$

In place of gamma function, we can also use the following formula:

$$\int_{0}^{\pi/2}\sin^{m}x\cos^{n}x\,dx=\frac{\sqrt{\frac{m+1}{2}}\sqrt{\frac{n+1}{2}}}{2\sqrt{\frac{m+n+2}{2}}}$$

It is important to note that we multiply by $\left(\frac{\pi}{2}\right)$ when both *m* and *n* are even.

Walli's formula :

$$\int_{0}^{\pi/2}\sin^{n}x\,dx=\int_{0}^{\pi/2}\cos^{n}x\,dx=\begin{cases}\frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}.....\frac{2}{3}, & \text{when } n \text{ is odd}\\ \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}.....\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}, & \text{when } n \text{ is even}\end{cases}$$

EXERCISE - 2

Q.1 If $F(x)=\int_{x^2}^{e^x}\cos t\,dt$, find $F'(x)$.

Q.2 If $F(t)=\int_{2}^{3}\sin(x+t^2)dx$, find $F'(t)$.

Q.3 If $I=\int_{0}^{1}\frac{dx}{1+x^{3/2}}$, prove that, ln 2 < | < $\frac{\pi}{4}$.

Q.4 Evaluate : $\int_{0}^{100}\{x\}dx$, where $\{x\}$ denotes the fraction part of x.

Q.5 Evaluate : $\int_{0}^{41\pi/2}\sin x\,dx$

Q.6 Evaluate : $\int_{0}^{2}[2x]dx$, [.] denotes the greatest integer function.

Q.7 Find $\lim_{n\to\infty}\sum_{r=1}^{2n}\frac{1}{n+r}$

Q.8 Find $\lim_{n\to\infty}\left(\frac{n!}{n^n}\right)^{1/n}$

Q.9 Find the interval in which $F(x)=\int_{-1}^{x}(e^t-1)(2-t)dt,(x>-1)$ is increasing.

Q.10 Prove that $\int_{0}^{\infty}e^{-x}x^n dx=n!$ $(n\in N)$.

Answer Key

1. cos $(e^x)e^x$ – cos$(x^2)2x$ 2. $\int_{2}^{3}\cos(x+t^2)2t\,dx$ 4. 50 5. 1 6. 3

7. ln 3 8. 1/e 9. [0, 2]

11. APPLICATION OF INTEGRATION TO AREAS

Definite integral is used to evaluate areas bounded by curves. To do problems under this heading, one must be able to draw a rough figure of the curve when the equation is given. Some rules about drawing curves are given below. Familiar curves likes lines, circles and conics are not discussed here.

Guidelines

(i) Check whether the curve is symmetrical about the x-axis or not. The curve is symmetrical about the x-axis, if its equation is unchanged when y is replaced by $-y$.

(ii) The curve is symmetrical about the y-axis if its equation is unchanged when x is replaced by $-x$.

(iii) Put $y = 0$ in the equation of the curve. This will give the points where it cuts the x-axis.

(iv) Put $x = 0$ in the equation of the curve. This will give the points where it cuts the y-axis.

(v) The curve is symmetrical about the line $y = x$ if its equation does not change when x and y are interchanged.

(vi) Find the turning points of the graph by equating $\frac{dy}{dx} = 0$.

(vii) Find the intervals of curve in which it increases and decreases if required.

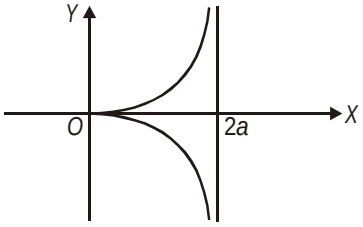
(viii) Use periodicity wherever possible.

(ix) Check behaviour at $x \to \pm\infty$ and $y \to \pm\infty$.

***Illustration 25*:** Trace the curve $y^2(2a - x) = x^3$, $a > 0$.

Soluiton : Note that the curve passes through the origin and is symmetrical about the x-axis

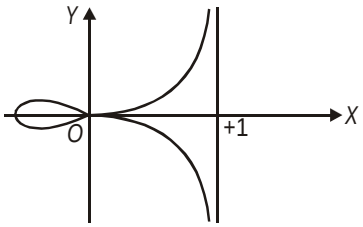
$$y^2 = \frac{x^3}{2a - x}$$



L.H.S. is positive. If x is greater than $2a$, R.H.S. becomes negative. Hence the curve lies only in the interval 0 to $2a$. When $x \to 2a$, $y \to \infty$. Therefore the line $x = 2a$ is an asymptote for the curve. A rough Figure is shown.

***Illustration 26*:** Trace the curve $y^2 = \frac{x^2(1+x)}{1-x}$.

Solution : The curve passes through the origin and is symmetrical about the x-axis. It intersects the x-axis at $x = -1$ anx $x = 0$. If the $x < -1$ or $x > 1$ the curve is non-existent. As $x \to 1$, $y \to \pm\infty$ *a* rough diagram is shown below.



The curve has a loop between –1 and 0.

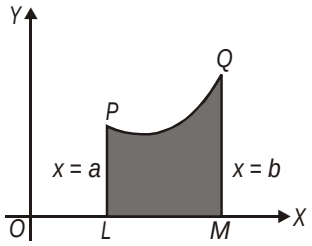
Student Task

Can you sketch the region bounded by the curves $x = 2y^2 - 2y$ and $x = 12y^2 - 12y^3$?

12. ESTIMATION OF AREAS

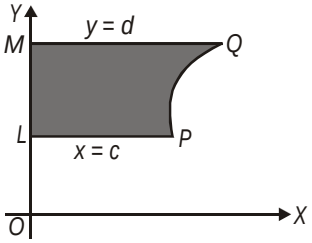
Four cases are discussed below :

Case I : *PQ* is an arc of a curve whose equation is $y = f(x)$. We have an area bounded by *PQ* on one side; by the x-axis on another and the two parallel lines $x = a$ and $x = b$ (shown by *PL* and *QM*), $a < b$.



The area $PLMQ = \int_{x=a}^{x=b} y\,dx = \int_{a}^{b} f(x)dx$

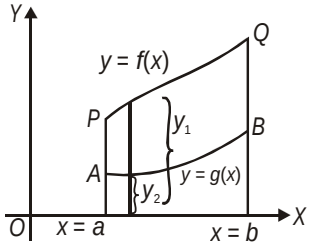
Case II : *PQ* is an arc of a curve whose equation is $y = f(x)$ or $x = f(y)$.
In this case y-axis is one boundary and the other two are the lines $y = c$ and $y = d$.



The area $LPQM = \int_{y=c}^{y=d} y\,dx = \int_{c}^{d} f(y)dy$

In this case the integration is with respect to *y*.

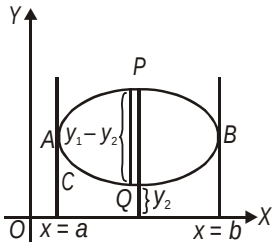
Case III : The figure encloses an area between two curves one of which is represented by *PQ* with equation $y = f(x)$ and the other by *AB* with the equation $y = g(x)$.



Area $PABQ = \int_{a}^{b} (y_1 - y_2)dx$ where $y_1 = f(x)$ and $y_2 = g(x)$ $= \int_{a}^{b} \{f(x) - g(x)\}dx$

Case IV : The figure represents the region bounded by a closed curve *ACQBP*.

The area of the region bounded by a closed curve *ACQBP* is $\int_{a}^{b} (y_1 - y_2)dx, y_1 > y_2$



The values of y_1 and y_2 are obtained by solving the equation of the curve as a quadratic in y whose large root y_1 and smaller root y_2 are functions of x.

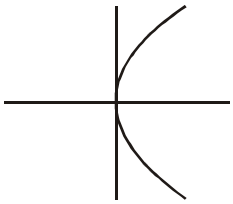
13. CURVE TRACING

To find the approximate shape of a curve, the following procedure is adopted in order.

(a) Symmetry :

(i) Symmetry about *x*-axis:

If all the powers of 'y' in the equation are even then the curve is symmetrical about the x-axis.

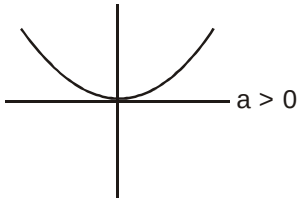


a > 0

E.g. : $y^2 = 4ax$.

(ii) Symmetry about *y*-axis:

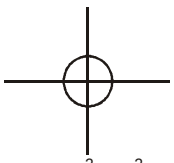
If all the powers of 'x' in the equation are even then the curve is symmetrical about the y-axis.



E.g. : $x^2 = 4ay$.

(iii) Symmetry about both axis:

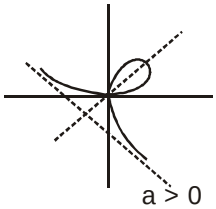
If all the powers of 'x' and 'y' in the equation are even, the curve is symmetrical about the axis of 'x' as well as 'y'.



E.g. : $x^2 + y^2 = a^2$

(iv) Symmetry about the line $y = x$:

If the equation of the curve remains unchanged on interchanging 'x' and 'y', then the curve is symmetrical about the line $y = x$.

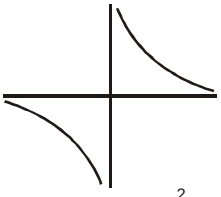


a > 0

E.g. : $x^3 + y^3 = 3axy$

(v) Symmetry in opposite quadrants:

If the equation of the curve ramins unlatered when 'x' and 'y' are replaced by $-x$ and $-y$ respectively, then there is symmetry in opposite quadrants.



E.g. : $xy = c^2$

(b) Find the points where the curve crosses the x-axis and also the *y*-axis

(c) Find $\frac{dy}{dx}$ and equate it to zero to find the points on the curve where you have horizontal tangents.

(d) Examine if possible the intervals when $f(x)$ is increasing or decreasing.

(e) Examine that happens to 'y' when $x \to \infty$ or $x \to -\infty$.

(f) Asymptotes :

Asymptote(s) is (are) line (s) whose distance from the curve tends to zero as point on curve moves towards infinity along branch of curve.

(i) If $\lim_{x \to a} f(x) = \infty$ or $\lim_{x \to a} f(x) = -\infty$, then $x = a$ is asymptote of $y = f(x)$

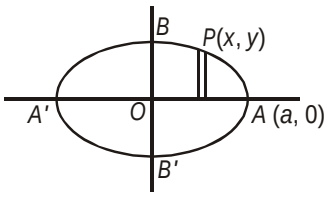
(ii) If $\lim_{x \to +\infty} f(x) = k$ or $\lim_{x \to -\infty} f(x) = k$, then $y = k$ is asymptote of $y = f(x)$

(iii) If $\lim_{x \to \infty} \frac{f(x)}{x} = m_1, \lim_{x \to \infty} (f(x) - m_1 x) = c,$ then $y = m_1 x + c_1$ is an asymptote. (inclined to right)

(iv) If $\lim_{x \to -\infty} \frac{f(x)}{x} = m_2, \lim_{x \to -\infty} (f(x) - m_2 x) = c_2,$ then $y = m_2 x + c_2$ is an asymptote (inclined to left)

***Illustration 27*:** Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$

Solution : The ellipse is symmetrical about both axes and hence the area enclosed = 4 (area of the quadrant)



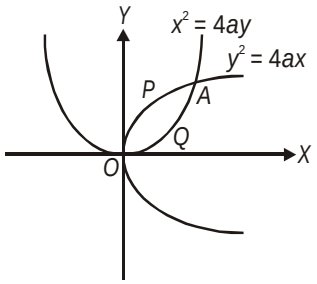
$$= 4\int_0^a y\,dx = 4\int_0^a b\sqrt{1 - \frac{x^2}{a^2}}\,dx = \frac{4b}{a}\int_0^a \sqrt{(a^2 - x^2)}\,dx = \frac{4b}{a}\left[\frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_0^a$$

$$= \frac{4b}{a}\left[\frac{a^2\pi}{4}\right] = \pi ab \text{ sq. units}$$

***Illustration 28*:** Find the area included between the parabola $y^2 = 4ax$ and $x^2 = 4ay$.

Solution :

The two parabolas intersect at $O(0, 0)$ and $A(4a, 4a)$. The area included between the two curves = area $OQAP$.



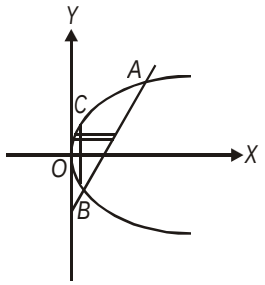
$$= \int_{x=0}^{x=4a} (y_1 - y_2)dx = \int_0^{4a} \left(2\sqrt{a}\sqrt{x} - \frac{x^2}{4a} \right) dx = \left[2\sqrt{a}\frac{2}{3}x^{3/2} - \frac{x^3}{12a} \right]_0^{4a}$$

$$= 4\frac{\sqrt{a}}{3}8.a^{3/2} - \frac{64a^3}{12a} = \frac{16a^2}{3} \text{ sq. units}$$

Note :- Sometimes it is better to use the formula $\int_c^d x\,dy$ instead of $\int_a^b y\,dx$ in the computation of area to simplify calculations, as the following illustration shows.

***Illustration 29*:** Find the area of the segment cut off from the parabola $y^2 = 2x$ by the line $y = 4x - 1$.

Solution : The line $y = 4x - 1$ intersects the parabola $y^2 = 2x$ at A and B



$2x = (4x - 1)^2 \Rightarrow 16x^2 - 10x + 1 = 0$

$\Rightarrow (8x - 1)(2x - 1) = 0$

$\therefore \quad A = \left(\frac{1}{2}, 1 \right)$ and $B = \left(\frac{1}{8}, -\frac{1}{2} \right)$

If the formula $\int y\,dx$ is to be used than the area will have to split up as OBC and CBA.

Instead the problem can be done directly by using the formula $\int (x_2 - x_1)dy$

Area required = $\int_{y=-1/2}^{1} (x_2 - x_1)dy$

$$= \int_{-1/2}^{1}\left(\frac{y+1}{4}-\frac{y^2}{2}\right)dy = \left[\frac{y^2}{8}+\frac{y}{4}-\frac{y^3}{6}\right]_{-1/2}^{1}$$

$$= \left(\frac{1}{8}+\frac{1}{4}-\frac{1}{6}\right)-\left(\frac{1}{32}-\frac{1}{8}+\frac{1}{48}\right)$$

$$= \frac{(3+6-4)}{24}-\frac{(3-12+2)}{96} = \frac{5}{24}+\frac{7}{96} = \frac{27}{96} = \frac{9}{32} \text{ sq. units}$$

***Illustration 30*:** Find the area bounded by the curve $xy^2 = a^2(a-x)$ and $x-$axis.

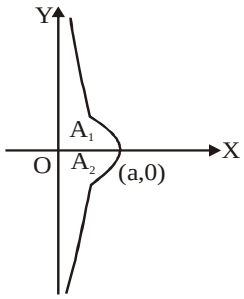
Solution: $xy^2 = a^2(a-x) \Rightarrow y^2 = a^2(a-x)/x$ or $y = a\sqrt{\frac{a-x}{x}}$

Here the power of *y* is even, so curve is symmetric w.r.t $x-$axis. Putting $x = a$ in equation (1) we get $y = 0$ and at $x = 0$, we get $y = \infty$.

$\therefore$ Required area $A = A_1 + A_2 = A_1$

$$A = 2\int_{x=0}^{x=a} y\,dx = 2\int_0^a a\sqrt{\frac{a-x}{x}}\,dx \qquad \text{(from Eq. (1))}$$

Putting $x = a\sin^2\theta$, $dx = 2a\sin\theta\cos\theta d\theta$



When $x = 0 \quad \Rightarrow \theta = 0 \qquad \Rightarrow x = a \Rightarrow \theta = \frac{\pi}{2}$

$$\therefore \text{ area } = 2\int_0^{\pi/2} a\,\frac{a\cos\theta}{a\sin\theta}.2a\cos\theta\sin\theta\,d\theta = 4a^2\int_{02}^{\pi/2}\cos^2\theta\,d\theta = 4a^2.\frac{1}{2}.\frac{\pi}{2} = \pi a^2$$

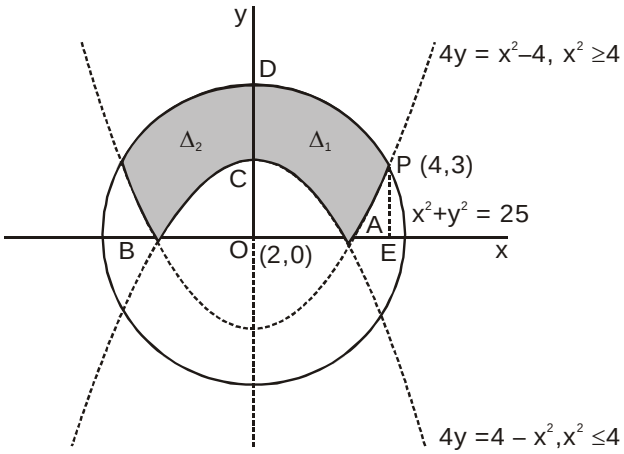
***Illustration 31*:** Find the area bounded by the curves $x^2+y^2 = 25, 4y = |4-x^2|$ and $x = 0$, above the $x-$axis.

Solution: Here the curves are

(i) $x^2+y^2 = 25$ which is a circle of radius 5 and its centre is (0,0),

(ii) $4y = 4-x^2 \le 4$. This is the parabola $x^2 = 4(1-y)$ whose vertex is (0,1), the axis is $x = 0$ and at all points $y \le 1$. As $x^2 \le 4$, only that part of the parabola is to be taken for which $-2 \le x \le 2$.

(iii) $4y = x^2-4, x^2 \ge 4$, this is the parabola $x^2 = 4(1+y)$ whose vertex is $(0,-1)$, the axis is $x = 0$ and at all points $y \ge -1$. As $x^2 \ge 4$, only that part of the parabola is to be taken for which $x \ge 2$ or $x \le -2$.



Clearly, there areas two bounded by the curves and the y-axis, and by symmetry they are equal. So $\Delta_1 = \Delta_2$. Now, $\Delta_1 = ar(ODPEO) - ar(OCAO) - ar(APEA)$ .....(1)

Solving $x^2 + y^2 = 25$ and $4y = x^2 - 4$, we get $4 + 4y + y^2 = 25$ or $y^2 + 4y - 21 = 0$

So $(y + 7)(y - 3) = 0$

$\therefore$ $y = 3, -7$ But y cannot be – 7.

So $y = 3$ and $x = 4$.

$\therefore$ $P = (4, 3)$.

Solving $4y = 4 - x^2$ and $y = 0$, we get $4 - x^2 = 0$, i.e., $x = \pm 2$. So $A = (2, 0)$ and $B = (-2, 0)$.

Now, $ar(ODPEO) = \int_0^4 (y)_{circle}\, dx$

$$= \int_0^4 \sqrt{25 - x^2}\, dx = \int_0^4 \sqrt{5^2 - x^2} dx = \left[\frac{25}{2}\sin^{-1}\frac{x}{5} + \frac{x\sqrt{25 - x^2}}{2}\right]_0^4 = \frac{25}{2}\sin^{-1}\frac{4}{5} + 6.$$

$ar(OCAO) = \int_0^2 y dx$, where $4y = 4 - x^2$

$$= \int_0^2 \left(1 - \frac{x^2}{4}\right) dx = \left[x - \frac{x^3}{12}\right]_0^2 = 2 - \frac{8}{12} = \frac{4}{3}$$

$ar\ (APEA) = \int_2^4 y dx$, where $4y = x^2 - 4$

$$= \int_2^4 \left(\frac{x^2}{4} - 1\right) dx = \left[\frac{x^3}{12} - x\right]_2^4 = \left(\frac{64}{12} - 4\right) - \left(\frac{8}{12} - 2\right) = \frac{8}{3}.$$

$\therefore$ by (1), $\Delta_1 = \frac{25}{2}\sin^{-1}\frac{4}{5} + 6 - \frac{4}{3} - \frac{8}{3} = \frac{25}{2}\sin^{-1}\frac{4}{5} + 2$.

Also, $\Delta_2 = \Delta_1 = \frac{25}{2}\sin^{-1}\frac{4}{5} + 2$.

***Illustration 32*:** Find the area of the region R given by $R = \{(x, y): 4 \le x^2 + y^2 \le 2(|x| + |y|)\}$

Solution: Given curve $4 \le x^2 + y^2 \le 2(|x| + |y|)$ is symmetrical about x-axis as well as about y-axis, therefore required area is 4 times the area of the region in 1st quadrant. Now in

1st quadrant, given inequalities consist of two inequalities :

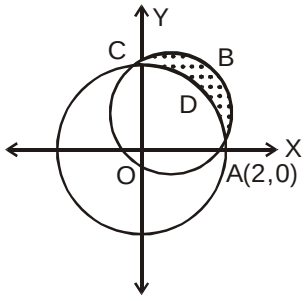
$x^2 + y^2 \geq 4$ .....(1)

and $x^2 + y^2 - 2x - 2y \leq 0$ .....(2)

(1) represents exterior (and boundary) of circle $x^2 + y^2 = 4$ and (2) represents interior (and boundary) of circle $(x-1)^2 + (y-1)^2 = 2$.

Therefore required region in 1st quadrant is ABCDA.

Area = Area of OABCO – Area of OACDA.



$$= \int_0^2 \left\{1 + \sqrt{2-(x-1)^2}\right\} dx - \int_0^2 \sqrt{4-x^2}\,dx$$

$$= [x]_0^2 + \left[\sin^{-1}\left(\frac{x-1}{\sqrt{2}}\right) + \frac{(x-1)}{2}\sqrt{2-(x-1)^2}\right]_0^2 - \left[2\sin^{-1}\frac{x}{2} + \frac{x}{2}\sqrt{4-x^2}\right]_0^2$$

$$= 2 + \sin^{-1}\frac{1}{\sqrt{2}} + \frac{1}{2} - \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) + \frac{1}{2}2\sin^{-1}(1)$$

$$= 2 + 2.\frac{\pi}{4} + 1 - 2\frac{\pi}{2} = \left(3 - \frac{\pi}{2}\right)$$

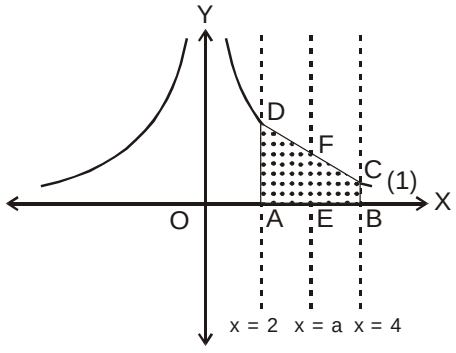
Therefore required area $= 4\left(3 - \frac{\pi}{2}\right)$

***Illustration 33*:** Find the area bounded by the *x*–axis, part of the curve $y = 1 + \frac{8}{x^2}$ and the ordinate at $x = 2$ and $x = 4$. If the ordinate at $x = a$ divides the area into two equal parts, find a.

Solution: It can be seen easily that $y \geq 1 \ \forall x$.

Curve, $y = 1 + \frac{8}{x^2}$.....(1) is above the x-axis

Now, required region is ABCDA.



Area $= \int_2^4 \left(1 + \frac{8}{x^2}\right) dx = \left[x - \frac{8}{x}\right]_2^4 = (4 - 2) - (2 - 4)$

$= 2 + 2 = 4$ sq. units.

Since ordinate $x = a$ divides area into two equal parts, therefore area between ordinates $x = 2$ and $x = a$ is equal to 2 sq. units.

Now $\int_2^a \left(1 + \frac{8}{x^2}\right) dx = 2$

$\Rightarrow \quad \left(a - \frac{8}{a}\right) + 2 = 2 \Rightarrow a - \frac{8}{a} = 0$

$\Rightarrow a^2 = 8 \qquad \Rightarrow a = \pm 2\sqrt{2}$

but $2 < a < 4$ therefore $a = 2\sqrt{2}$.

Student Task

Q. Can you give an example of a curve which tends to an infinite value close to a certain value of independent variable but the area bounded by the curve between that value of independent variable and some other value is finite?

EXERCISE-03

Q.1 The total area bounded by $y = f(x)$, $x = a$ and $x = b$ and the x-axis is always equal to $\int_a^b f(x)dx$. **[True/False]**

Q.2 Area bounded by the curve $y = \sin x$ and x-axis from $x = 0$ to 2π is zero. **[True/False]**

Q.3 The area of the region enclosed by $y = 2 - x^2$ and $y = -x$ is _____.

Q.4 Obtain area bounded by $x = y^2$ and x-axis between $x = 0$ and $x = 9$.

Q.5 Find the area bounded by x-axis and the curve $y = -\sqrt{25 - x^2}$.

Q.6 Sketch the curves $y = x^2 - 4$ and $y = -x^2 - 2x$ and find area of region bounded by them.

Q.7 Find area bounded by $y = \sin\left(\frac{\pi x}{2}\right)$ and $y = x$ in first quadrant.

Q.8 Sketch $y^2 = x^2 (a^2 - x^2)$.

Q.9 Does the equation $|x| + |y| = 10$ decide a closed region?

Q.10 Sketch the region enclosed by $y = \sqrt{|x|}$ and $5y = x + 6$.

Q.11 The area between the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the straight line $\frac{x}{a} + \frac{y}{b} = 1$ is-

(a) $\frac{1}{4}\pi ab - \frac{1}{2}ab$ (b) $\frac{1}{4}ab$ (c) $\frac{1}{2}\pi ab$ (d) None of these

Q.12 The area of the figure bounded by the curves $y = e^x, y = e^{-x}$ and the straight line $x = 1$ is

(a) $e + \frac{1}{e}$ (b) $e - \frac{1}{e}$ (c) $e + \frac{1}{e} - 2$ (d) none of these

Q.13 The area of the figure bounded by the curves $y = |x-1|$ and $y = 3-|x|$, is

(a) 2 sq units (b) 3 sq. units (c) 4 sq. units (d) 1 sq. units

Q.14 The whole area between the curve $y^2(a-x) = x^2(a+x), a > 0$ and its asymptotes is equal to

(a) $2a^2\left(1+\frac{\pi}{4}\right)$ (b) $2a^2\left(1+\frac{\pi}{3}\right)$ (c) $a^2\left(1+\frac{\pi}{4}\right)$ (d) $a^2\left(1+\frac{\pi}{3}\right)$

Q.15 The area bounded by $y = xe^{|x|}$ and lines $|x| = 1$ and $y = 0$ is

(a)4 (b) 6 (c) 1 (d) 2

Answers :

1. False 2. False 3. $\frac{9}{2}$ sq. units 4. 18 sq. units

5. $\frac{25\pi}{2}$ sq. units 6. 9 sq. units 7. $\left(\frac{2}{\pi}-\frac{1}{2}\right)$ sq. units 9. Yes

11. () 12. () 13. () 14. ()

15. ()

MISCELLANEOUS EXAMPLES

Example 1 : $\int_0^{\pi} \frac{x\sin x}{2-\sin^2 x}dx =$

(a) $\frac{\pi^2}{4}$ (b) $\frac{\pi^2}{2}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

Solution : **(a)** $I = \int_0^{\pi} \frac{x\sin x}{2-\sin^2 x}dx = \int_0^{\pi} \frac{(\pi-x)\sin(\pi-x)}{2-\sin^2(\pi-x)}dx$, (using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$)

$\Rightarrow \quad 2I = \pi\int_0^{\pi} \frac{\sin x}{2-\sin^2 x}dx$

$\therefore \quad I = \frac{\pi}{2}\int_0^{\pi} \frac{-d(\cos x)}{1+\cos^2 x} = \left[-\frac{\pi}{2}\tan^{-1}(\cos x)\right]_0^{\pi}$

$= \left(\frac{-\pi}{2}\right)\left(\frac{-\pi}{4}\right) + \frac{\pi}{2}\left(\frac{\pi}{4}\right) = \frac{\pi^2}{4}$

Example 2 : $\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}}\,dx =$

(a) 2 (b) 0 (c) –1 (d) 1

Solution : **(a)** $\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}}dx = \int_0^{\pi} \sqrt{\cos^2 x}\,dx = \int_0^{\pi} |\cos x|\,dx$

$$= \int_0^{\pi/2} \cos x\, dx + \int_{\pi/2}^{\pi} -\cos x\, dx$$

$$= [\sin x]_0^{\pi/2} - [\sin x]_{\pi/2}^{\pi} = 1 - (-1) = 2$$

Example 3 : $\int_0^{\pi} e^{\cos^2 x}.\cos^3(2n+1)x\, dx$, (n is an integer) is equal to

(a) π (b) 0 (c) 1 (d) $\pi/2$

Solution : **(b)** $I = \int_0^{\pi} e^{\cos^2 x}\cos^3(2n+1)x\, dx = \int_0^{\pi} e^{\cos^2(\pi-x)}.\cos^3(2n+1)(\pi - x)dx$

$$= \int_0^{\pi} e^{\cos^2 x}.\{-\cos(2n+1)x\}^3\, dx = -I$$

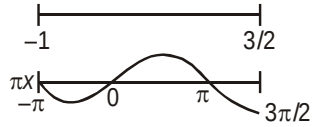
Hence $I = 0$

Example 4 : $\int_{-1}^{3/2} |\sin \pi x|\, dx =$

(a) $\frac{2}{\pi}$ (b) $\frac{3}{\pi}$ (c) $\frac{4}{\pi}$ (d) $\frac{5}{\pi}$

Solution : **(d)** $\int_{-1}^{3/2} |\sin \pi x|\, dx = \int_{-1}^{0} -\sin(\pi x)dx + \int_0^1 \sin(\pi x)dx + \int_1^{3/2} -\sin(\pi x)dx$

$$= \left[\frac{\cos \pi x}{\pi}\right]_{-1}^{0} + \left[\frac{-\cos \pi x}{\pi}\right]_0^1 + \left[\frac{\cos \pi x}{\pi}\right]_1^{3/2}$$



Graph of sin πx

Example 5 : If $f : R \to R$ and $g : R \to R$ be two continuous functions, the value of

$$\int_{-\frac{\pi}{2}}^{\pi/2} \{f(x) + f(-x)\}\{g(x) - g(-x)\}dx =$$

(a) π (b) 1 (c) –1 (d) 0

Solution : **(d)** If $\phi(x) = \{f(x) + f(-x)\}\{g(x) - g(-x)\}$, then
$\phi(-x) = \{f(-x) + f(x)\}\{g(-x) - g(x)\} = -\phi(x)$

Hence the integrand $\phi(x)$ is an odd function and therefore $\int_{-\pi/2}^{\pi/2} \phi(x)dx = 0$

Example 6 : $\int_0^{\pi/2} \log(\tan x)dx =$

(a) 0 (b) 1/2 (c) 1 (d) 2

Solution : **(a)** $I = \int_0^{\pi/2} \log(\tan x)dx = \int_0^{\pi/2} \log\left(\tan\left(\frac{\pi}{2} - x\right)\right)dx = \int_0^{\pi/2} \log(\cot x)dx$

$$= \frac{1}{2}\int_0^{\pi/2} (\log(\tan x) + \log(\cot x)dx$$

$$= \frac{1}{2}\int_0^{\pi/2} \log(\tan x.\cot x)dx = \frac{1}{2}\int_0^{\pi/2} \log 1\, dx$$

$$\therefore \quad I = \frac{1}{2}\int_0^{\pi/2} 0\, dx = 0$$

Example 7 : $\int_{\pi/8}^{7\pi/8} \frac{x\,dx}{\cos^2 x - \sin^2 x} =$

(a) $\frac{\pi}{2}\log\left(3-2\sqrt{2}\right)$ (b) $\frac{\pi}{4}\log\left(3-2\sqrt{2}\right)$ (c) $\pi\log\left(\sqrt{2}-1\right)$ (d) $\pi\log\left(\sqrt{2}+1\right)$

Solution : **(b)** $I = \int_{\pi/8}^{7\pi/8} \frac{x\,dx}{\cos^2 x - \sin^2 x} = \int_{\pi/8}^{7\pi/8} \frac{\pi - x}{\cos 2(\pi - x)}dx$

$$= \int_{\pi/8}^{7\pi/8} \frac{(\pi - x)dx}{\cos 2x}$$

$$= \pi\int_{\pi/8}^{7\pi/8} \frac{dx}{\cos 2x} - I$$

$$\therefore \quad 2I = \frac{\pi}{2}\left[\log(\sec 2x + \tan 2x)\right]_{\pi/8}^{7\pi/8}$$

$$I = \frac{\pi}{4}\left[\log\left(\sec\frac{7\pi}{4} + \tan\frac{7\pi}{4}\right) - \log\left(\sec\frac{\pi}{4} + \tan\frac{\pi}{4}\right)\right]$$

$$= \frac{\pi}{4}\left[\log\left(\sqrt{2}-1\right) - \log\left(\sqrt{2}+1\right) = \frac{\pi}{4}\log\left(3-2\sqrt{2}\right)\right]$$

Example 8 : $\int_0^{100\pi} \sqrt{1-\cos 2x}\, dx =$

(a) 0 (b) $100\sqrt{2}$ (c) $200\sqrt{2}$ (d) 200

Solution : **(c)** $\int_0^{100\pi} \sqrt{1-\cos 2x}\, dx = \int_0^{100\pi} \sqrt{2\sin^2 x}\, dx = \sqrt{2}\int_0^{100\pi} |\sin x|\, dx$

Now $\int_0^{100\pi} |\sin x|\, dx = \int_0^{\pi} |\sin x|\, dx + \int_{\pi}^{2\pi} |\sin x|\, dx = 2\int_0^{\pi} |\sin x|\, dx$

Since the function $f(x) = |\sin x|$ is periodic with period π, so

$$I = \sqrt{2}\times 100\int_0^{\pi} \sin x\, dx = 100\sqrt{2}[-\cos x]_0^{\pi}$$

$$= 100\sqrt{2}[1-(-1)] = 200\sqrt{2}$$

Example 9 : $\lim_{n\to\infty}\left\{\frac{1}{1-n^2} + \frac{2}{1-n^2} + \frac{3}{1-n^2} + ... + \frac{n}{1-n^2}\right\} =$

(a) 0 (b) –1/2 (c) 1/2 (d) ∞

Solution : **(b)** $\lim\limits_{n\to\infty}\left\{\frac{1}{1-n^2}+\frac{2}{1-n^2}+\frac{3}{1-n^2}+....+\frac{n}{1-n^2}\right\}$

$$=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r}{1-n^2}=\lim_{n\to\infty}\frac{1}{\left(\frac{1}{n^2}-1\right)}\sum_{r=1}^{n}\frac{1}{n}\left(\frac{r}{n}\right)$$

$$=\lim_{n\to\infty}\frac{1}{\left(\frac{1}{n^2}-1\right)}\int_0^1 x\,dx=\lim_{n\to\infty}\frac{1}{2}\times\frac{1}{\left(\frac{1}{n^2}-1\right)}=\frac{-1}{2}$$

Example 10 : $\int_0^{\infty}\left[\frac{2}{e^x}\right]dx$ is, [where [ ] represents the greatest integer function]

(a) 0 (b) $2/e$ (c) e^2 (d) $\log_e 2$

Solution : **(d)** $\int_0^{\infty}\left[\frac{2}{e^x}\right]dx$

$x \in (0, \log_e 2)$, $\frac{2}{e^x}$ varies 2 to 1 and hence $\left[\frac{2}{e^x}\right]=1$

$x \in (0, \log_e 2, \infty)$, $\frac{2}{e^x}$ varies from 1 to 0 and hence $\left[\frac{2}{e^x}\right]=0$

$$\therefore \quad \int_0^{\infty}\left[\frac{2}{e^x}\right]dx=\int_0^{\log_e 2}1\,dx+\int_{\log_e 2}^{\infty}0\,dx$$

$$=[x]_0^{\log_e 2}+0=\log_e 2$$

Example 11 : Evaluate $\int_0^{\pi/2}\ln(\sin x)dx$

Solution : Let $I=\int_0^{\pi/2}\ln(\sin x)dx$

$$\Rightarrow \quad I=\int_0^{\pi/2}\ln\left(\sin\left(\frac{\pi}{2}-x\right)\right)dx$$

$$=\int_0^{\pi/2}\ln(\cos x)dx$$

Adding both, we get

$$2I=\int_0^{\pi/2}(\ln(\sin x)+\ln(\cos x)dx=\int_0^{\pi/2}\ln\left(\frac{\sin 2x}{2}\right)dx$$

$$=\int_0^{\pi/2}\ln(\sin 2x)dx-\int_0^{\pi/2}\ln 2dx$$

$$= I_1 - \frac{\pi}{2}\ln 2$$

$$I_1 = \int_0^{\pi/2} \ln(\sin 2x)dx$$

Put $2x = t \Rightarrow dx = \frac{dt}{2}$

$$I_1 = \frac{1}{2}\int_0^{\pi} \ln(\sin t)dt$$

$$= \frac{1}{2}\left[\int_0^{\pi/2} \ln(\sin t)dt + \int_0^{\pi/2} \ln(\sin(\pi - t)dt\right] = \int_0^{\pi/2} \ln(\sin t)dt$$

$$= \int_0^{\pi/2} \ln(\sin x)dx = I$$

So $2I = I - \frac{\pi}{2}\ln 2 \Rightarrow I = \frac{-\pi}{2}\ln 2 = \frac{\pi}{2}\ln\left(\frac{1}{2}\right)$

Example 12 : Evaluate $\int_0^1 (tx + 1 - x)^n dx$ when $n \in N$ and t is a parameter independent of x, hence or otherwise prove that $\int_0^1 x^m (1-x)^{n-m} dx = \frac{1}{(n+1).{}^nC_m}$ $(0 \le m \le n)$.

Solution : $I = \int_0^1 (tx + 1 - x)^n dx$

$$= \int_0^1 ((t-1)x + 1)^n dx = \frac{((t-1)x+1)^{n+1}}{(n+1)(t-1)}\Bigg|_0^1 = \frac{1}{(n+1)}\left[\frac{t^{n+1}-1}{t-1}\right]$$

$$\frac{1}{n+1}\left[1 + t + t^2 + ... + t^n\right] \qquad .... (i)$$

Now $I = \int_0^1 (tx + 1 - x)^n dx$

$$= \int_0^1 ((tx + (1-x))^n dx$$

$$= \int_0^1 \sum_{m=0}^{n} {}^nC_m (tx)^m (1-x)^{n-m} dx$$

$$I = \sum_{m=0}^{n} {}^nC_m \left(\int_0^1 x^m (1-x)^{n-m} dx\right) t^m \qquad ... (ii)$$

Equate the coefficient of t^m in (i) and (ii) we get $\int_0^1 x^m(1-x)^{n-m}dx = \frac{1}{(n+1)\,{}^nC_m}$

Example 13 : Evaluate $\int_0^{\pi/4} \frac{\sin^2 x \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx$

Solution : $\int_0^{\pi/4} \frac{\sin^2 x \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx = \int_0^{\pi/4} \frac{\tan^2 x \sec^2 x}{(1+\tan^3 x)^2} dx$

dividing numerator and denominator by $\cos^6 x$.

Put $z = 1 + \tan^3 x$ $\quad\therefore\quad dz = 3\tan^2 x \sec^2 x\, dx$

When $x = 0$, $z = 1$ and when $x = \frac{\pi}{4}, z = 2$

$$\therefore \quad \int_0^{\pi/4} \frac{\sin^2 x \cos^2 x}{(\sin^3 x + \cos^2 x)^2} dx = \int_{z=1}^{2} \frac{1}{3}\frac{dz}{z^2} = \left[\frac{1}{3}\left(\frac{-1}{z}\right)\right]_1^2 = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$$

Example 14 : If $f(x) = \sqrt{x-m+1}$, $x \in [m-1, m)$, $(m \in I)$, then evaluate $\int_0^n f(x)dx, n(\in N)$.

Solution : $\int_0^n f(x)dx = \sum_{m=1}^{n} \int_{m-1}^{m} f(x)dx = \sum_{m=1}^{n} \int_{m-1}^{m} \left(\sqrt{x-m+1}\right)dx$

$$= \sum_{m=1}^{n} \frac{2}{3}(x-m+1)^{3/2}\Big|_{m-1}^{m} = \sum_{m=1}^{n} \frac{2}{3} = \frac{2n}{3}$$

Example 15 : Evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{\cos^2 x + \sin^4 x} dx$

Solution : Observe that $\sin x + \cos x = \frac{d}{dx}(\sin x - \cos x)$

Therefore express the denominator as a function of $\sin x - \cos x$

$$\cos^2 x + \sin^4 x = \frac{1+\cos 2x}{2} + \left(\frac{1-\cos 2x}{2}\right)^2$$

$$= \frac{3}{4} + \frac{\cos^2 2x}{4} = \frac{4-\sin^2 2x}{4} = 1 - \sin^2 x \cos^2 x$$

$$= 1 - \left\{\frac{1-(\sin x - \cos x)^2}{2}\right\}^2$$

Let $z = \sin x - \cos x$; $dz = (\cos x + \sin x)\, dx$

When $x = 0$, $z = -1$, when $x = \pi/4$, $z = 0$

$$I = \int_{-1}^{0} \frac{dz}{1 - \left(\frac{1-z^2}{2}\right)^2} = \int_{-1}^{0} \frac{4dz}{2^2 - (1-z^2)^2}$$

$$= \int_{-1}^{0} \frac{4\,dz}{(3-z^2)(1+z^2)}$$

$$= \int_{-1}^{0} \left\{ \frac{1}{1+z^2} + \frac{1}{3-z^2} \right\} dz$$

$$= \left[\tan^{-1} z + \frac{1}{2\sqrt{3}} \log \frac{\sqrt{3}+z}{\sqrt{3}-2} \right]_{-1}^{0}$$

$$= 0 - \left\{ \tan^{-1}(-1) + \frac{1}{2\sqrt{3}} \log \left(\frac{\sqrt{3}-1}{\sqrt{3}-1} \right) \right\}$$

$$= \frac{\pi}{4} - \frac{1}{2\sqrt{3}} \log \left(\frac{\sqrt{3}-1}{\sqrt{3}+1} \right) = \frac{\pi}{4} - \frac{1}{2\sqrt{3}} \log \frac{\left(\sqrt{3}-1\right)^2}{2}$$

$$= \frac{\pi}{4} - \frac{1}{\sqrt{3}} \log \left(\frac{\sqrt{3}-1}{\sqrt{2}} \right)$$

Example 16 : Prove that $\frac{1}{2}\int_0^x e^{zx} e^{-z^2}\,dz = e^{x^2/4} \int_0^{x/2} e^{-z^2}\,dz$

Solution : Put $z = \frac{x+2u}{2}$ $\therefore\ dz = du$

$z = 0 \Rightarrow u = -\frac{x}{2}$ and $z = x \Rightarrow u = \frac{x}{2}$

$\therefore$ L.H.S. = $I = \frac{1}{2} \int_{u=-x/2}^{x/2} e^{\left(\frac{x+2u}{2}\right)\left(x-\frac{x+2u}{2}\right)}.du$

$$= \frac{1}{2} \int_{-x/2}^{x/2} e^{\frac{x^2-4u^2}{4}}\,du = \frac{1}{2} e^{\frac{x^2}{4}} \int_{-\frac{x}{2}}^{\frac{x}{2}} e^{-u^2}\,du$$

$$= \frac{1}{2}.2e^{\frac{x^2}{4}} \int_0^{\frac{x}{2}} e^{-u^2}\,du = e^{x^2/4} \int_0^{x/2} e^{-z^2}\,dz = \text{ R.H.S.}$$

Example 17 : Evaluate $\int_{-\pi/2}^{\pi/2} \sqrt{\cos x - \cos^3 x}\,dx$

Solution :

$$I = \int_{-\frac{\pi}{2}}^{\pi/2} \sqrt{\cos x - \cos^3 x}\,dx$$

$$= 2\int_0^{\pi/2} \sqrt{\cos x - \cos^3 x}\,dx \quad \text{(as cos } x \text{ is an even function)}$$

$$= 2\int_0^{\pi/2} \sqrt{\cos x}\sqrt{\sin^2 x}\, dx$$

$$= 2\int_0^{\pi/2} \sqrt{\cos x}\ \sin x\, dx \text{ since } x \text{ is positive in } (0, \pi/2)$$

$$= \left[-2(\cos x)^{3/2}.\frac{2}{3}\right]_0^{\pi/2} = \frac{2}{3}$$

Example 18 : Show that $\int_0^{\pi} \frac{x\sin 2x \sin\left\{\left(\frac{\pi}{2}\cos x\right)\right\}dx}{2x-\pi} = \frac{8}{\pi^2}$.

Solution : $I = \int_0^{\pi} \frac{x\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)dx}{2x-\pi}$

$$= \int_0^{\pi} \frac{(\pi - x)\sin(2\pi - 2x)\sin\left\{\frac{\pi}{2}\cos(\pi - x)\right\}}{2(\pi - x) - \pi} dx \text{ using property 1}$$

$$= \pi\int_0^{\pi} \frac{(-\sin 2x)\sin\left(-\frac{\pi}{2}\cos x\right)}{\pi - 2x} dx - \int_0^{\pi} \frac{x(-\sin 2x)\sin\left(-\frac{\pi}{2}\cos x\right)dx}{\pi - 2x}$$

$$= \pi\int_0^{\pi} \frac{\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)}{\pi - 2x} dx + \int_0^{\pi} \frac{x\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)}{2x - \pi} dx$$

$$= \pi\int_0^{\pi} \frac{\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)}{\pi - 2x} dx + I \qquad \text{.... (i)}$$

This gives $\int_0^{\pi} \frac{\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)}{\pi - 2x} = 0$ .... (ii)

Now $2I = \int_0^{\pi} \frac{2x\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)dx}{2x-\pi}$

$$= \int_0^{\pi} \frac{\{(2x - \pi) + \pi\}\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)dx}{2x - \pi}$$

$$= \int_0^{\pi} \sin 2x \sin\left(\frac{\pi}{2}\cos x\right) dx + \pi \int_0^{\pi} \frac{\sin 2x \sin\left(\frac{\pi}{2}\cos x\right) dx}{2x - \pi}$$

$$= \int_0^{\pi} \sin 2x \sin\left(\frac{\pi}{2}\cos x\right) dx \text{ since the second integral is zero by (2)}$$

$$= 2\int_0^{\frac{\pi}{2}} \sin 2x \sin\left(\frac{\pi}{2}\cos x\right) dx \text{ using property (4) since } f(2a - x) = f(x)$$

$$= 4 \int_0^{\pi/2} \sin x \cos x \sin\left(\frac{\pi}{2}\cos x\right) dx$$

Now put $\frac{\pi}{2}\cos x = t \ \therefore\ -\frac{\pi}{2}\sin x\, dx = dt$ and $x = 0 \Rightarrow t = \frac{\pi}{2}$; $x = \frac{\pi}{2} \Rightarrow t = 0$

$$\therefore \quad 2I = 4 \int_{t=\frac{\pi}{2}}^{t=0} \sin t(-2/\pi).\frac{2t}{\pi} dt$$

$$= \frac{16}{\pi^2}\int_0^{\frac{\pi}{2}} t \sin t\, dt = \frac{16}{\pi^2}[t(-\cos t) + \sin t]_0^{\pi/2} = \frac{16}{\pi^2}$$

$$\therefore \quad I = \frac{8}{\pi^2}$$

Example 19 : Evaluate : $\int_0^{3\pi/2} (\log|\sin x|)(\cos(2nx))dx, n \in N$.

Solution : Let $I(n) = \int_0^{3\pi/2} (\log|\sin x|)(\cos 2nx)dx$, using integration by parts,

$$I(n) = \left(\log|\sin x|.\frac{\sin 2nx}{2n}\right)_0^{3\pi/2} - \int_0^{3\pi/2} \cot x.\frac{\sin 2nx}{2n}.dx$$

$$I(n) = 0 - \frac{1}{2n}\int_0^{3\pi/2} \frac{\sin 2nx.\cos x}{\sin x}.dx = -\frac{1}{2n} I_1(n) \quad \text{... (i)}$$

Let, $I_1(n) = \int_0^{3\pi/2} \frac{\sin(2nx)\cos x}{\sin x}.dx$

$$I_1(n+1) = \int_0^{3\pi/2} \frac{\sin(2n+2)x.\cos x.dx}{\sin x}$$

$$\therefore \quad I_1(n+1) - I_1(n) = \int_0^{3\pi/2} \frac{(\sin(2n+2)x - \sin 2nx).\cos x}{\sin x}.dx$$

$$= \int_0^{3\pi/2} \frac{2\cos(2n+1)x.\sin x.\cos x}{\sin x}.dx$$

$$= \int_0^{3\pi/2} (\cos(2n+2)x + \cos 2nx)dx$$

$$= \left(\frac{\sin(2n+2)x}{(2n+2)} + \frac{\sin(2n)x}{2n} \right)_0^{3\pi/2} = 0 + 0$$

$\Rightarrow \quad I_1(n+1) = I_1(n) = \ldots\ldots\ldots = I_1(1)$

$\therefore \qquad I_1(1) = \int_0^{3\pi/2} \frac{\sin 2x.\cos x}{\sin x} dx = \int_0^{3\pi/2} \frac{2\sin x\cos x.\cos x}{\sin x} dx$

$$= \int_0^{3\pi/2} 2\cos^2 x.dx = \int_0^{3\pi/2} (1+\cos 2x)dx = \left(x + \frac{\sin 2x}{2} \right)_0^{3\pi/2}$$

$\Rightarrow \quad I_1(1) = \frac{3\pi}{2} \Rightarrow I(n) = -\frac{1}{2n} I_1(n)$ {using (i)}

$$I(n) = \frac{3\pi}{4n}$$

Example 20 : Find $\lim_{n\to\infty} \left\{ \left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)\left(1+\frac{3^2}{n^2}\right).....\left(1+\frac{n^2}{n^2}\right) \right\}^{\frac{1}{n}}$.

Solution : Let $A = \lim_{n\to\infty} \left\{ \left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)\left(1+\frac{3^2}{n^2}\right).....\left(1+\frac{n^2}{n^2}\right) \right\}^{\frac{1}{n}}$

$$\Rightarrow \quad \log_e A = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n} \log\left(1+\frac{r^2}{n^2}\right)$$

$$= \lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} \log\left\{1+\left(\frac{r}{n}\right)^2\right\} = \int_0^1 \log(1+x^2)dx$$

$$= \left[x\log(1+x^2) \right]_0^1 - \int_0^1 \frac{x.2x\,dx}{1+x^2}$$

$$= 1\log 2 - \int_0^1 \frac{2(1+x^2 1)-2}{1+x^2} dx$$

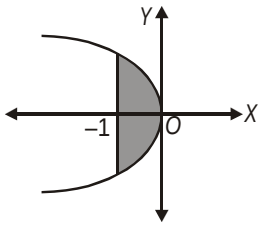
$$= \log 2 - \left[2x - 2\tan^{-1} x \right]_0^1 = \log 2 - \left[2 - \frac{2\pi}{4} \right] = \log 2 + \frac{\pi-4}{2}$$

$$\Rightarrow \quad A = e^{\log 2 + \frac{\pi-4}{2}} = 2e^{\frac{\pi-4}{2}}$$

Example 21 : The area bounded by $y^2 = -4x$ and its latus rectum is

(a) Not defined (b) 1 sq. units (c) 2/3 sq. units (d) None of these

Solution : **(d)** The required area $= 2\int_{-1}^{0}\sqrt{-4x}\,dx$



$$= -\frac{8}{3}(-x)^{3/2}\Big|_{-1}^{0} = \frac{8}{3} \text{ sq. units}$$

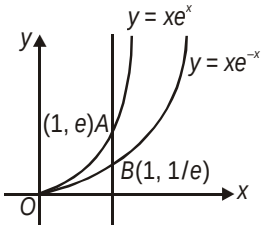
Example 22 : The area between $y = xe^x$ and $y = xe^{-x}$ and the line $x = 1$ is

(a) $2e$ (b) e (c) $2/e$ (d) $1/e$

Solution : **(c)** The lines $x = 1$ meets the curves in $A(1, e)$ and $B(1, 1/e)$. Both the curves pass through origin.

The required area

$$A = \int_0^1 (y_1 - y_2)dx = \int_0^1 (xe^x - xe^{-x})dx$$



$$= \left[x\left\{e^x + e^{-x}\right\}\right]_0^1 - \int_0^1 (e^x + e^{-x})dx$$

$$= \left(e + \frac{1}{e}\right) - \left[e^x - e^{-x}\right]_0^1 = \left(e + \frac{1}{e}\right) - \left(e - \frac{1}{e}\right) = \frac{2}{e} \text{ sq. units}$$

Example 23 : The area bounded by the curve, $y = f(x) = x^4 - 2x^3 + x^2 + 3$, the x-axis and the ordinates corresponding to the minimum of function $f(x)$ is

(a) 1 (b) 91/30 (c) 30/9 (d) 4

Solution : **(b)** We have $f(x) = x^4 - 2x^3 + x^2 + 3$

To find minimum

$f'(x) = 4x^3 - 6x^2 + 2x = 0$

$\Rightarrow$ $2x(x - 1)(2x - 1) = 0 \Rightarrow x = 0, 1, \frac{1}{2}$

$f''(x) = 12x^2 - 12x + 2 = 2(6x^2 - 6x + 1)$

$f''(x) < 0 \Rightarrow$ minimum exists at $x = 0$

$f''(x) < 0 \Rightarrow$ maximum exists at $x = \frac{1}{2}$

$f''(x)_{x=1} > 0 \Rightarrow$ minimum exists at $x = 1$

$\Rightarrow$ The curve is bounded by the ordinates $x = 0$ and $x = 1$

$\therefore$ Required area $= \int_0^1 (x^4 - 2x^3 + x^2 + 3)dx = \left(\frac{x^5}{5} - \frac{x^4}{2} + \frac{x^3}{3} + 3x\right)\Big|_0^1 = \frac{91}{30}$ sq. units

Example 24 : The area of the region bounded by the curve $y = \frac{1}{1 + (\tan x)^{1/2}}$ and the x-axis between the ordinates $x = \frac{\pi}{6}$ and $x = \frac{\pi}{3}$ is

(a) $\pi/4$ (b) $\pi/2$ (c) $\pi/8$ (d) None of these

Solution : **(d)** The given curve is $y = \frac{1}{1 + (\tan x)^{1/2}}$

The given curve y is positive and continuous in the given interval $x = \frac{\pi}{6}$ and $\frac{\pi}{3}$

$$\Rightarrow \quad y = \frac{(\cos x)^{1/2}}{(\sin x)^{1/2} + (\cos x)^{1/2}}$$

Area bounded between the intervals $x = \frac{\pi}{6}$ and $x = \frac{\pi}{3}$ is given by

$$A = \int_{\pi/6}^{\pi/3} \frac{(\cos x)^{1/2}}{(\sin x)^{1/2} + (\cos x)^{1/2}} dx$$

Also $A = \int_{\pi/6}^{\pi/3} \frac{(\sin x)^{1/2}}{(\sin x)^{1/2} + (\cos x)^{1/2}} dx$, using formula $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

Adding $2A = \int_{\pi/6}^{\pi/3} dx \quad \Rightarrow \quad A = \frac{1}{2}\left[\frac{\pi}{3} - \frac{\pi}{6}\right] = \frac{\pi}{12}$ sq. units

Example 25 : The area bounded by $y = \frac{\sin x}{x}$, x-axis and the ordinate x = 0, $x = \frac{\pi}{4}$ is

(a) Less than or equal to $\frac{\pi}{4}$ (b) Less than $\int_0^{\pi/4} \frac{\tan x}{x} dx$

(c) Greater than $\frac{\pi}{4}$ (d) None of these

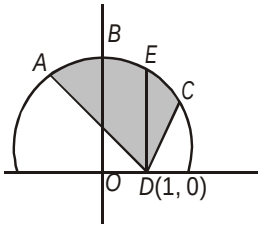
Solution : **(a) and (b)** Since $\frac{\sin x}{x} \le 1$

$$\Rightarrow \quad \int_0^{\pi/4} \frac{\sin x}{x} dx \le \int_0^{\pi/4} 1 dx \quad \Rightarrow \quad \int_0^{\pi/4} \frac{\sin x}{x} dx \le \frac{\pi}{4}$$

Also $\int_0^{\pi/4} \frac{\sin x}{x} dx \le \int_0^{\pi/4} \frac{\tan x}{x} dx$ (in the similar way).

Example 26 : Sketch the region bounded by the curves, $y = \sqrt{5 - x^2}$ and y = $|x - 1|$ and find its area.

Solution : $y = \sqrt{5 - x^2}$ represents the upper part of the circle $y^2 + x^2 = 5$



$y = |x - 1|$ consists of two equations $y = x - 1$ or $x > 1$ and $y = 1 - x$ for $x < 1$. Points of intersection $A = (-1, 2)$ and $C = (2, 1)$

Required area = Area *ADEA* + Area *EDCE*

$$= \int_{-1}^{+1} \left\{\sqrt{5 - x^2} - (1 - x)\right\} dx + \int_1^2 \left\{\sqrt{5 - x^2} - (x - 1)\right\} dx$$

Note : The ordinate of point on the segment *AD* is $1 - x$ and on the segment *DC* is $x - 1$.

$$2\int_0^1\left(\sqrt{5-x^2}-1\right)dx+\int_1^2\left(\sqrt{5-x^2}-(x-1)\right)dx \quad \text{(Note: } x \text{ is an odd function)}$$

$$=2\left[\frac{x}{2}\sqrt{5-x^2}+\frac{5}{2}\sin^{-1}\frac{x}{\sqrt{5}}-x\right]_0^1+\left[\frac{x}{2}\sqrt{5-x^2}+\frac{5}{2}\sin^{-1}\frac{x}{\sqrt{5}}-\frac{x^2}{2}+x\right]_1^2$$

$$=2\left[\frac{2}{2}+\frac{5}{2}\sin^{-1}\frac{1}{\sqrt{5}}-1\right]+\left[1+\frac{5}{2}\sin^{-1}\frac{2}{\sqrt{5}}-2+2\right]-\left[\frac{2}{2}+\frac{5}{2}\sin^{-1}\frac{1}{\sqrt{5}}-\frac{1}{2}+1\right]$$

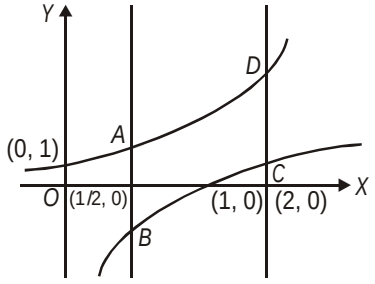
$$=\frac{5}{2}\left[\sin^{-1}\frac{2}{\sqrt{5}}+\sin^{-1}\frac{1}{\sqrt{5}}\right]-\frac{1}{2}$$

Example 27 : Sketch the curves and identify the region bounded by $x=\frac{1}{2}$, $x = 2$, $y = \log_e x$ and $y = 2^x$. Find the area of the region.

Solution : (i) $y = \log_e x$

This curve passes through (1, 0); for $x > 1$, $\log_e x > 0$ and the graph is above the X-axis. For $x < 1$, $\log_e x < 0$ and the corresponding graph is below the x-axis. As $x \to 0$, $\log_e x \to -\infty$.

Therefore the curve is asymptotic with negative Y-axis.



It meets $x=\frac{1}{2}$ at $B\left(\frac{1}{2},\log_e\frac{1}{2}\right)$ and $x = 2$ at C $(2, \log_e 2)$

(ii) $y = 2^x$

This curve passes through (0, 1). 2^x is always positive and therefore the graph is above the x-axis.

As $x \to -\infty$, $2^x \to 0$. Therefore the negative X-axis is an asymptote for the curve.

It meets $x=\frac{1}{2}$ at $A\left(\frac{1}{2},2^{1/2}\right)$ and $x = 2$ at $D(2, 4)$

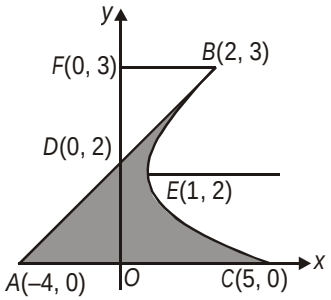
Required = area of region $ABCD$

$$=\int_{1/2}^{2}(2^x-\log_e x)dx=\left[\frac{2^x}{\ln 2}-(x\ln x-x)\right]_{1/2}^{2}$$

$$=\frac{4-\sqrt{2}}{\ln 2}-2\ln 2+2-\frac{1}{2}+\frac{1}{2}\ln\left(\frac{1}{2}\right)=\frac{4-\sqrt{2}}{\ln 2}-\frac{5}{2}\ln 2+\frac{3}{2}$$

Example 28 : Find the area of the region bounded by the parabola $(y-2)^2 = (x-1)$, the tangent drawn to it at the point whose ordinate is 3 and the X-axis.

Solution : $(y-2)^2 = (x-1)$ represents a parabola with vertex at $E(1, 2)$. Point on the parabola with ordinate 3 is (2, 3)



Differentiating

$$2(y-2)\frac{dy}{dx}=1$$

$\therefore$ Slope of the tangent at $B(2, 3)$ is $\frac{1}{2}$

Equation of the tangent at B is

$$y-3=\frac{1}{2}(x-2)$$

i.e. $x - 2y + 4 = 0$

This meets the x-axis at $A(-4, 0)$ and y-axis at $D(0, 2)$

The parabola itself meets the x-axis at $C(5, 0)$

Required area = shaded in the figure

= Area of ΔAOD + Area of region $OCEBDO$

$$= \frac{1}{2} \times 4 \times = [\text{Area of } OCEBFO - \text{Area of } \Delta DBF]$$

$$=4-\frac{1}{2}\times 2\times 1+\int_{y=0}^{3}\{(y-2)^2+1\}dy \text{ using the formula } \int x\,dy$$

$$=3+\int_{0}^{3}(y^2-4y+5)dy$$

$$=3+\left[\frac{y^3}{3}-2y^2+5y\right]_0^3=3+(9-18+15)=3+6=9$$

Example 29 : Find the total area enclosed by the curve $a^2y^2 = x^2(a^2 - x^2)$.

Solution : The following points may be noted with reference to the curve.

(i)Changing x to $-x$ does not alter the equation. Hence the curve has symmetry about the y-axis.

(ii) Equally, it has symmetry about the x-axis also.

(iii) The curve intersects the x-axis at $x = -a$, $x = 0$, $x = a$.

(iv) $a^2y^2 > 0$ and hence $x^2(a^2 - x^2) > 0$

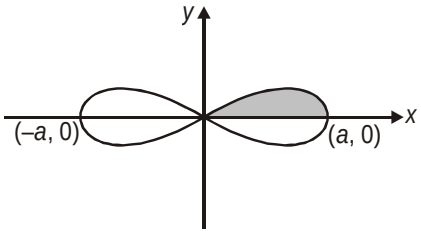
$\therefore$ $x^2 < a^2$ i.e., $-a < x < a$

$\therefore$ The curve gets bounded at $(-a, 0)$ and $(a, 0)$ beyond which on either side of the x-axis the curve does not exist.

(v) The only point where it intersects the y-axis is at the origin.

(vi) Hence the curve may be found to have two parts (each part is called a loop) on either side of the y-axis.

The curve is as shown below :



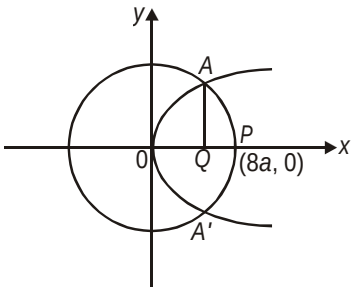
Area of both the loops = 4 × shaded area (by symmetry)

$$= 4\int_0^a y\,dx = \frac{4}{a}\int_0^a x\sqrt{a^2 - x^2}\,dx$$

$$= -\frac{2}{a}.\frac{2}{3}\left[(a^2 - x^2)^{3/2}\right]_0^a = \frac{4}{3a}.a^3 = \frac{4a^2}{3}$$

Example 30 : Show that the area enclosed by the circle $x^2 + y^2 = 64a^2$ and the parabola $y^2 = 12ax$ is $a^2\left(\frac{16}{\sqrt{3}} + \frac{64\pi}{\sqrt{3}}\right)$.

Solution : Solving these equations we get the points of intersection as $A(4a, 4\sqrt{3}a)$ and $A(4a, -4\sqrt{3}a)$



Required area = *OAPA'O*

= 2[Area *OPAO*]

= [area *OQAO* + area *QPAQ*]

$$= 2\left[\int_0^{4a} 2\sqrt{3ax}\,dx + \int_{4a}^{8a}\sqrt{64a^2 - x^2}\,dx\right]$$

$$= 4\sqrt{3}\sqrt{a}\,\frac{2}{3}\left[x^{3/2}\right]_0^{4a} + 2\left\{\frac{x}{2}\sqrt{64a^2 - x^2} + \frac{64a^2}{2}\sin^{-1}\left(\frac{x}{8a}\right)\right\}_{4a}^{8a}$$

$$= \frac{8.8\sqrt{3a}}{3}a\sqrt{a} + 2\left[-2.4\sqrt{3}a^2 + 32a^2\left(\frac{\pi}{2} - \frac{\pi}{6}\right)\right]$$

$$= \frac{64a^2\sqrt{3}}{3} - 16a^2\sqrt{3} + 64a^2\frac{2\pi}{6}$$

$$= \frac{16a^2}{\sqrt{3}} + \frac{64a^2\pi}{3} = a^2\left(\frac{16}{\sqrt{3}} + \frac{64\pi}{3}\right) \text{ sq. units}$$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

1. Definite Integral

Q.1 Let a, b, c be non-zero real numbers such that $\int_0^1 (1+\cos^8 x)(ax^2+bx+c)dx = \int_0^2 (1+\cos^8 x)(ax^2+bx+c)dx$

Then, the quadratic equation $ax^2 + bx + c = 0$ has

(a) No root in (0, 2)
(b) At least one root in (1, 2)
(c) A double root in (0, 2)
(d) Two imaginary roots

Q.2 The value of the integral $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x}+\sqrt{\tan x}}dx$ is

(a) $\pi/4$ (b) $\pi/2$
(c) π (d) None of these

Q.3 Let $f : R \to R$ and $g : R \to R$ be continuous functions. Then, the value of the integral $\int_{-\pi/2}^{\pi/2}[f(x)+f(-x)][g(x)-g(-x)]dx$ is

(a) π (b) 1
(c) –1 (d) 0

Q.4 If $f(x) = A\sin\left(\frac{\pi x}{2}\right) + B$, $f'\left(\frac{1}{2}\right) = \sqrt{2}$ and $\int_0^1 f(x)dx = \frac{2A}{\pi}$, then constants A and B are

(a) $\frac{\pi}{2}$ and $\frac{\pi}{2}$ (b) $\frac{2}{\pi}$ and $\frac{3}{\pi}$
(c) 0 and $-\frac{\pi}{4}$ (d) $\frac{4}{\pi}$ and 0

Q.5 The value of $\int_0^{2\pi}[2\sin x]dx$ where [.] represents the greatest integral functions, is

(a) $-\frac{5\pi}{3}$ (b) $-\pi$
(c) $\frac{5\pi}{3}$ (d) -2π

Q.6 If $g(x) = \int_0^x \cos^4 t\,dt$, then $g(x + \pi)$ equals

(a) $g(x) + g(\pi)$ (b) $g(x) - g(\pi)$
(c) $g(x)g(\pi)$ (d) $\frac{g(x)}{g(\pi)}$

Q.7 $\int_{\pi/4}^{3\pi/4} \frac{dx}{1+\cos x}$ is equal to

(a) 2 (b) –2
(c) 1/2 (d) –1/2

Q.8 Let $f : (0, \infty) \to R$ and $F(x) = \int_0^x f(t)dt$. If $F(x^2) = x^2(1 + x)$, then $f(4)$ equals

(a) 5/4 (b) 7
(c) 4 (d) 2

Q.9 Let $T > 0$ be a fixed real number. Suppose, f is a continuous function such that for all $x \in R$. $f(x + T) = f(x)$. If $I = \int_0^T f(x)dx$, then the value of $\int_3^{3+3T} f(2x)dx$ is

(a) $\frac{3}{2}I$ (b) I
(c) $3I$ (d) $6I$

Q.10 If $f(x) = \int_{x^2}^{x^2+1} e^{-t^2}dt$, then $f(x)$ increases in

(a) (2, 2) (b) No value of x
(c) $(0, \infty)$ (d) $(-\infty, 0)$

Q.11 The value of the integral $\int_0^1 \sqrt{\frac{1-x}{1+x}}dx$ is

(a) $\frac{\pi}{2}+1$ (b) $\frac{\pi}{2}-1$
(c) –1 (d) 1

Q.12 The value of $\int_{-2}^{0}[x^3+3x^2+3x+3+(x+1)\cos(x+1)]dx$ is

(a) 0 (b) 3
(c) 4 (d) 1

Q.13 If $I = \int_0^{\pi} \frac{\cos x}{(x+2)^2} dx$, then $\int_0^{\pi} \frac{\sin 2x}{x+1} dx$ is equal to

(a) 21 (b) $\frac{1}{\pi+2} - \frac{1}{2} - 1$

(c) 0 (d) $\frac{1}{\pi+2} + \frac{1}{2} - 1$

Q.14 If $\int_2^3 \frac{x^2 dx}{\sqrt{x^4 - x^2 + 1}} = 1$, then the value of $\int \frac{x dx}{\sqrt{1 + \left(x - \frac{1}{x}\right)^2}} - \int_{1/2}^{1/3} \frac{dx}{\sqrt{x^2 + \frac{1}{x^2} - 1}}$ is equal to

(a) $-I$ (b) 0

(c) I (d) $2I$

Q.15 If $\int_0^{\pi/2} \frac{dx}{\cos^2 x + b^2 \sin^2 x} = \frac{\pi}{2ab}$, then the value of $\int_0^{\pi/2} \frac{dx}{(4\cos^2 x + 9\sin^2 x)^2}$ is equal to

(a) $\frac{11\pi}{864}$ (b) $\frac{13\pi}{864}$

(c) $\frac{17\pi}{864}$ (d) None of these

Q.16 I f $\int_0^{\infty} \frac{\cos x}{x} dx = \frac{\pi}{2}$, then $\int_0^{\pi} \frac{1}{x}(1 - \sin^2 x)^{3/2} dx$ is equal to

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{6}$ (d) $\frac{3\pi}{2}$

Q.17 The value of $\int_{-\pi/2}^{\pi/2} \left(x^2 + \log \frac{\pi - x}{\pi + x}\right) \cos x$

(a) 0 (b) $\frac{\pi^2}{2} - 4$

(c) $\frac{\pi^2}{2} + 4$ (d) $\frac{\pi^2}{2}$

Q.18 If $\int_0^1 \frac{(1-x^2)dx}{1+x^2+x^4} = \ln b$, the $2b^2$ is

(a) 5 (b) 6

(c) 7 (d) 8

Q.19 Let the function f satisfies $f(-x)f'(x) = f(x)$ $f'(-x)$ for all x and $f(0) = 3$ then $\int_{-2010}^{2010} \frac{dx}{3 + f(x)} =$

(a) 0 (b) 2010

(c) 4020 (d) 670

Q.20 If $\int_0^x e^{tx - t^2} dt = f(x) \int_0^x e^{\frac{-t^2}{4}} dt$, then $f(x) =$

(a) $e^{\frac{x^2}{2}}$ (b) $e^{\frac{x^2}{4}}$

(c) $e^{\frac{-x^2}{2}}$ (d) $e^{\frac{-x^2}{4}}$

Q.21 $\lim_{n\to\infty} \left\{\frac{n!}{(kn)^n}\right\}^{1/n}$, where $k \neq 0$ is a constant and $n \in N$, is equal to

(a) ke (b) $k^{-1}.e$

(c) ke^{-1} (d) $k^{-1}.e^{-1}$

Q.22 $\lim_{n\to\infty} \frac{2^k + 4^k + 6^k + ... + (2n)^k}{n^{k+1}}, k \neq -1$, is equal to

(a) 2^k (b) $\frac{2^k}{k+1}$

(c) $\frac{1}{k+1}$ (d) none of these

Q.23 If $af(x) + bf\left(\frac{1}{x}\right) = \frac{1}{x} - 5, x \neq 0, a \neq 0$, than $\int_1^2 f(x) dx$ equals

(a) $\frac{(\log 2 - 5)a + \frac{13}{2}b}{a^2 - b^2}$

(b) $\frac{(\log 2 - 5)a + \frac{17b}{2}}{a^2 - b^2}$

(c) $\frac{(5 - \log 2)a + \frac{7b}{2}}{a^2 - b^2}$

(d) none of these

Q.24 If $a_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin x} dx$ then

$a_2 - a_1, a_3 - a_2, a_4 - a_3, ...$ are in

(a) AP (b) GP
(c) HP (d)none of these

Q.25 Let $a_n = \int_0^{\pi/2} \tan^n x \, dx$. then

$a_2 + a_4, a_3 + a_5, a_4 + a_6$ are in

(a) AP (b) GP
(c) HP (d) none of these

Q.26 Let $\frac{d}{dx}F(x) = \frac{e^{\sin x}}{x}, x > 0.$ If

$\int_1^4 \frac{2e^{\sin x^2}}{x} dx = F(k) - F(1)$ then one of the possible values of k is

(a) 4 (b) -4
(c) 16 (d) none of these

Q.27 If $f(x) = \int_{-1}^{1} \frac{\sin x}{1+t^2} dt$ then $f'\left(\frac{\pi}{3}\right)$ is

(a) nonexistent (b) $\pi/4$
(c) $\pi\sqrt{3}/4$ (d) none of these

Q.28 If $f\left(\frac{1}{x}\right) + x^2 f(x) = 0, x > 0,$ and

$I = \int_{1/x}^{x} f(z) dz, \frac{1}{2} \le x \le 2,$ then I is

(a) $f(2) - f(1/2)$ (b) $f(1/2) - f(2)$
(c) 0 (d) none of these

Q.29 $\lim_{n\to\infty} \frac{1^{99} + 2^{99} + ... + n^{99}}{n^{100}}$ is equal to

(a) $\frac{99}{100}$ (b) $\frac{1}{100}$

(c) $\frac{1}{99}$ (d) $\frac{1}{101}$

Q.30 $\int_0^{400\pi} \sqrt{1 - \cos 2x} \, dx$ is equal to-

(a) $400\sqrt{2}$ (b) $800\sqrt{2}$
(c) 0 (d) None of these

Q.31 If $[x]$ stands for the greatest integer function, the value of $\int_5^{10} \frac{[x^2]}{[x^2 - 30x + 225] + [x^2]} dx$ is

(a) 3 / 2 (b) 1
(c) 5 / 2 (d) none of these

Q.32 Let $\frac{d}{dx}(F(x)) = \frac{e^{\sin x}}{x}, x > 0.$

If $\int_1^4 \frac{2e^{\sin x^2}}{x} dt = F(k) - F(1)$, then one of the possible values of k is

(a) 4 (b) 8
(c) 16 (d) 32

Q.33 The expression $\frac{\int_0^n [x] dx}{\int_0^n \{x\} dx}$, where $[x]$ and $\{x\}$ are integral and fractional parts of x and $n \in N$ is equal to

(a) $\frac{1}{n-1}$ (b) $\frac{1}{n}$
(c) n (d) $n - 1$

Q.34 If $f(x) = Min\{|x-1|, |x|, |x+1|\}$, then $\int_{-1}^{1} f(x) dx$ equals

(a) 1 (b) 0
(c) 2 (d) none of these

Q.35 The value of $\sum_{n=1}^{1000} \int_{n-1}^{n} e^{x-[x]} dx$ is ($[x]$ is the greatest integer function

(a) $\frac{e^{1000} - 1}{1000}$ (b) $\frac{e^{1000} - 1}{e - 1}$

(c) $1000\,(e-1)$ (d) $\frac{e-1}{1000}$

Q.36 For any integer n, the integral $\int_0^{\pi} e^{\cos^2 x} \cos^3(2n+1)x \, dx$ has the value

(a) π (b) 1
(c) 0 (d) none of these

Q.37 If $\int_a^b \frac{x^n}{x^n + (16-x)^n} dx = 6$, then

(a) $a = 4, b = 12, n \in R$
(b) $a = 2, b = 14, n \in R$
(c) $a = -4, b = 20, n \in R$
(d) $a = 2, b = 8, n \in R$

Q.38 If $I_1 = \int_0^1 2^{x^2}dx, I_2 = \int_0^1 2^{x^3}dx, I_3 = \int_1^2 2^{x^2}dx$ and $I_4 = \int_1^2 2^{x^3}dx$ then

(a) $I_3 = I_4$ (b) $I_3 > I_4$
(c) $I_2 > I_1$ (d) $I_1 > I_2$

Q.39 The value of the integral $\int_0^1 e^{x^2}dx$ is

(a) greater than e (b) less than e
(c) less than 1 (d) none of these

Q.40 If $\int_{10}^{19} \frac{\sin x}{1+x^a}dx < \frac{1}{9}$ then the minimum value of odd integer $a > 1$ is

(a) 3 (b) 5
(c) 7 (d) 11

2. Question Based Area Under the Curve

Q.41 The area bounded by the curve $y = \sqrt{x}$, the line $2y + 3 = x$ and the x-axis in the first quadrant is

(a) 9 (b) $\frac{27}{4}$
(c) 36 (d)18

Q.42 Let $f(x)$ be a continuous function such that the area bounded by the curve $y = f(x)$, the x-axis and the two ordinates x = 0 and x = a is $\frac{a^2}{2} + \frac{a}{2}\sin a + \frac{\pi}{2}\cos a$. Then $f\left(\frac{\pi}{2}\right)$ is

(a) $\frac{1}{2}$ (b) $\frac{\pi^2}{8} + \frac{\pi}{4}$
(c) $\frac{\pi+1}{2}$ (d) $\frac{\pi}{2}$

Q.43 The area bounded by the curve f(x) = ce^x (c>0), the x-axis and the two ordinates x = p and x = q is proportional to

(a) $f(p).f(q)$ (b) $|f(p) - f(q)|$
(c) $f(p) + f(q)$ (d) $\sqrt{f(p)f(q)}$

Q.44 The ordinate x =a divides the area bounded by the x-axis, the curve $y = 1 + \frac{8}{x^2}$ and the ordinates x = 2 and x = 4 into equal parts. Then a is

(a) 3 (b) $\frac{7}{2}$
(c) $2\sqrt{2}$ (d) $\frac{5}{2}$

Q.45 The area of the ellipse $\frac{(x+1)^2}{4} + y^2 = 1$ falling in the first quadrant is

(a) $\frac{1}{6}\left(4\pi - 3\sqrt{3}\right)$ (b) $\frac{1}{12}\left(4\pi - 3\sqrt{3}\right)$
(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{3}\left(\pi - \sqrt{3}\right)$

Q.46 The area bounded by the curves $y = f(x)$, the x-axis and the ordinates $x = 1$ and $x = b$ is $(b - 1)\sin(3b + 4)$. Then, $f(x)$ is
(a) $(x - 1)\cos(3x + 4)$
(b) $8\sin(3x + 4)$
(c) $\sin(3x + 4) + 3(x - 1)\cos(3x + 4)$
(d) None of these

Q.47 The triangle formed by the tangent to the curve $f(x) = x^2 + bx - b$ at the point (1, 1) and the coordinate axes, lies in the first quadrant. If its area is 2, then the value of b is
(a) –1 (b) 3
(c) –3 (d) 1

Q.48 The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is
(a) 1 (b) 2
(c) $2\sqrt{2}$ (d) 4

Q.49 The area of the quadrilateral formed by the tangents at the end points of latusrectum to the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$, is
(a) 27/4 sq. unit (b) 9 sq. unit
(c) 27/2 sq. unit (d) 27 sq. unit

Q.50 The area bounded by the curves $y = \sqrt{x}, 2y + 3 = x$ and x-axis in the Ist quadrant is
(a) 9 sq. unit (b) 27/4 sq. unit
(c) 36 sq. unit (d) 18 sq. unit

Q.51 The area enclosed between the curves $y = ax^2$ and $x = ay^2 (a > 0)$ is 1 sq. unit. Then, the value of a is
(a) $1/\sqrt{3}$ (b) 1/2
(c) 1 (d) 1/3

Q.52 The area of the equilateral triangle, in which three coins of radius 1 cm are placed, as shown in the figure, is



(a) $(6 + 4\sqrt{3})$ sq cm (b) $(4\sqrt{3} - 6)$ sq cm

(c) $(7 + 4\sqrt{3})$ sq cm (d) $4\sqrt{3}$ sq cm

Q.53 The area bounded by the curves $y = (x-1)^2$, $y = (x+1)^2$ and $y = \frac{1}{4}$ is

(a) 1/3 sq unit (b) 2/3 sq unit

(c) 1/4 sq unit (d) 1/5 sq unit

Q.54 The area of the region between the curves $y = \sqrt{\frac{1+\sin x}{\cos x}}$ and $y = \sqrt{\frac{1-\sin x}{\cos x}}$ and bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$ is

(a) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}}dt$

(b) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

(c) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

(d) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}}dt$

Q.55 Let the straight line $x = b$ divide the area enclosed by $y = (1-x)^2$, $y = 0$ and $x = 0$ into two parts $R_1(0 \le x \le b)$ and $R_2(b \le x \le 1)$ such that $R_1 - R_2 = \frac{1}{4}$. Then b equals

(a) 3/4 (b) 1/2

(c) 1/3 (d) 1/4

Q.56 Let $f : [-1, 2] \to [0, \infty)$ be a continuous function such that $f(x) = f(1-x)$ for all $x \in [-1, 2]$. Let $R_1 = \int_{-1}^{2} xf(x)dx$ and R_2 be the area of the region bounded by $y = f(x)$, $x = -1$, $x = 2$ and the x-axis. Then

(a) $R_1 = 2R_2$ (b) $R_1 = 3R_2$

(c) $2R_1 = R_2$ (d) $3R_1 = R_2$

Q.57 The area bounded by the curve $f(x) = x + \sin x$ and its inverse function between the ordinates $x = 0$ to $x = 2\pi$ is

(a) 4π sq units (b) 8π sq units

(c) 4 sq units (d) 8 sq units

Q.58 Area of the region bounded by the curves $y = x^2$ and $y = \frac{2}{1+x^2}$ is

(a) $\pi + \frac{2}{3}$ (b) $\pi - \frac{2}{3}$

(c) $\frac{\pi}{2} + \frac{1}{3}$ (d) $2\pi - \frac{2}{3}$

Q.59 The area of the region bounded by the curve $C : y = \frac{x+1}{x^2+1}$ and the line $L : y = 1$ is

(a) $1 - \frac{1}{2}\ln 2 + \frac{\pi}{4}$ (b) $\ln 2 - \frac{\pi}{4} + 1$

(c) $\frac{1}{2}\ln 2 + \frac{\pi}{4} - 1$ (d) $\ln 2 + \frac{\pi}{4} + 1$

Q.60 Let $f(x) = \min\{x + 1, \sqrt{1-x}\}$ for all $x \le 1$. Then area bounded by $y = f(x)$ and x-axis is

(a) 1/6 sq units (b) 5/6 sq units

(c) 7/6 sq units (d) 11/6 sq units

Q.61 The area of the smaller portion enclosed by the curves $x^2 + y^2 = 9$ and $y^2 = 8x$ is

(a) $\frac{\sqrt{2}}{3} + \frac{9\pi}{4} - \frac{9}{2}\sin^{-1}\left(\frac{1}{3}\right)$

(b) $2\left(\frac{\sqrt{2}}{3} + \frac{9\pi}{4} - \frac{9}{2}\sin^{-1}\left(\frac{1}{3}\right)\right)$

(c) $2\left(\frac{\sqrt{2}}{3} + \frac{9\pi}{4} + \frac{9}{2}\sin^{-1}\left(\frac{1}{3}\right)\right)$

(d) $\frac{\sqrt{2}}{3} + \frac{9\pi}{4} + \frac{9}{2}\sin^{-1}\left(\frac{1}{3}\right)$

Q.62 The area of the loop of the curve $y^2 = x^4(x + 2)$ is [in square units]

(a) $\frac{32\sqrt{2}}{105}$ (b) $\frac{64\sqrt{2}}{105}$

(c) $\frac{128\sqrt{2}}{105}$ (d) $\frac{256\sqrt{2}}{105}$

Q.63 A curve $y = f(x)$ passes through the origin and lies entirely in the first quadrant. Through any point $P(x, y)$ on the curve, lines are drawn parallel to the coordinate axes. If the curve divides the area formed by these lines and

coordinates axes in $m : n$, then the value of $f(x)$ is equal to

(a) cx^{m+n} (b) cx^{m-n}

(c) $x^{m/n}$ (d) $cx^{m/n}$

Q.64 The area bounded by the curve $y = f(x)$, $y = x$ and the lines $x = 1$, $x = t$ is $(t + \sqrt{1+t^2}) - \sqrt{2} - 1$ sq unit, for all $t > 1$. If $f(x)$ satisfying $f(x) > x$ for all $x > 1$, then $f(x)$ is equal to

(a) $x + 1 + \frac{x}{\sqrt{1+x^2}}$ (b) $x + \frac{x}{\sqrt{1+x^2}}$

(c) $1 + \frac{x}{\sqrt{1+x^2}}$ (d) $\frac{x}{\sqrt{1+x^2}}$

Q.65 A square is constructed by joining the points $A(-1, 1)$, $B(-1, -1)$, $C(1, -1)$ and $D(1, 1)$. Considering *A, B, C* and *D* as centres, four circles each of 2 unit radius are described on the square. Then the area common to these circles is

(a) $3\left(\frac{\pi}{4} - \sqrt{3}\right)$ sq unit

(b) $4\left(\frac{\pi}{3} - \sqrt{3}\right)$ sq unit

(c) $\left(\frac{\pi}{3} - \sqrt{3}\right)$ sq unit

(d) $\left(\frac{\pi}{4} - \sqrt{3}\right)$ sq unit

Q.66 If $y = f(x)$ makes positive intercepts of 2 and 1 unit on x and y coordinate axes and encloses an area of 3/4 sq unit with the axes, then $\int_0^2 x\, f'(x)dx$ is

(a) 3/2 (b) 1

(c) 5/4 (d) –3/4

Section-B
(Previous Years Question)

Q.1 The following integral $\int_{\pi/4}^{\pi/2} (2\text{cosec}x)^{17}\, dx$ is equal to

(a) $\int_0^{\log(1+\sqrt{2})} 2\left(e^u + e^{-u}\right)^{16} du$

(b) $\int_0^{\log(1+\sqrt{2})} \left(e^u + e^{-u}\right)^{17} du$

(c) $\int_0^{\log(1+\sqrt{2})} \left(e^u - e^{-u}\right)^{17} du$

(d) $\int_0^{\log(1+\sqrt{2})} 2\left(e^u - e^{-u}\right)^{16} du$

Q.2 Let $f:[0,2] \to R$ be a function which is continuous on [0,2] and is differentiable on (0,2) with f(0) =1. Let $F(x) = \int_0^{x^2} f\left(\sqrt{t}\right)dt$ for $x \in [0,2]$. If $F'(x) = f'(x)$ for all $x \in (0,2)$, then $F(2)$ equals

(a) e^2-1 (b) e^4-1

(c) e-1 (d) e^4

Q.3 The area enclosed by the curves y= sinx+cosxand y= |cosx-sinx| over the interval $\left[0, \frac{\pi}{2}\right]$ is

(a) $4\left(\sqrt{2}-1\right)$ (b) $2\sqrt{2}\left(\sqrt{2}-1\right)$

(c) $2\left(\sqrt{2}+1\right)$ (d) $2\sqrt{2}\left(\sqrt{2}+1\right)$

Q.4 The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals (for some arbitrary constant K)

(a) $-\frac{1}{(\sec x + \tan x)^{11/2}}\left\{\frac{1}{11} - \frac{1}{7}(\sec x + \tan x)^2\right\} + K$

(b) $\frac{1}{(\sec x + \tan x)^{11/2}}\left\{\frac{1}{11} - \frac{1}{7}(\sec x + \tan x)^2\right\} + K$

(c) $-\frac{1}{(\sec x + \tan x)^{11/2}}\left\{\frac{1}{11} + \frac{1}{7}(\sec x + \tan x)^2\right\} + K$

(d) $\frac{1}{(\sec x + \tan x)^{11/2}}\left\{\frac{1}{11} + \frac{1}{7}(\sec x + \tan x)^2\right\} + K$

Q.5 The value of the integral $\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln\frac{\pi + x}{\pi - x}\right)\cos x dx$ is

(a) 0 (b) $\frac{\pi^2}{2}-4$

(c) $\frac{\pi^2}{2}+4$ (d) $\frac{\pi^2}{2}$

Q.6 The valule of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x\sin x^2}{\sin x^2+\sin\left(\ln 6-x^2\right)}dx$ is

(a) $\frac{1}{4}\ln\frac{3}{2}$ (b) $\frac{1}{2}\ln\frac{3}{2}$

(c) $\ln\frac{3}{2}$ (d) $\frac{1}{6}\ln\frac{3}{2}$

Q.7 Let f be a non-negative function defined on the interval $[0,1]$. If $\int_0^x \sqrt{1-\left(f'(t)\right)^2}dt=\int_0^x f(t)dt, 0\le x\le 1,$ and $f(0)=0,$ then

(a) $f\left(\frac{1}{2}\right)<\frac{1}{2}$ and $f\left(\frac{1}{3}\right)>\frac{1}{3}$

(b) $f\left(\frac{1}{2}\right)>\frac{1}{2}$ and $f\left(\frac{1}{3}\right)>\frac{1}{3}$

(c) $f\left(\frac{1}{2}\right)<\frac{1}{2}$ and $f\left(\frac{1}{3}\right)<\frac{1}{3}$

(d) $f\left(\frac{1}{2}\right)>\frac{1}{2}$ and $f\left(\frac{1}{3}\right)<\frac{1}{3}$

Q.8 Let $I=\int\frac{e^x}{e^{4x}+e^{2x}+1}dx, J=\int\frac{e^{-x}}{e^{-4x}+e^{-2x}+1}dx.$ Then, for an arbitrary constant C, the value of J-I equals

(a) $\frac{1}{2}\log\left(\frac{e^{4x}-e^{2x}+1}{e^{4x}+e^{2x}+1}\right)+C$

(b) $\frac{1}{2}\log\left(\frac{e^{2x}+e^{x}+1}{e^{2x}-e^{x}+1}\right)+C$

(c) $\frac{1}{2}\log\left(\frac{e^{2x}-e^{x}+1}{e^{2x}+e^{x}+1}\right)+C$

(d) $\frac{1}{2}\log\left(\frac{e^{4x}+e^{2x}+1}{e^{4x}-e^{2x}+1}\right)+C$

Q.9 Let $f(x)=\frac{x}{\left(1+x^n\right)^{1/n}}$ for $n\ge 2$ and

$g(x)=\underbrace{(f\circ f\circ f\circ\ldots\circ f)}_{f \text{ occurs times}}(x)$

Then $\int x^{n-2}g(x)dx$ equals

(a) $\frac{1}{n(n-1)}\left(1+nx^n\right)^{1-\frac{1}{n}}+K$

(b) $\frac{1}{n-1}\left(1+nx^n\right)^{1-\frac{1}{n}}+K$

(c) $\frac{1}{n(n+1)}\left(1+nx^n\right)^{1+\frac{1}{n}}+K$

(d) $\frac{1}{n+1}\left(1+nx^n\right)^{1+\frac{1}{n}}+K$

Q.10 $\int\frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}dx=$

(a) $\frac{\sqrt{2x^4-2x^2+1}}{x^2}+c$

(b) $\frac{\sqrt{2x^4-2x^2+1}}{x^3}+c$

(c) $\frac{\sqrt{2x^4-2x^2+1}}{x}+c$

(d) $\frac{\sqrt{2x^4-2x^2+1}}{2x^2}+c$

Q.11 If f(x) is differentiable and $\int_0^{t^2}xf(x)dx=\frac{2}{5}t^2,$ then $f\left(\frac{4}{25}\right)$ equals

(a) 2/5 (b) -5/2

(c) 1 (d) 5/2

Q.12 If $f(x)=\int_{x^2}^{x^2+1}e^{-t^2}dt,$ then f(x) increases in

(a) (2, 2) (b) no value of x

(c) $(0,\infty)$ (d) $(-\infty,0]$

Q.13 If $I_{m,n}=\int_0^1 t^m(1+t)^n dt,$ then the expression for $I_{(m,n)}$ in terms of $I_{m+1,n-1}$ is

(a) $\frac{2^n}{n+1}-\frac{n}{m+1}I_{m+1,n-1}$ (b) $\frac{n}{m+1}I_{m+1,n-1}$

(c) $\frac{2^n}{m+1}+\frac{n}{m+1}I_{m+1,n-1}$ (d) $\frac{m}{n+1}I_{m+1,n-1}$

Q.14 The integral $\int_{\frac{-1}{2}}^{\frac{1}{2}}\left[[x]+\ln\left(\frac{1+x}{1-x}\right)\right]dx$ equals

(a) $\frac{-1}{2}$ (b) 0

(c) 1 (d) $2\ln\left(\frac{1}{2}\right)$

Q.15 Let $T>0$, be a fixed real number. Suppose f is a continuous function. Such that for all $x\in R.\ f(x+T)=f(x)$.

If $I=\int_0^T f(x)dx$, then the value of $\int_3^{3+3T} f(2x)dx$ is

(a) $\frac{3}{2}I$ (b) I

(c) 3I (d) 6I

Q.16 Let $f(x)=\int_1^x \sqrt{2-t^3}dt$, then the real roots of the equation $x^2-f'(x)=0$ are

(a) ± 1 (b) $\pm\frac{1}{\sqrt{2}}$

(c) $\pm\frac{1}{2}$ (d) 0 and 1

Q.17 The value of $\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^x}dx, a>0$ is

(a) π (b) $a\pi$

(c) $\pi/2$ (d) 2π

Q.18 Let $f:(0,\infty)\to R$ and $F(x)=\int_0^x f(t)dt$. If $F(x^2)=x^2(1+x)$, then $f(4)$ equals

(a) 5 / 4 (b) 7

(c) 4 (d) 2

Q.19 The value of the integral $\int_{e^{-1}}^{e^2}\left|\frac{\log_e x}{x}\right|dx$ is

(a) 3 / 2 (b) 5 / 2

(c) 3 (d) 5

Q.20 If for a real number y, [y] is the greateger integer less than or equal to y, then the value of the integral $\int_{\pi/2}^{3\pi/2}[2\sin x]dx$ is

(a) $-\pi$ (b) 0

(c) $-\pi/2$ (d) $\pi/2$

Q.21 Let f be a positive function. Let

$I_1=\int_{1-k}^{k} xf[x(1-x)]dx, I_2=\int_{1-k}^{k} f[x(1-x)]dx,$

where $2k-1>0$, Then $\frac{I_1}{I_2}$ is

(a) 2 (b) k

(c) 1/ 2 (d) 1

Q.22 If $g(x)=\int_0^x \cos^4 t\,dt$ then $g(x+\pi)$ equals

(a) $g(x)+g(\pi)$ (b) $g(x)-g(\pi)$

(c) $g(x)-g(\pi)$ (d) $g(x)/g(\pi)$

Q.23 $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}\frac{r}{\sqrt{n^2+r^2}}$

(a) $1+\sqrt{5}$ (b) $-1+\sqrt{5}$

(c) $-1+\sqrt{2}$ (d) $1+\sqrt{2}$

Q.24 $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}\frac{r}{\sqrt{n^2+r^2}}$

(a) $1+\sqrt{5}$ (b) $-1+\sqrt{5}$

(c) $-1+\sqrt{2}$ (d) $1+\sqrt{2}$

Q.25 The value of the integral

$\int\frac{\cos^3 x+\cos^5 x}{\sin^2 x+\sin^4 x}dx$ is

(a) $\sin x-6\tan^{-1}(\sin x)+c$

(b) $\sin x-2(\sin x)^{-1}+c$

(c) $\sin x-2(\sin x)^{-1}-6\tan^{-1}(\sin x)+c$

(d) $\sin x-2(\sin x)^{-1}+5\tan^{-1}(\sin x)+c$

Q.26 The value of $\int_{\pi}^{2\pi}[2\sin x]dx$ where [.] represents the greatest integer function is

(a) $\frac{-5\pi}{3}$ (b) $-\pi$

(c) $\frac{5\pi}{3}$ (d) -2π

Q.27 Let a, b, c be non-zero real numbers such that $\int_0^1(1+\cos^8 x)(ax^2+bx+c)dx$

$=\int_0^2(1+\cos^8 x)(ax^2+bx+c)dx$ then

$ax^2+bx+c=0$ has

(a) no root in (0, 2)

(b) at least one root in (1,2)

(c) a double root in (0, 2)

(d) two imaginary roots

Q.28 $\int_0^1 \left(1+e^{-x^2}\right)dx$

(a) -1 (b) 2

(c) $1+e^{-1}$ (d) none of these

Q.29 Let $f:(0,\infty)\to R$ be given by $f(x)=\int_{1/x}^{x} e^{-\left(t+\frac{1}{t}\right)}\frac{dt}{t}$. Then

(a) $f(x)$ is monotonically increasing on $[1,\infty)$

(b) $f(x)$ is monotonically decreasing on $(0,1)$

(c) $f(x)+f\left(\frac{1}{x}\right)=0$, for all $x\in(0,\infty)$

(d) $f\left(2^x\right)$ is an odd function of x on R

Q.30 The value (s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2}dx$ is (are)

(a) $\frac{22}{7}-\pi$ (b) $\frac{2}{105}$

(c) 0 (d) $\frac{71}{15}-\frac{3\pi}{2}$

Q.31 Let f be a real - valued function defined on the interval $(0,\infty)$ by $f(x)=\ln x+\int_0^x \sqrt{1+\sin t}\,dt$. Then which of the following statement (s) is (are) true?

(a) $f''(x)$ exists for all $x\in(0,\infty)$

(b) $f'(x)$ exists for all $x\in(0,\infty)$ and f' is continuous on $(0,\infty)$, but not differentiable on $(0,\infty)$

(c) there exists $\alpha>1$ such that $|f'(x)|<|f(x)|$ for all $x\in(\alpha,\infty)$

(d) there exists $\beta>0$ such taht $|f(x)|+|f'(x)|\le\beta$ for all $x\in(0,\infty)$

Q.32 If $I_n=\int_{-\pi}^{\pi}\frac{\sin nx}{(1+\pi^x)\sin x}dx$, n = 0, 1,2,....then

(a) $I_n=I_{n+2}$ (b) $\sum_{m=1}^{10} I_{2m+1}=10\pi$

(c) $\sum_{m=1}^{10} I_{2m}=0$ (d) $I_n=I_{n+1}$

Q.33 Area of the region bounded by the curve $y = e^x$ and lines x = 0 and y = e is

(a) e-1 (b) $\int_1^e \ln(e+1-y)dy$

(c) $e-\int_1^1 e^x dx$ (d) $\int_1^1 \ln y\,dy$

Q.34 Let $S_n=\sum_{k=1}^{n}\frac{n}{n^2+kn+k^2}$ and

$T_n=\sum_{k=0}^{n-1}\frac{n}{n^2+kn+k^2}$, n = 1,2,3,..... Them,

(a) $S_n<\frac{\pi}{3\sqrt{3}}$ (b) $S_n>\frac{\pi}{3\sqrt{3}}$

(c) $T_n<\frac{\pi}{3\sqrt{3}}$ (d) $T_n>\frac{\pi}{3\sqrt{3}}$

Q.35 Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty,\infty)$ such that f(x) = f (1-x)and $f'\left(\frac{1}{4}\right)=0$ Then,

(a) $f''(x)$ vanishes at least twice on [0, 1]

(b) $f'\left(\frac{1}{2}\right)=0$

(c) $\int_{-1/2}^{1/2} f\left(x+\frac{1}{2}\right)\sin x\,dx=0$

(d) $\int_0^{1/2} f(t)e^{\sin\pi t}dt=\int_{1/2}^{1}(1-t)e^{\sin\pi t}dt$

Q.36 **Statement- I :** The value of the integral $\int_{\pi/6}^{\pi/3}\frac{dx}{1+\sqrt{\tan x}}$ is equal to $\pi/6$

Statement-II : $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$.

(a) Statement -I is true, Statement-II is true, Statement-II is not a correct explanation for Statement-I

(b) Statement -I is true, Statement-II is false.

(c) Statement -I is false, Statment - II is true.

(d) Statement-I is true, Statement - II is true, Statement-II is a correct explanation for Statement-I.

Q.37 If the integral $\int \frac{5\tan x}{\tan x-2}dx = x + a\ln$ $|\sin x - 2\cos x| + k$, then a is equal to

(a) 1 (b) 2
(c) –1 (d) –2

Q.38 $\int_0^{\pi} [\cot x]dx$, where [.] denotes the greatest integer function, is equal to

(a) 1 (b) –1
(c) $-\pi/2$ (d) $\pi/2$

Q.39 Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}}dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}}dx$. Then which one of the following is true?

(a) $I > \frac{2}{3}$ and $J < 2$ (b) $I > \frac{2}{3}$ and $J > 2$

(c) $I < \frac{2}{3}$ and $J < 2$ (d) $I < \frac{2}{3}$ and $J > 2$

Q.40 Let $F(x) = f(x) + f\left(\frac{1}{x}\right)$, where

$f(x) = \int_1^x \frac{\log t}{1+t}dt$, Then $F(e)$ equals

(a) 1 (b) 2
(c) 1/2 (d) 0

Q.41 The value of $\int_1^a [x]f'(x)dx, a > 1$, where [x] denotes the greatest integer not exceeding x is

(a) $a\,f(a) - \{f(1) + f(2) + ... + f([a])\}$

(b) $[a]\,f(a) - \{f(1) + f(2) + ... + f([a])\}$

(c) $[a]f([a]) - \{f(1) + f(2) + ... + f(a)\}$

(d) $af([a]) - \{f(1) + f(2) + ... + f(a)\}$

Q.42 $\int_{-3\pi/2}^{-\pi/2} \left[(x+\pi)^3 + \cos^2(x+3\pi)\right]dx$ is equal to

(a) $\frac{\pi^4}{32}$ (b) $\frac{\pi^4}{32} + \frac{\pi}{2}$

(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{2} - 1$

Q.43 $\lim_{n\to\infty}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2} + \frac{2}{n^2}\sec^2\frac{4}{n^2} + ... + \frac{1}{n}\sec^2 1\right]$ equals

(a) $\frac{1}{2}\cos ex1$ (b) $\frac{1}{2}\sec 1$

(c) $\frac{1}{2}\tan 1$ (d) $\tan 1$

Q.44 The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x}dx, a > 0$, is

(a) $\pi/2$ (b) $a\pi$
(c) 2π (d) π/a

Q.45 If $I_1 = \int_0^1 2^{x^2}dx, I_2 = \int_0^1 2^{x^3}dx, I_3 = \int_1^2 2^{x^2}dx$ and $I_4 = \int_1^2 2^{x^3}dx$ then

(a) $I_1 > I_2$ (b) $I_2 > I_1$
(c) $I_3 > I_4$ (d) $I_3 = I_4$

Q.46 Let $F: R \to R$ be a differntiable function having $f(2) = 6,\ f'(2) = \left(\frac{1}{48}\right)$. Then $\lim_{x\to 2} \int_6^{f(x)} \frac{4t^3}{x-2}dt$ equals

(a) 36 (b) 24
(c) 18 (d) 12

Q.47 If $f(a+b-x) = f(x)$, then $\int_a^b xf(x)dx$ is equal to

(a) $\frac{a+b}{2}\int_a^b f(x)dx$ (b) $\frac{a-b}{2}\int_a^b f(x)dx$

(c) $\frac{a+b}{2}\int_a^b f(a+b-x)dx$ (d) $\frac{a+b}{2}\int_a^b f(b-x)dx$

Q.48 If $f(y) = e^y, g(y) = y; y > 0$ and $F(t) = \int_0^t f(t-y)g(y)dy$, then

(a) $F(t) = e^t - (1+t)$ (b) $F(t) = te^t$
(c) $F(t) = te^{-t}$ (d) $F(t) = 1 - e^t(1+t)$

Q.49 $I_n = \int_0^{\pi/4} \tan^n x\,dx$, then $\lim_{n\to\infty} n[I_n + I_{n-2}]$ equals

(a) 1/2 (b) 1
(c) ∞ (d) 0

Q.50 $\int_0^{\sqrt{2}} \left[x^2\right]dx$ is

(a) $2 - \sqrt{2}$ (b) $2 + \sqrt{2}$
(c) $\sqrt{2} - 1$ (d) $\sqrt{2} - 2$

Section-C
(Board Pattern)

Q.1 Evaluate $\int_0^{\pi/4} \sqrt{1-\sin 2x}\,dx$

Q.2 If $\int_0^a \sqrt{x}\,dx = 2a\int_0^{\pi/2} \sin^3 x\,dx$, find the value of $\int_a^{a+1} x\,dx$

Q.3 Evaluate $\int_0^1 \left\{ xe^x + \sin\frac{\pi x}{4} \right\} dx$

Q.4 Evaluate $\int_0^\infty \frac{1}{(x^2+a^2)(x^2+b^2)}\,dx$

Q.5 Evaluate $\int_{\pi/4}^{\pi/2} \cos 2x \log\sin x\,dx$

Q.6 Evaluate $\int_1^2 \left(\frac{x-1}{x^2} \right) e^x dx$

Q.7 Evaluate : (i) $\int_0^{1/\sqrt{2}} \frac{\sin^{-1} x}{(1-x^2)^{3/2}}\,dx$,

(ii) $\int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2} \right) dx$

Q.8 Evaluate $\int_0^{\pi/2} (\sqrt{\tan x} + \sqrt{\cot x})\,dx$

Q.9 Evaluate $\int_0^{\pi/2} \frac{\sin 2x}{\sin^4 x + \cos^4 x}\,dx$

Q.10 If $I_n = \int_0^{\pi/4} \tan^n x\,dx$, prove that $I_n + I_{n+2} = \frac{1}{n+1}$.

Q.11 Evaluate (i) $\int_0^a \sin^{-1}\sqrt{\frac{x}{a+x}}\,dx$,

(ii) $\int_0^{\pi/4} \frac{1}{a^2\sin^2 x + b^2\cos^2 x}\,dx$

Q.12 Evaluate : (i) $\int_1^4 (|x-1| + |x-2| + |x-3|)\,dx$,

(ii) $\int_{-1}^2 |x^3 - x|\,dx$

Q.13 Evaluate $\int_{\pi/6}^{\pi/3} \frac{1}{1+\sqrt{\cot x}}\,dx$

Q.14 Evaluate (i) $\int_0^{\pi/2} \log\tan x\,dx$,

(ii) $\int_0^{\pi/4} \log(1+\tan x)\,dx$

Q.15 Evaluate $\int_0^{\pi/2} (2\log\sin x - \log\sin 2x)\,dx$

Q.16 Evaluate $\int_0^1 \cot^{-1}(1-x+x^2)\,dx$

Q.17 Evaluate $\int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx$

Q.18 Evaluate $\int_0^\pi \frac{x\tan x}{\sec x + \sin x}\,dx$

Q.19 Prove that $\int_0^\pi \frac{x}{1-\cos\alpha\sin x}\,dx = \frac{\pi(\pi-\alpha)}{\sin\alpha}$

Q.20 Evaluate $\int_{-\pi/2}^{\pi/2} \sin^7 x\,dx$

Q.21 Evaluate $\int_{-a}^a \sqrt{\frac{a-x}{a+x}}\,dx$

Q.22 Evaluate $\int_{-1}^{3/2} |x\sin\pi x|\,dx$

Q.23 Evaluate : $\int_0^\pi \tan^{-1}\left(\frac{2x-1}{1+x-x^2} \right) dx$

Q.24 Evaluate : $\int_0^\pi \log(1+\cos x)\,dx$

Q.25 Evaluate : $\int_0^{100\pi} \sqrt{1-\cos 2x}\,dx$

Q.26 Evaluate :

$$\lim_{n\to\infty}\left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \ldots + \frac{1}{8n} \right] = \frac{3}{8}$$

Q.27 Evaluate :

$$\lim_{n\to\infty} \frac{1}{n}\left[\sec^2\frac{\pi}{4n} + \sec^2\frac{2\pi}{4n} + \ldots + \sec^2\frac{n\pi}{4n} \right] = \frac{4}{\pi}$$

Q.28 Evaluate

$$\int_0^{\pi/4} [\sin x + [\cos x + [\tan x + [\sec x]]]]\, dx,$$

where [.] denotes greatest integer function.

Q.29 Evaluate : $\int_0^1 x\sqrt{\frac{1-x^2}{1+x^2}}dx$.

Q.30 Evaluate : $\int_0^{\pi/2} \frac{\sin 2x}{\sin^4 x + \cos^4 x}dx$

Q.31 Evaluate : $\int_2^e \left(\frac{1}{\log x} - \frac{1}{(\log x)^2}\right)dx$

Q.32 Evaluate : $\int_0^1 \frac{\log(1+x)}{1+x^2}dx$

Q.33 Evaluate : $\int_{\pi/4}^{\pi/2} (\cos 2x)\log(\sin x)\,dx$.

Q.34 Evaluate $\int_0^3 (2x^2 + 3x + 5)\,dx$ as the limit of sum.

Q.35 Find the value of $\int_{-\pi/2}^{\pi/2} \sin^{31} x\,dx$.

Q.36 Evaluate the following integrals as limit of sums $\int_0^2 e^x\,dx$

Q.37 Evaluate the integrals as limit of sums $\int_0^3 (2x^2 + 3x + 5)dx$

Q.38 Find the area of the region included between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$, where $a > 0$.

Q.39 Find the area of the region $\{(x, y) : x^2 \le y \le x\}$.

Q.40 Find the area of the region $\{(x, y) : x^2 \le y \le |x|\}$

Q.41 Find the area of the smaller region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the straight line $\frac{x}{a} + \frac{y}{b} = 1$.

Q.42 Find the area of the region $\{(x, y) : y^2 < 4x, 4x^2 + 4y^2 \le 9\}$.

Q.43 Find the area of the circle $4x^2 + 4y^2 = 9$ which is interior to the parabola $y^2 = 4x$.

Q.44 Find the area of the region enclosed between the two circles $x^2 + y^2 = 1$ and $(x-1)^2 + y^2 = 1$.

Q.45 Using integration, find the area of the triangle *ABC* whose vertices have coordinates *A*(2, 5), *B*(4, 7) and *C*(6, 2).

Q.46 Sketch region common to the circle $x^2 + y^2 = 16$ and the parabola $x^2 = 6y$. Also find the area of the region using integration.

Q.47 If the area enclosed between the curves $y = ax^2$ and $x = ay^2$ is 159 unit. Find the value of a.

Q.48 Prove that the curves $y^2 = 4x$ and $x^2 = 4y$ divide the area of the square bounded by $x = 0$, $y = 0$, $x = 4$ and $y = 4$ into three equal parts.

Q.49 Using integration, find the area of the triangular region, the equations of whose sides are $y = 2x + 1$, $y = 3x + 1$ and $x = 4$.

Q.50 Find the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the ordinates $x = ae$ and $x = 0$, where $b^2 = a^2(1 - e^2)$ and $e < 1$.

Q.51 Using integration, find the area of the triangle *ABC* coordinates of whose vertices are *A*(4, 1), *B*(6, 6) and *C*(8, 4).

Q.52 Find the area of the region in the first quadrant enclosed by x-axis, the line $x = \sqrt{3}y$ and the circle $x^2 + y^2 = 4$.

Q.53 Using integration, find the area of the region : $\{(x, y) : |x - 1| \le y \le \sqrt{5 - x^2}\}$

CHAPTER 2

DIFFERENTIAL EQUATIONS

1. ORDER AND DEGREE OF DIFFERENTIAL EQUATIONS

A differential equation is an equation involving an unknown function and its derivatives.

For example, the following are differential equations involving the unknown function y.

$$\frac{dy}{dx} = 7x + 11$$

$$4\left(\frac{d^3y}{dx^3}\right) + (\cos x)\frac{d^2y}{dx^2} + 7xy = 0$$

$$\left(\frac{d^2y}{dx^2}\right)^3 + 5y\left(\frac{dy}{dx}\right)^7 + y^3\left(\frac{dy}{dx}\right)^2 = 3x$$

A differential equation in an ordinary differential equation if the unknown function depends on only one independent variable.

If the unknown function depends on two or more independent variables the differential equation is a partial differential equation. We will confine ourselves to the ordinary differential equations only in this chapter.

The **order** of a differential equation is the order of the highest derivative which occurs.

The **degree** of a differential equation which **can be** written as a polynomial in the derivatives is the degree of the highest order derivative which then occurs.

Thus $3\left(\frac{dy}{dx}\right)^5 + \left(\frac{dy}{dx}\right)^2 + 5y = mx$ is a differential equation of first order and fifth degree, while

$\frac{d^4y}{dx^4} + \left(\frac{d^3y}{dx^3}\right)^8 + 3\left(\frac{d^2y}{dx^2}\right)^5 + 8y = 9x$ is a differential equation of fourth order and first degree.

Illustration 1: Determine the order and degree of each of the following differential equations.

(i) $y''' - 5xy' = e^{2x} + 3$

(ii) $t\ddot{y} + t^2\dot{y} + (\sin t)\sqrt{y} = t^2 - 3t + 5$

(iii) $\frac{d^2t}{ds^2} + 2st\frac{dt}{ds} = s$

(iv) $\left(\frac{d^4b}{dp^4}\right)^5 + 3\left(\frac{db}{dp}\right)^{10} + b^7 - 2b^5 = p^2$

Solution : First a word about the notation. The expression y', y'', y''', $y^{(4)}$ ....., $y^{(n)}$ are often used to represent respectively the 1st, 2nd, 3rd, 4th, ..., n^{th} order derivatives of y with respect to independent variable under consideration. Thus y'' represents $\frac{d^2y}{dx^2}$ if x is the

independent variable, and y'' represents $\frac{d^2y}{dp^2}$ if p is the independent variable.

Note that parentheses are used $y^{(n)}$ to distinguish it from y^n the nth power of y. When the independent variable happens to be time, denoted by t, primes are often replaced by dots. $\dot{y}, \ddot{y}, \dddot{y}$ then represent $\frac{dy}{dt}, \frac{d^2y}{dt^2}, \frac{d^3y}{dt^3}$ respectively.

(i) As the highest order derivative, y''' is of 3rd order the order of the differential equation is 3. The exponent of highest order derivatives, y''' is one. Hence the degree is 1.

(ii) Here independent variable is t. As the highest order derivative, $\ddot{y}$, is of 2nd order - the order of the differential equation is 2. The exponent of the highest order derivative, $\ddot{y}$ is one. Hence the degree is 1.

(iii) Here independent variable is s. As the highest order derivative, $\frac{d^2t}{ds^2}$, is of 2nd order, the order of the differential equation is 2. The exponent of the highest order derivative, $\frac{d^2t}{ds^2}$, is one. Hence the degree is 1.

(iv) Here independent variable is p. As the highest order derivative, $\frac{d^4b}{dp^4}$, is of 4th order, the order of the differential equation is 4. The exponent of the highest order derivative, $\frac{d^4b}{dp^4}$, is five. Hence the degree is 5.

Illustration 2 : Find the order and degree of the following differential equations.

(i) $\left\{1+\left(\frac{dy}{dx}\right)^2\right\}^{\frac{3}{2}} = k\frac{d^2y}{dx^2}$

(ii) $\frac{dy}{dx} + m^2y + \int 3x^2dx = k$

(iii) $\frac{d^2y}{dx^2} + a\frac{dy}{dx} + b\int y\,dx = c$

Solution : (i) The highest derivative occurring in the differential equation $\left\{1+\left(\frac{dy}{dx}\right)^2\right\}^{\frac{3}{2}} = k\frac{d^2y}{dx^2} = k\frac{d^2y}{dx^2}$ is $\frac{d^2y}{dx^2}$ which is of order 2. Hence the order of the differential equation is 2. Recall that the definition of degree of a differential equation requires the differential equation to be written as a polynomial in derivatives. That means we have to recast, if possible, the equation $\left\{1+\left(\frac{dy}{dx}\right)^2\right\}^{\frac{3}{2}} = k\frac{d^2y}{dx^2}$ as a polynomial derivatives.

Raising both sides to the power 2 we have,

$$\left\{1+\left(\frac{dy}{dx}\right)^2\right\}^3 = \left(k\frac{d^2y}{dx^2}\right)^2$$

The exponent of the highest order derivative, i.e., $\frac{d^2y}{dx^2}$, is 2. Thus the degree of the equation is 2.

(ii) $\frac{d^2y}{dx^2}+a\frac{dy}{dx}+b\int y\,dx = c$ is an integro-differential equation and to make the equation free of integral we have to differentiate it to obtain.

$$\frac{d^3y}{dx^3}+a\frac{d^2y}{dx^2}+by = 0$$

Now, it is obvious that order is 3 due to presence of the term $\frac{d^3y}{dx^3}$ the highest order derivative and degree is 1 for this is the exponent of $\frac{d^3y}{dx^3}$.

Note :- Although (ii) and (iii) equations resemble in their make up, the fine difference is that in (ii) we have $\int 3x^2dx$, a term like $\int f(x)dx$, while in (iii) we have $\int y\,dx$, the integral of unknown function w.r.t. x.

2. FORMATION OF ORDINARY DIFFERENTIAL EQUATION (ODE)

A relation between the variables which involves n essential arbitrary constants, as $y = ax^2 + bx$ or $y = x^4 + kx$ is called a **primitive**. These n constants are called essential inthe sense that they cannot be replaced by a smaller number of constants.

Consider for example $y = a\cos x + b\sin(x + C)$

It seems at first that the primitive has 3 essential arbitrary constants. But recast the equation as below :

$y = a\cos x + b\sin x\cos C + b\cos x\sin C$

$= (a + b\sin C)\cos x + (b\cos C)\sin x$

$= \alpha\cos x + \beta\sin x$

where $\alpha = a + b\sin C$ and $\beta = b\cos C$. Now it is obvious that the equation has in fact only 2 essential arbitrary constants. In general, a primitive involving n essential arbitrary constants will give rise to a differential equation of order n. This equation is obtained by eliminating the n constants between the $(n + 1)$ equation consisting of primitive and the n equation obtained by differentiating the primitive n times with respect to the independent variable.

Illustration 3 : Form the differential equation whose primitive is $y = e^x(A\cos x + B\sin x)$, where A and B are essential arbitrary constants.

Solution : **Ist method :**

$y = e^x(A\cos x + B\sin x)$

Differentiating the above w.r.t. x,

$\frac{dy}{dx} = e^x(-A\sin x + B\cos x) + e^x(A\cos x + B\sin x)$

$= e^x(A\cos x + B\sin x) + \left\{\frac{dy}{dx} - y\right\} + \frac{dy}{dx}$ .... (i) $[\because\ y = e^x(A\cos x + B\sin x)]$

Differentiating again w.r.t. x, we have,

$\frac{d^2y}{dx^2} = e^x(-A\cos x - B\sin x) + e^x(-A\sin x + B\cos x) + \frac{dy}{dx}$

$= -e^x(A\cos x + B\sin x) + \left\{\frac{dy}{dx} - y\right\} + \frac{dy}{dx}$ .... (ii)

$\left[\text{from (i) } e^x(-A\sin x + B\cos x) = \frac{dy}{dx} - y\right]$

$\Rightarrow \quad \frac{d^2y}{dx^2} = -y + \frac{dy}{dx} - y + \frac{dy}{dx}$ $[\because\ y = e^x(A\cos x - B\sin x)]$

$\Rightarrow \quad \frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$

This is the required differential equation

2nd method :

Rewrite $y = e^x(A\cos x + B\sin x)$ as

$e^{-x}y = A\cos x + B\sin x$

Differentiating w.r.t. x, we have

$e^{-x}(y_1 - y) = -A\sin x + B\cos x$

Again differentiating w.r.t. x, we have

$e^{-x}(y_2 - y_1) - e^{-x}(y_1 - y) = -A\cos x - B\sin x$

$\Rightarrow \quad e^{-x}(y_2 - 2y_1 + y) = -(A\cos x + B\sin x) = -e^{-x}y \quad [\because\ e^{-x}y = A\cos x + B\sin x]$

Cancelling e^{-x} on both sides and bringing everything to the left hand side

We have, $y_2 - 2y_1 + 2y = 0$

i.e., $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$, as before

Observe that the order of the resulting differential equation is the same as the number of essential arbitrary constants in the primitive.

***Illustration 4*:** Form the differential equation whose primitive is $y = ae^x + be^{2x} + ce^{3x}$

Solution : **1st method :**

$y = ae^x + be^{2x} + ce^{3x}$

$\Rightarrow \quad ye^{-x} = a + be^x + ce^{2x}$

Differentiating w.r.t. x we have

$e^{-x}(y_1 - y) = be^x + 2ce^{2x}$

$\Rightarrow \quad e^{-2x}(y_1 - y) = b + 2ce^x$

Differentiating w.r.t. x again, we have

$e^{-2x}\{y_2 - y_1\} - 2e^{-2x}\{y_1 - y\} = 2ce^x$

$\Rightarrow \quad e^{-2x}\{y_2 - 3y_1 + 2y\} = 2ce^x$

$\Rightarrow \quad e^{-3x}\{y_2 - 3y_1 + 2y\} = 2c$

Differentiating w.r.t. x again, we have

$e^{-3x}\{y_3 - 3y_2 + 2y_1\} - 3e^{-3x}\{y_2 - 3y_1 + 2y\} = 0$

$\Rightarrow \quad e^{-3x}\{y_3 - 6y_2 + 11y_1 - 6y\} = 0$

(as $e^{-3x} \neq 0$)

We have, $y_3 - 6y_2 + 11y_1 - 6y = 0$

Thus the required differential equation is

$$\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$$

2nd method :

$y = ae^x + be^{2x} + ce^{3x}$ gives

$ae^x + be^{2x} + ce^{3x} - y = 0$ ..... (i)

Differentiating the above relation in succession we have,

$ae^x + 2be^{2x} + 3ce^{3x} - y_1 = 0$ ..... (ii)

$ae^x + 4be^{2x} + 9ce^{3x} - y_2 = 0$ ..... (iii)

$ae^x + 8be^{2x} + 27ce^{3x} - y_3 = 0$ ..... (iv)

The eliminant of a, b and c can be written in the determinant form as

$$\begin{vmatrix} 1 & 1 & 1 & -y \\ 1 & 2 & 3 & -y_1 \\ 1 & 4 & 9 & -y_2 \\ 1 & 8 & 27 & -y_3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 & y \\ 1 & 2 & 3 & y_1 \\ 1 & 4 & 9 & y_2 \\ 1 & 8 & 27 & y_3 \end{vmatrix} = 0 \text{ (Performing } C_1 \to -C_1\text{)}$$

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 & y \\ 0 & 1 & 2 & y_1 - y \\ 0 & 3 & 8 & y_2 - y \\ 0 & 7 & 26 & y_3 - y \end{vmatrix} = 0 \text{ (Performing } R_i \to R_i - R_1,\ i = 2, 3, 4\text{)}$$

Expansion of determinant along 1st column yields

$$\begin{vmatrix} 1 & 2 & y_1 - y \\ 3 & 8 & y_2 - y \\ 7 & 26 & y_3 - y \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 2 & y_1 - y \\ 0 & 2 & (y_2 - y) - 3(y_1 - y) \\ 0 & 12 & (y_3 - y) - 7(y_1 - y) \end{vmatrix} = 0 \text{ (Performing } R_2 \to R_2 - 3R_1, R_3 \to R_3 - 7R_1)$$

$$\Rightarrow \begin{vmatrix} 1 & 1 & y_1 - y \\ 0 & 1 & y_2 - 3y_1 + 2y \\ 0 & 6 & y_3 - 7y_1 + 6y \end{vmatrix} = 0$$

$(y_3 - 7y_1 + 6y) - 6(y_2 - 3y_1 + 2y) = 0$

$\Rightarrow y_3 - 6y_2 + 11y_1 - 6y = 0$

$$\Rightarrow \frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$$

***Illustration 5*:** Form the differential equation of

(i) Family of circles of fixed radius *r* having centres on the x-axis.

(ii) Family of parabolas with foci at the origin and axis along the x-axis

Solution : (i) The equation of family of circles of radius *r* and centres on the x-axis is

$(x - k)^2 + y^2 = r^2$, where *k* is the parameter .... (i)

Differential w.r.t. *x* we have,

$2(x - k) + 2yy_1 = 0$

$\Rightarrow x - k = -yy_1$ .... (ii)

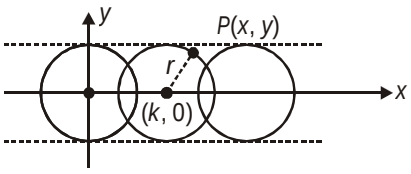
Substituting (ii) in (i), we have

$(yy_1)^2 + y^2 = r^2$

$\Rightarrow y^2(1 + y_1^2) = r^2$

Thus the required differential equatio is

$$y^2\left\{1 + \left(\frac{dy}{dx}\right)^2\right\} = r^2$$



Note that *r* shows up in the differential equation obtained.

(ii) Let us obtain the equation of a parabola with focus at origin and its axis as x-axis.

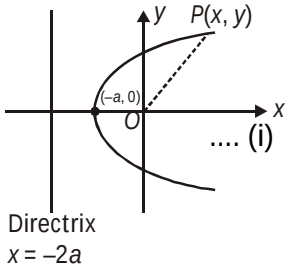
As distance from focus = distance from directrix

$x^2 + y^2 = (2a + x)^2$

$\Rightarrow x^2 + y^2 = 4a^2 + 4ax + x^2$

$\Rightarrow y^2 = 4a(a + x)$ .... (i)

This is the equation of all parabolas with foci at origin and axis along the x-axis, *a* being a parameter.



Differentiating (i) w.r.t. *x* we have,

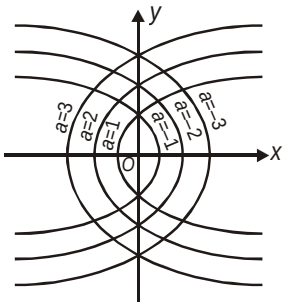
$2yy_1 = 4a$

$$\Rightarrow a = \frac{yy_1}{2} \quad \text{.... (ii)}$$

Eliminating *a* between (i) and (ii)

$$y^2 = 2yy_1\left(\frac{yy_1}{2} + x\right)$$

$= yy_1(yy_1 + 2x)$



$\Rightarrow y = y_1(yy_1 + 2x)$
$\Rightarrow y = y_1^2 y + 2xy_1$
Then the required differential equations is

$$y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$$

Then the required differential equation is

$$y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$$

Student Task

Q.1 Find order and degree of differential equation, $y = \frac{dy}{dx} - \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + \frac{1}{5!}\left(\frac{dy}{dx}\right)^5$.

Q.2 Find degree of differential equation $\left\{\frac{dy}{dx}\right\}^{\frac{2}{3}} = \left\{\frac{d^2y}{dx^2}\right\}^{\frac{3}{2}}$.

Q.3 Find the order of differential equation whose general solution is given by $y = (k_1 + k_2)\tan(x + k_3) - k_4 e^x + k_5$, where k_1, k_2, k_3, k_4, k_5 are arbitrary constants.

Q.4 Find differential equation corresponding to equation $y = a \sin(mx + b)$, where a, m, b all are arbitrary constants.

Q.5 Find the order of differential equation of all conics whose centre lies at origin.

Answer Key:

1. Order = 1, Degree = 1,
2. 9
3. 4
4. $y\left\{\frac{d^3y}{dx^3}\right\} - \left\{\frac{d^2y}{dx^2}\right\}\left\{\frac{dy}{dx}\right\} = 0$
5. 3

3. GENERAL AND PARTICULAR SOLUTION

In the earlier article we obtained the differential equation, when the primitive was given.
We now consider the inverse problem-that is, to recover the primitive when the differential equation is given. Put other way, solving a differential equation of order n amounts to finding a relation between the variables involving n essential arbitrary constants which together with the derivatives obtained from it satisfy the given differential equation. Some of the examples that we have done earlier are listed below :

Differential equation	**Primitive**
$y^2\left\{1 + \left(\frac{dy}{dx}\right)^2\right\} = r^2$	$(x - k)^2 + y^2 = r^2$

$\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$ $\quad y = ae^x + be^{2x} + ce^{3x}$

$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$ $\quad y = e^x(A \cos x + B \sin x)$

The primitive of a differential equation is usually called the **general solution** of the equation. We have already seen that, the general solution of the differential equation of order n must contain n essential arbitrary constants. A **particular solution** of a differential equation is a solution obtained from the gene7ral solution by assigning definite values to the arbitrary constants.

For example, while the curve $(x - k)^2 + y^2 = r^2$ is the general solution of the differential equation $y^2\left\{1 + \left(\frac{dy}{dx}\right)^2\right\} = r^2$, the curve $(x - 1)^2 + y^2 = r^2$, $(x + 2)^2 + y^2 = r^2$ etc are particular solutions obtained by setting $k = 1$, $k = -2$ respectively.

Geometrically, the primitive (general solution) is the equation of a family of curves and a particular solution is the equation of some one of the curves. These curves are called integral curves of the differential equation. A differential equation may have solutions which cannot be obtained form the primitive by any manipulation of the arbitrary constants. Such solutions, called singular solutions, are beyond the scope of our chapter.

Note :-

1. The order of a differential equation of a primitive is equal to the number of essential arbitrary constants in the primitive.
2. The general solution of a differential equation will contain as many arbitrary constants as the order of the equation.

4. METHODS OF SOLUTION OF ORDINARY DIFFERENTIAL EQUATION OF FIRST ORDER AND FIRST DEGREE

A differential equation of first order and first degree may be written in the form

$M(x, y)dx + N(x, y)dy = 0$

For example, $\frac{dy}{dx} + \frac{y + x}{y - x} = 0$ may be written as

$(y + x)dx + (y - x)dy = 0$ in which $M(x, y) + x$ and $N(x, y) = y - x$.

Similarly, $\frac{dy}{dx} = 1 + x^2y$ may be written as

$(1 + x^2y)dx - dy = 0$ in which $M(x, y) = 1 + x^2y$ and $N(x, y) = -1$

It is not always possible to solve a differential equation. We have at our disposal a large class of soluble differential equations that fall under known forms. In the pages that follow, we'll study various techniques of solving a differential equation.

(i) Variable Separable

The variables of the equation $M(x, y)dx + N(x, y)dy = 0$ are separable if the equation can be written in the form

$f_1(x).g_2(y)dx + f_2(x).g_1(y)dy = 0$

which upon division by $f^2(x).g_2(y)$ reduces to

$$\frac{f_1(x)}{f_2(x)}dx + \frac{g_1(y)}{g_2(y)}dy = 0$$

from which the primitive may be obtained by integration.

Illustration 6 : Solve $\sqrt{1+x^2+y^2+x^2y^2} + xy\frac{dy}{dx} = 0$

Solution : Note that $f(x, y) = \sqrt{1+x^2+y^2+x^2y^2} = \sqrt{(1+x^2)(1+y^2)}$

$\sqrt{1+x^2}.\sqrt{1+y^2} = g(x)h(y)$

Thus $f(x, y)$ can be written as a product of two functions, one of x alone and the other of y along.

$$\sqrt{1+x^2+y^2+x^2y^2} + xy\frac{dy}{dx} = 0$$

$$\Rightarrow \quad \sqrt{1+x^2}\sqrt{1+y^2} + xy\frac{dy}{dx} = 0$$

$$\Rightarrow \quad \frac{\sqrt{1+x^2}}{x}dx + \frac{y}{\sqrt{1+y^2}}dy = 0$$

Integration yields

$$\int \frac{1+x^2}{x\sqrt{1+x^2}}dx + \frac{1}{2}\int \frac{d(1+y^2)}{1+y^2} = k$$

$$\Rightarrow \quad \int \frac{1}{x\sqrt{1+x^2}}dx + \int \frac{x}{\sqrt{1+x^2}}dx + \frac{1}{2}\times 2\sqrt{1+y^2} = k$$

$$\Rightarrow \quad \int \frac{-\frac{1}{t^2}dt}{\left(\frac{1}{t}\right)\sqrt{1+\left(\frac{1}{t}\right)^2}} + \sqrt{1+x^2} + \sqrt{1+y^2} = k \qquad \text{[setting } x = 1/t \text{ in the first integral]}$$

$$\Rightarrow \quad -\int \frac{dt}{\sqrt{1+t^2}} + \sqrt{1+x^2} + \sqrt{1+y^2} = k$$

$$\Rightarrow \quad -\ln\left|t+\sqrt{1+t^2}\right| + \sqrt{1+x^2} + \sqrt{1+y^2} = k$$

$$\Rightarrow \quad -\ln\left|\frac{1}{x}+\sqrt{1+\frac{1}{x^2}}\right| + \sqrt{1+x^2} + \sqrt{1+y^2} = k$$

$$\Rightarrow \quad \sqrt{1+x^2} - \ln\left|\frac{x+\sqrt{1+x^2}}{x}\right| + \sqrt{1+y^2} = k$$

Which is the general solution of the differential equation.

Illustration 7 : Solve $(x^2 - yx^2)dy + (y^2 + xy^2)dx = 0$

Solution : $x^2(1 - y)dy + y^2(1 + x)dx = 0$

$$\Rightarrow \quad \frac{1-y}{y^2}dy+\frac{1+x}{x^2}dx=0$$

$$\Rightarrow \quad \frac{dy}{y^2}-\frac{dy}{y}+\frac{dx}{x}+\frac{dx}{x^2}=0$$

Integrating we get,

$$-\frac{1}{y}-\ln y-\frac{1}{x}+\ln x=k$$

$$\Rightarrow \quad \ln\left(\frac{x}{y}\right)-\frac{x+y}{xy}=k$$

where k is the constant of integration.

***Illustration 8*:** Solve $\cos y dx + (1 + 2e^{-x})\sin y dy = 0$

Solution : Separating the variables we get,

$$\frac{dx}{1+2e^{-x}}+\frac{\sin y}{\cos y}dy=0$$

$$\Rightarrow \quad \frac{e^x dx}{e^x+2}+\tan y dy=0$$

Integrating we get,

$\ln(e^x + 2) + \ln \sec y = \ln k$

$\Rightarrow \ln\{(e^x + 2)\sec y\} = \ln k$

$\Rightarrow \sec y(e^x + 2) = k$, k being the constant of integration.

(ii) Reducible to variables separable

Some differential equations can be reduced to variables separable form by a suitable substitution that is suggested by the make-up of the equation.

***Illustration 9*:** Solve $(x^2 + 2xy + y^2 + 1)\ \frac{dy}{dx}=2(x+y)$

Solution : Observe that $x^2 + 2xy + y^2 = (x + y)^2$

The equation becomes,

$$\{(x + y)^2 + 1\}\ \frac{dy}{dx} = 2(x + y)$$

Set $x + y = t$, so that $1+\frac{dy}{dx}=\frac{dt}{dx}$ (on differentiation w.r.t. x)

Now,

$$(t^2+1)\left(\frac{dt}{dx}-1\right)=2t$$

$$\Rightarrow \quad \frac{dt}{dx}=\frac{2t}{t^2+1}+1=\frac{t^2+2t+1}{t^2+1}=\frac{(t+1)^2}{t^2+1}$$

$$\Rightarrow \quad \frac{t^2+1}{(t+1)^2}dt = dx$$

$$\Rightarrow \quad \frac{(t+1)^2-2(t+1)+2}{(t+1)^2}dt = dx$$

$$\Rightarrow \quad dt - \frac{2}{t+1}dt + 2\frac{dt}{(1+t)^2} = dx$$

Integrating, we obtain,

$$t - 2\ln|t+1| - \frac{2}{t+1} = x + k$$

$$\Rightarrow \quad x + y - 2\ln|x+y+1| - \frac{2}{x+y+1} = x + k$$

$$\therefore \quad y - 2\ln|x+y+1| - \frac{2}{x+y+1} = k$$, k being the constant of integration

***Illustration 10*:** Solve $\frac{dy}{dx} = \sec(x+y)$

Solution : Setting $x + y = t$, we have on differentiation w.r.t. x

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

Our equation now reads

$$\frac{dt}{dx} - 1 = \sec t$$

$$\Rightarrow \quad \frac{dt}{dx} - 1 + \sec t$$

$$\Rightarrow \quad \frac{dt}{1+\sec t} = dx$$

$$\Rightarrow \quad \frac{\cos t\, dt}{1+\cos t} = dx$$

$$\Rightarrow \quad \frac{(1+\cos t)-1}{1+\cos t}dt = dx$$

$$\Rightarrow \quad dt - \frac{dt}{1+\cos t} = dx$$

$$\Rightarrow \quad dt - \frac{1}{2}\sec^2\frac{t}{2}dt = dx$$

Integrating, we get,

$$t - \tan\frac{t}{2} = x + k$$

$$\Rightarrow \quad x + y - \tan\frac{x+y}{2} = x + k$$

$\therefore \quad y - \tan\frac{x+y}{2} = k$, k being the constant of integration n.

***Illustration 11*:** Solve $\left(\frac{x+y-1}{x+y-3}\right)\frac{dy}{dx} = \left(\frac{x+y+1}{x+y+3}\right)$

Solution : Set $x + y = v$,on differentiation w.r.t. x we have

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$

Our equation becomes

$$\left(\frac{v-1}{v-3}\right)\left(\frac{dv}{dx} - 1\right) = \left(\frac{v+1}{v+3}\right)$$

$$\Rightarrow \quad \frac{dv}{dx} - 1 = \frac{(v+1)(v-3)}{(v+3)(v-1)} = \frac{v^2 - 2v - 3}{v^2 + 2v - 3}$$

$$\Rightarrow \quad \frac{dv}{dx} = 1 + \frac{v^2 - 2v - 3}{v^2 + 2v - 3} = \frac{2v^2 - 6}{v^2 + 2v - 3}$$

$$\Rightarrow \quad \frac{v^2 + 2v - 3}{v^2 - 3} dv = 2dx$$

$$\Rightarrow \quad \left(1 + \frac{2v}{v^2 - 3}\right) dv = 2dx$$

Integrating

$v + \ln |v^2 - 3| = 2x + k$

$\Rightarrow$ $x + y + \ln |(x + y)^2 - 3| = 2x + k$

$\Rightarrow$ $y + \ln |(x + y)^2 - 3| - x = k$, k being the constant of integration

(iii) Solution by Inspection

At times, by recognizing a certain group of terms as being part of an exact differential, we can solve the differential equation in which they occur by inspection. And whenever this method applies the solution is truly elegant. The following list will be help in finding a perfect differential made up of group terms.

$d(xy) = xdy + ydx$

$$d\left(\frac{y}{x}\right) = \frac{xdy - ydx}{x^2}$$

$$d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{x^2}$$

$$d\left(\ln\frac{y}{x}\right) = \frac{xdy - ydx}{xy}$$

$$d\left(\tan^{-1}\frac{y}{x}\right) = \frac{xdy - ydx}{x^2 + y^2}$$

$$d\left(\tan^{-1}\frac{x}{y}\right) = \frac{ydx - xdy}{x^2 + y^2}$$

$$d\left(\frac{y^2}{x}\right)=\frac{2xydy-y^2dx}{x^2}$$

$$d\left(\frac{x^2}{y}\right)=\frac{2xydx-x^2dy}{y^2}$$

$$d\left(\frac{1}{xy^2}\right)=-\frac{y^2dx+2xydy}{x^2y^4}$$

$$d\left\{\sin^{-1}(xy)\right\}=\frac{xdy+ydx}{\sqrt{1-x^2y^2}}$$

***Illustration 12*:** Solve

(i) $xdx+ydy+\frac{xdy-ydx}{x^2+y^2}=0$

(ii) $y(1+xy)dx-xdy=0$

(iii) $(1+x^2)\frac{dy}{dx}+2xy=4x^2$

Solution : (i) Identifying $\frac{xdy-ydx}{x^2+y^2}$ with $d\left(\tan^{-1}\frac{y}{x}\right)$, we have,

$$xdx+ydy+d\left(\tan^{-1}\frac{y}{x}\right)=0$$

Integrating we obtain,

$\frac{x^2}{2}+\frac{y^2}{2}+\tan^{-1}\frac{y}{x}=k$, k being the constant of integration

$\therefore\ x^2+y^2+2\tan^{-1}\frac{y}{x}=\lambda$, λ being another constant

(ii) $y(1+xy)dx-xdy=0$

$\Rightarrow\ (ydx-xdy)+xy^2dx=0$

$$\Rightarrow\ \frac{ydx-xdy}{y^2}+xdx=0$$

$$\Rightarrow\ d\left(\frac{x}{y}\right)+xdx=0$$

Integrating we get,

$\frac{x}{y}+\frac{x^2}{2}=k$, k being the constant of integration

(iii) **Ist method :**

Rewrite the equation as

$(1+x^2)dy+2xy\,dx=4x^2dx$

$\Rightarrow\ dy+(x^2dy+2xydx)=4x^2dx$

$\Rightarrow\ dy+d(x^2y)=4x^2dx$

Integrating

$y + x^2y = \frac{4x^3}{3} + k$, k being a constant of integration

2nd method :

One can also view this equation as a linear equation in y by writing it in the form

$$\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{4x^2}{1+x^2}$$

I.F. = $e^{\int \frac{2x}{1+x^2}dx} = e^{\int \frac{d(1+x^2)}{1+x^2}} = e^{\ln(1+x^2)} = 1 + x^2$

Multiplying the equation with I.F. and integrating

$y(1+x^2) = \int \frac{4x^2}{1+x^2}.(1+x^2)dx + k$, k being the constant of integration

$y(1+x^2) = \frac{4x^2}{3} + k$, as before

Observe that the solution by inspection is at once and elegant.

***Illustration 13*:** Solve

(i) $(2x - y + 1)dx + (2y - x + 1)dy = 0$

(ii) $x^2\frac{dy}{dx} + xy + \sqrt{1-x^2y^2} = 0$

Solution : (i) We have,

$2xdx - ydx + dx + 2ydy - xdy + dy = 0$

$\Rightarrow 2xdx - (\underbrace{ydx + xdy}) + dx + 2ydy + dy = 0$

$\Rightarrow 2xdx - d(xy) + dx + 2ydy + dy = 0$

Integrating we obtain,

$x^2 - xy + x + y^2 + y = k$, k being the constant of integration.

(ii) $x^2\frac{dy}{dx} + xy + \sqrt{1-x^2y^2} = 0$

$\Rightarrow x^2\frac{dy}{dx} + xy + \sqrt{1-x^2y^2} = 0$

$\Rightarrow x(\underbrace{xdy + ydx}) + \sqrt{(1-x^2y^2)}dx = 0$

Dividing by $x\sqrt{1-x^2y^2} = 0$

$$\frac{xdy + ydx}{\sqrt{1-x^2y^2}} + \frac{dx}{x} = 0$$

$$\Rightarrow \frac{d(xy)}{\sqrt{1-x^2y^2}} + \frac{dx}{x} = 0$$

Integrating we get,

$\sin^{-1}(xy) + \ln x = k$, k being the constant of integration.

(iv) Homogeneous Equations

A function $f(x, y)$ is called homogeneous of degree n if

$f(\lambda x, \lambda y) = \lambda^n f(x, y)$

For example, $f(x, y) = x^4 + x^3y$ is homogeneous of degree 4 since $f(\lambda x, \lambda y) = (\lambda x)^4 + (\lambda x)^3(\lambda y) = \lambda^4(x^4 + x^3y) = \lambda^4 f(x, y)$.

The differential equation $M(x, y)dx + N(x, y)dy = 0$ is called homogeneous if $M(x, y)$ and $N(x, y)$ are homogeneous and of this same degree.

The transformation $y = vx$, so that $dy = vdx + xdv$ will reduce any homogeneous equation to the form

$P(x, v)dx + Q(x, v)dy = 0$

in which the variables becomes separable.

***Illustration 14*:** Solve $(x^2 + y^2)\frac{dy}{dx} = 2xy$

Solution : $\frac{dy}{dx} = \frac{2xy}{x^2 + y^2}$

Set $y = vx$, so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$

Now, $v + x\frac{dv}{dx} = \frac{2x.vx}{x^2 + v^2x^2} = \frac{2v}{1 + v^2}$

$\Rightarrow \quad x\frac{dv}{dx} = \frac{2v}{1 + v^2} - v = \frac{2v - v - v^3}{1 + v^2} = \frac{v - v^3}{1 + v^2}$

$\Rightarrow \quad \frac{dx}{x} = \frac{1 + v^2}{v(1 - v^2)}dv$

$\Rightarrow \quad \frac{dx}{x} = \left(\frac{1}{v} + \frac{1}{1 - v} - \frac{1}{1 + v}\right)dv$ (Using partial fractions)

Integrating

$\ln x = \ln v - \ln|1 - v| - \ln|1 + v| + \ln k$

$\Rightarrow \quad x = \frac{v}{(1 - v)(1 + v)}k$

$\Rightarrow \quad x = \frac{\frac{y}{x}}{1 - \frac{y^2}{x^2}}k$

$\Rightarrow \quad x = \frac{xy}{x^2 - y^2}k$

$\Rightarrow \quad ky = (x^2 - y^2)$ where k is the constant of integration.

***Illustration 15*:** Integrate $x\cos\left(\frac{y}{x}\right)(ydx + xdy) = y\sin\left(\frac{y}{x}\right)(xdy - ydx)$

Solution : $x\cos\frac{y}{x}\left(y + x\frac{dy}{dx}\right) = y\sin\frac{y}{x}\left(x\frac{dy}{dx} - y\right)$

$$\Rightarrow \quad x\frac{dy}{dx}\left(x\cos\frac{y}{x}-y\sin\frac{y}{x}\right)=-y\left(y\sin\frac{y}{x}+x\cos\frac{y}{x}\right)$$

$$\Rightarrow \quad \frac{dy}{dx}=-\frac{y}{x}.\frac{y\sin\frac{y}{x}+x\cos\frac{y}{x}}{x\cos\frac{y}{x}-y\sin\frac{y}{x}}$$

Set $y = vx$, so that

$$\frac{dy}{dx}=v+x\frac{dv}{dx}$$

$$\Rightarrow \quad v+x\frac{dv}{dx}=v.\frac{vx\sin v+x\cos v}{vx\sin v-x\cos v}$$

$$\Rightarrow \quad x\frac{dv}{dx}=\frac{v(v\sin v+\cos v)}{v\sin v-\cos v}-v=\frac{2v\cos v}{v\sin v-\cos v}$$

$$\Rightarrow \quad 2\frac{dx}{x}=\frac{v\sin v-\cos v}{v\cos v}dv=\frac{\sin v}{\cos v}dv-\frac{dv}{v}$$

Integrating we get,

$2\ln x + \ln k = \ln \sec v - \ln v$

$$\Rightarrow \quad \ln kx^2=\ln\frac{\sec v}{v}$$

$$\Rightarrow \quad kx^2=\frac{\sec v}{v}=\frac{\sec\frac{y}{x}}{\frac{y}{x}}$$

$$\Rightarrow \quad kxy=\sec\frac{y}{x}$$

$$\Rightarrow \quad xy\cos\frac{y}{x}=\lambda \text{ , } \lambda \text{ being a constant of integration}$$

***Illustration 16*:** Solve $\frac{dy}{dx}=\frac{y^3+y^2\sqrt{y^2-x^2}}{x^3}$

Solution : **I^st^ method**

Set $y = vx$, so that $\frac{dy}{dx}=v+x\frac{dv}{dx}$

Equation becomes

$$v+x\frac{dv}{dx}=\frac{v^3x^3+v^2x^2\sqrt{v^2x^2-x^2}}{x^3}$$

$$\Rightarrow \quad x\frac{dv}{dx}=v^3+v^2\sqrt{v^2-1}-v=v(v^2-1)+v^2\sqrt{v^2-1}$$

$$\Rightarrow \quad \frac{dv}{(v^2-1)v+v^2\sqrt{v^2-1}}=\frac{dx}{x}$$

$$\Rightarrow \quad \frac{dv}{v\sqrt{v^2-1}\left[v+\sqrt{v^2-1}\right]}=\frac{dx}{x}$$

$$\Rightarrow \quad \frac{v-\sqrt{v^2-1}}{v\sqrt{v^2-1}}dv=\frac{dx}{x}$$ (Rationalizing the denominator on the LHS)

$$\Rightarrow \quad \frac{dv}{\sqrt{v^2-1}}-\frac{dv}{v}=\frac{dx}{x}$$

Integrating

$$\ln|v+\sqrt{v^2-1}|-\ln v=\ln x+\ln k$$

$$\Rightarrow \quad \frac{v+\sqrt{v^2-1}}{v}=kx$$

$$\Rightarrow \quad \frac{\frac{y}{x}+\sqrt{\frac{y^2}{x^2}-1}}{\frac{y}{x}}=kx$$

$$\Rightarrow \quad \frac{y+\sqrt{y^2-x^2}}{y}=kx$$

$\therefore \quad kxy=y+\sqrt{y^2-x^2}$, k being a constant integration.

2nd method

Let $y=x\sec\theta \Rightarrow \frac{dy}{dx}=x\sec\theta\tan\theta\frac{d\theta}{dx}+\sec\theta$

Our equation becomes,

$$x\sec\theta\tan\theta\frac{d\theta}{dx}+\sec\theta=\frac{x^3\sec^3\theta+x^3\sec^2\theta\tan\theta}{x^3}$$

$$\Rightarrow \quad x\sec\theta\tan\theta\frac{d\theta}{dx}+\sec\theta=\sec^3\theta+\sec^2\theta\tan\theta$$

$$\Rightarrow \quad x\tan\theta\frac{d\theta}{dx}+1=\sec^2\theta+\sec\theta\tan\theta$$

$$\Rightarrow \quad x\tan\theta\frac{d\theta}{dx}=\tan^2\theta+\sec\theta\tan\theta$$

$$\Rightarrow \quad x\frac{d\theta}{dx}=\tan\theta+\sec\theta$$

$$\Rightarrow \quad \frac{d\theta}{\sec\theta+\tan\theta}=\frac{dx}{x}$$

$$\Rightarrow \quad (\sec\theta-\tan\theta)d\theta=\frac{dx}{x}$$

$$\Rightarrow \quad \sec\theta d\theta-\tan\theta d\theta=\frac{dx}{x}$$

Integrating we get,

$\ln|\sec\theta + \tan\theta| - \ln|\sec\theta| = \ln x + \ln k$

$$\Rightarrow \quad \frac{\sec\theta + \tan\theta}{\sec\theta} = kx$$

$$\Rightarrow \quad 1 + \frac{\tan\theta}{\sec\theta} = kx$$

$$\Rightarrow \quad 1 + \frac{\sqrt{y^2 - x^2}}{y} = kx$$

$$\Rightarrow \quad y + \sqrt{y^2 - x^2} = kxy \text{ , as obtained before.}$$

(v) Non-homogeneous equation of first degree in *x* and *y*

The equation $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$, can be reduced to the homogeneous form by suitable substitution.

We have three possibilites

Case : I

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

Then, set $x = u + h$, $y = v + k$

These give, $\frac{dy}{dx} = \frac{dv}{du}$

Our equation becomes,

$$\frac{dv}{du} = \frac{a_1u + b_1v + (a_1h + b_1k + c_1)}{a_2u + b_2v + (a_2h + b_2k + c_2)}$$

Suppose h, k are chosen such that

$$\left.\begin{aligned} a_1h + b_1k + c_1 = 0 \\ \text{and } a_2h + b_2k + c_2 = 0 \end{aligned}\right\} \ldots\ldots\ldots (* *)$$

then $\frac{dv}{du} = \frac{a_1u + b_1v}{a_2u + b_2v}$

which is homogeneous and can be solved by $v = tu$. Note that the simultaneous equation (* *) possess a solution because $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ i.e., $a_1b_2 - a_2b_1 \neq 0$

Case : II

$\frac{a_1}{a_2} = \frac{b_1}{b_2}$ then we can't have a unique solution to (* *)

Now, let $\frac{a_1}{a_2} = \frac{b_1}{b_2} = m$

Then our equation becomes,

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{\frac{1}{m}(a_1x + b_1y) + c_2}$$

Set $a_1x + b_1y = v$ so that $\frac{dy}{dx} = \frac{1}{b_1}\left(\frac{dy}{dx} - a_1\right)$

And our equation becomes

$\frac{1}{b_1}\left(\frac{dv}{dx} - a_1\right) = \frac{v + c_1}{\frac{v}{m} + c_2}$, which can be solved by the method of separation of variables

Case : III

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = I \text{ (say)}$$

This is a trivial case and our equation reads

$$\frac{dy}{dx} = \frac{I(a_2x + b_2y + c_2)}{a_2x + b_2y + c_2}$$

$\Rightarrow$ $dy = Idx$

$\therefore$ $y = Ix + k$, which is the required solution.

***Illustration 17*:** Solve $\frac{dy}{dx} = \frac{x + 2y - 3}{2x + y - 3}$

Solution : Here $a_1 = 1, b_1 = 2$ $\therefore$ $\frac{a_1}{b_1} = \frac{1}{2}$

$a_2 = 2, b_2 = 1$ $\therefore$ $\frac{a_2}{b_2} = 2$

Now, $\frac{a_1}{b_1} \neq \frac{a_2}{b_2} = 2$

Thus case-I applies, we have by setting,

$$\left.\begin{aligned} x &= u + h \\ y &= v + k \end{aligned}\right\}$$

$$\frac{dv}{du} = \frac{(u + 2v) + (h + 2k - 3)}{(2u + v) + (2h + k - 3)}$$

Choose h, k such that

$$\left.\begin{aligned} h + 2k - 3 &= 0 \\ 2h + k - 3 &= 0 \end{aligned}\right\}$$

The solution is $h = 1$, $k = 1$

Set $v = tu$, $\Rightarrow$ $\frac{dv}{du} = t + \frac{dt}{du}xu$

Our equation reduces to

$$t+u\frac{dt}{du}=\frac{u+2tu}{2u+tu}=\frac{1+2t}{2+t}$$

$$\Rightarrow \quad u\frac{dt}{du}=\frac{1+2t}{2+t}-t=\frac{1+2t-2t-t^2}{2+t}=\frac{1-t^2}{2+t}$$

$$\Rightarrow \quad \frac{2+t}{1-t^2}dt=\frac{du}{u}$$

$$\Rightarrow \quad \frac{2}{1-t^1}dt+\frac{1}{2}.2\frac{t}{1-t^2}dt=\frac{du}{u}$$

On integrating, we get

$$\ln\frac{1+t}{1-t}-\frac{1}{2}\ln|1-t^2|=\ln u+\ln k$$, k being the constant of integration.

$$\Rightarrow \quad \ln\left(\frac{1+\frac{v}{u}}{1-\frac{v}{u}}\right)-\frac{1}{2}\ln\left(1-\frac{v^2}{u^2}\right)=\ln(uk)$$

$$\Rightarrow \quad \ln\left(\frac{u+v}{u-v}\right)-\ln\frac{\sqrt{u^2-v^2}}{u}=\ln(uk)$$

$$\Rightarrow \quad \ln\left(\frac{u+v}{u-v}.\frac{u}{\sqrt{u^2-v^2}}\right)=\ln(uk)$$

$$\Rightarrow \quad \frac{u(\sqrt{u+v}}{(u-v)^{\frac{3}{2}}}=uk$$

$$\Rightarrow \quad k(u-v)^{\frac{3}{2}}=\sqrt{u+v}$$

$$\Rightarrow \quad k(x-1-y+1)^{\frac{3}{2}}=\sqrt{x-1+y-1}$$

$$\Rightarrow \quad k(x-y)^{\frac{3}{2}}=\sqrt{x+y-2}$$

$\therefore \quad k^2(x-y)^3 = (x+y-2)$

$\Rightarrow \quad (x-y)^3 = \lambda(x+y-2)$, λ being another constant.

***Illustration 18*:** Solve $\frac{dy}{dx}=\frac{x+2y-3}{2x+4y-3}$

Solution : Here, $\frac{a_1}{b_1}=\frac{1}{2}$ and $\frac{a_2}{b_2}=\frac{2}{4}=\frac{1}{2}$

Then, $\frac{a_1}{b_1}=\frac{a_2}{b_2}$. Hence case-II applies.

Let $x + 2y = v$

On differentiation, $1+2\frac{dy}{dx}=\frac{dv}{dx}$

$$\therefore \quad \frac{dy}{dx}=\frac{1}{2}\left(\frac{dv}{dx}-1\right)$$

Our equation becomes

$$\frac{1}{2}\left(\frac{dv}{dx}-1\right)=\frac{v-3}{2v-3}$$

$$\Rightarrow \quad \frac{dv}{dx}-1=\frac{2v-6}{2v-3}$$

$$\Rightarrow \quad \frac{dv}{dx}=1+\frac{2v-6}{2v-3}=\frac{4v-9}{2v-3}$$

$$\Rightarrow \quad \frac{2v-3}{4v-9}dv=dx$$

$$\Rightarrow \quad \frac{1}{2}.\frac{4v-6}{4v-9}dv=dx$$

$$\Rightarrow \quad \frac{1}{2}\left\{\frac{(4v-9)+3}{4v-9}\right\}dv=dx$$

$$\Rightarrow \quad \frac{1}{2}\left[dv+\frac{3}{4v-9}dv\right]=dx$$

Integrating we get,

$\frac{1}{2}v+\frac{3}{2}.\frac{1}{4}\ln(4v-9)=x+k$, k being a constant of integration.

$$\Rightarrow \quad \frac{1}{2}(x+2y)+\frac{3}{8}\ln[4(x+2y)-9]=x+k$$

$$\Rightarrow \quad \frac{1}{2}(x+2y)+\frac{3}{8}\ln(4x+8y-9)=x+k$$

$\Rightarrow$ $4(x + 2y) + 3\ln(4x + 8y – 9) = (x + k)8$

$\Rightarrow$ $4(2y – x) + 3\ln(4x + 8y – 9) = 8k$

$\therefore$ $4(2y – x) + 3\ln(4x + 8y – 9) = \lambda$, λ being another constant

Student Task

Q.1 Solve $y-x\frac{dy}{dx}=a\left\{y^2+\frac{dy}{dx}\right\}$.

Q.2 Solve the differential equation, $\frac{xdy+ydy}{xdy-ydx}=\sqrt{\frac{a^2-x^2-y^2}{x^2+y^2}}$.

Q.3 Solve the differential equation $2\frac{dy}{dx}=\frac{y}{x}+\frac{y^2}{x^2}$.

Q.4 Show that differential equation of the form $yf(xy)\,dx + x.g(xy)\,dy = 0$ can be converted into variable separable form by substituting $xy = v$

Answers:

1. $y = c(a + x)(1 - ay)$ 2. $\sqrt{x^2+y^2} = a\sin\left\{c + \tan^{-1}\left(\frac{y}{x}\right)\right\}$ 3. $y - x = cy\,x$,

4. $\frac{dx}{x} + \frac{g(v)dv}{v\{f(v) - g(v)\}} = 0$ variable sepaarable form

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(vi) Linear Differential Equation (Lagrange's Linear Differential Equation)

An equation of the form $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x alone or constants, is called a linear differential equation of first order in y.
Multiplying both sides by $e^{\int Pdx}$ we get,

$$\frac{dy}{dx}e^{\int Pdx} + Pe^{\int Pdx}y = Qe^{\int Pdx}$$

$$\Rightarrow \quad \frac{d}{dx}(ye^{\int Pdx}) = Qe^{\int Pdx}$$

$$\Rightarrow \quad d(ye^{\int Pdx}) = Qe^{\int Pdx}dx$$

Integrating both sides

$$ye^{\int Pdx} = \int Qe^{\int Pdx} + k$$, k being the constant of integration .... (i)

The factor $e^{\int Pdx}$ which makes it possible for the left hand side to be integrable is called an 'Integrating Factor' (I.F.)
The solution (i) can be written as
$y \,.\, IF = \int Q.\, IF\, dx + k$

Note :- If the differential equation of first order in x is $\frac{dx}{dy} + Rx = S$, where R and S are functions of y alone or constant, then $IF = e^{\int Rdy}$
Here x becomes the dependent variable and y the independent variable.

***Illustration 19*:** Solve $(1 + y^2)dx = (\tan^{-1}y - x)dy$
Solution : Rewrite the equation as,

$$\frac{dx}{dy} = \frac{\tan^{-1}y}{1+y^2} - \frac{x}{1+y^2}$$

$$\Rightarrow \quad \frac{dx}{dy} + \frac{1}{1+y^2}x = \frac{\tan^{-1}y}{1+y^2}$$

This is linear in x, with form being $\frac{dx}{dy} + Rx = S$

$$IF = e^{\int Rdy} = e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1}y}$$

The solution is

x . $IF = \int S$. $IF\, dy + k$, (k being the constant of integration)

$$\Rightarrow \quad x.e^{\tan^{-1}y} = \int e^{\tan^{-1}y}.\frac{\tan^{-1}y}{1+y^2}dy + k \qquad \text{.... (i)}$$

Put $\tan^{-1}y = t \;\Rightarrow\; \dfrac{dy}{1+y^2} = dt$

$$\therefore \quad \int e^{\tan^{-1}y}\frac{\tan^{-1}y}{1+y^2}dy = \int te^t dt = t\int e^t dt - \int e^t 1dt$$

$= te^t - e^t = (t-1)e^t$.

(Using Integration by parts)

From (i)

$$x.e^{\tan^{-1}y} = (\tan^{-1}y - 1)e^{\tan^{-1}y} + k$$

$$\therefore \quad x = ke^{-\tan^{-1}y} + \tan^{-1}y - 1$$

***Illustration 20*:** Solve $\dfrac{dy}{dx} + \dfrac{1-2x}{x^2}y = 1$

Solution : The given equation is of the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$I.F. = e^{\int Pdx} = e^{\int \frac{1-2x}{x^2}} = e^{\int \frac{dx}{x^2} - 2\frac{dx}{x}} = e^{-\frac{1}{x} - 2\ln x}$$

$$= e^{-\frac{1}{x}}.e^{-2\ln x}$$

$$= \frac{1}{x^2}e^{-\frac{1}{x}}$$

Multiplying the given equation by IF and integrating we get,

y . $IF = \int Q$. $IF\, dx + k$, k being the constant of integration

$$\Rightarrow \quad ye^{-\frac{1}{x}}.\frac{1}{x^2} = \int e^{-\frac{1}{x}}.\frac{1}{x^2}dx + k$$

$$\Rightarrow \quad ye^{-\frac{1}{x}}\frac{1}{x^2} = \int e^{\frac{-1}{x}}d\left(\frac{1}{x}\right) + k$$

$$\Rightarrow \quad y^{-\frac{1}{x}}.\frac{1}{x^2} = -e^{\frac{-1}{x}} + k$$

$$\Rightarrow \quad y = x^2\left(1 + ke^{\frac{1}{x}}\right)$$

***Illustration 21*:** Solve $x\cos x\dfrac{dy}{dx} + y(x\sin x + \cos x) = 1$

Solution : We have,

$$\frac{dy}{dx} + \frac{x\sin x + \cos x}{x\cos x}y = \frac{1}{x\cos x}$$

Which is of the form $\frac{dy}{dx} + P(x)y = Q(x)$

$IF = e^{\int P dx} = e^{\int \frac{x\sin x + \cos x}{x\cos x} dx} = e^{\int \tan x dx + \int \frac{dx}{x}}$

$= e^{\ln \sec x + \ln x} = e^{\ln(x \sec x)} = x \sec x$

Multiplying the given equation by *IF* and integrating

$y \times IF = \int Q.IF\, dx + k$, k being a constant of integration

$$\Rightarrow \quad y.x\sec x = \int x\sec x.\frac{1}{x\cos x}dx + k$$

$= \int \sec^2 x dx + k$

$= \tan x + k$

$\therefore \quad xy \sec x = \tan x + k$

(vii) Bernoulli's Equation (Reducible to Linear Equations)

The differential equation

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, n \neq 1,\ n \notin R \qquad \text{.... (i)}$$

where P and Q are function of x alone or constant is called Bernoulli's equation. This can be reduced to the linear form by the substitution $y^{-n+1} = v$,

So that, $(-n + 1)y^{-n}\frac{dy}{dx} = \frac{dv}{dx}$

Equation (i) then reduces to

$$\frac{1}{-n+1}y^n\frac{dv}{dx} + P(x)y = Q(x)y^n$$

$$\Rightarrow \quad \frac{1}{-n+1}\frac{dv}{dx} + P(x)y^{1-n} = Q(x)$$

$$\Rightarrow \quad \frac{dv}{dx} + (1-n)Py^{1-n} = (1-n)Q$$

$$\Rightarrow \quad \frac{dv}{dx} + (1-n)P.V. = (1-n)Q$$

Which is linear in v and hence solved by multiplying by its *IF* and integrating.

***Illustration 22*:** Solve : $\frac{dy}{dx} - \frac{2y}{x} = y^4$

Solution : On dividing the equation by y^4, we have

$$\frac{1}{y^4}\frac{dy}{dx} - \frac{2}{y^3}\frac{1}{x} = 1$$

Set $\frac{1}{y^3} = v$, so that, $-\frac{3}{y^4}\frac{dy}{dx} = \frac{dv}{dx}$

Our equation reads,

$$-\frac{1}{3}\frac{dv}{dx} - \frac{2v}{x} = 1$$

$\Rightarrow \quad \frac{dv}{dx}+\frac{6}{x}v=-3$

Which is the linear differential equation in v.

$IF = e^{6\int\frac{dx}{x}} = e^{6\ln x} = x^6$

Multiplying the equation by IF and integrating we get,

$vx^6 = -3\int x^6 dx + k$, k being a constant of integration

$$= -3\frac{x^7}{7}+k$$

$$\Rightarrow \quad \frac{1}{y^3}x^6 = -\frac{3}{7}x^7 + k$$

$$\Rightarrow \quad \frac{3}{7}x^7 + \frac{x^6}{y^3} = k$$

$$\therefore \quad x^6\left(3x+\frac{7}{y^3}\right) = \lambda \text{ , } \lambda \text{ being another constant,}$$

***Illustration 23*:** Solve $\frac{dy}{dx}+\left(\frac{2\sin y\cos y}{\cos^2 y}\right)x = x^3$

Solution : $\Rightarrow \quad \sec^2 y\frac{dy}{dx}+(2\tan y)x = x^3$

Set $\tan y = v$, So that $\sec^2 y\frac{dy}{dx} = \frac{dv}{dx}$

Our equation becomes

$$\frac{dv}{dx}+2vx = x^3$$

which is linear in v

$IF = e^{\int 2xdx} = e^{x^2}$

Multiplying the equation by IF and integrating

$ve^{x^2} = \int e^{x^2}.x^2 dx + k$, k being the constant of integration

$$= \int e^{x^2}.x^2.xdx + k \quad (\text{set } t = x^2)$$

$$= \frac{1}{2}\int e^t.tdt + k$$

$$= \frac{1}{2}e^t.(t-1)+k$$

$$\Rightarrow \quad \tan y e^{x^2} = \frac{1}{2}e^{x^2}(x^2-1)+k$$

$\Rightarrow \quad 2\tan y e^{x^2} = e^{x^2}(x^2-1)+\lambda$, λ being another constant.

$\therefore \quad 2\tan y = (x^2-1)+\lambda e^{-x^2}$

***Illustration 24*:** Solve : $x\frac{dy}{dx}+y=y^2\ln x$

Solution : Rewrite the equation as

$$\frac{1}{y^2}\frac{dy}{dx}+\frac{1}{xy}=\frac{1}{x}\ln x$$

Set $\frac{1}{y}=v$ so that, $-\frac{1}{y^2}\frac{dy}{dx}=\frac{dv}{dx}$

Our equation becomes

$$-\frac{dv}{dx}+\frac{v}{x}=\frac{1}{x}\ln x$$

$$\Rightarrow \quad \frac{dv}{dx}-\frac{v}{x}=-\frac{1}{x}\ln x$$

Which is linear in *v*.

$$IF=e^{-\int\frac{dx}{x}}=e^{-\ln x}=e^{\ln x^{-1}}=\frac{1}{x}$$

Multiplying the equation by *IF* and integrating, we get

$\frac{1}{x}v=-\int\frac{1}{x^2}\ln x dx+k$, *k* being a constant of integration.

$$=-\int\frac{1}{x}(\ln x)\frac{dx}{x}+k$$

$$=-\int te^{-t}dt+k \qquad \text{(setting } \ln x = t\text{)}$$

$$= (t + 1)e^{-t} + k \qquad \text{(using integration by parts)}$$

$$=(1+\ln x)\frac{1}{x}+k$$

$$\therefore \quad \frac{1}{xy}=\frac{1}{x}(1+\ln x)+k$$

Student Task

Q.1 Solve $\frac{dy}{dx}+\frac{y}{x}=\log x$

Q.2 Solve $\cos^2 x\frac{dy}{dx}-y\tan 2x=\cos^4 x$, where $|x|<\frac{\pi}{4}$ and $y\left(\frac{\pi}{6}\right)=\frac{3\sqrt{3}}{8}$.

Q.3 Solve $\frac{dy}{dx}+y\phi'(x)=\phi(x).\phi'(x)$, where $\phi(x)$ is a given function.

Answers :

1. $yx=\frac{x^2}{2}\log x-\frac{x^2}{4}+c$ 2. $y=\frac{\sin 2x}{2(1-\tan^2 x)}$ 3. $y = ce^{-\phi(x)} + \phi(x) - 1$

5. GEOMETRICAL AND PHYSICAL APPLICATIONS

The following results will often be used to solve problems having geometrical flavour.
For the curve $y = f(x)$ or $f(x, y) = 0$

(i) Slope of tangent to the curve at (x, y) is $\frac{dy}{dx}$

(ii) Slope of normal to the curve at (x, y) is $-\frac{dx}{dy} = -\frac{1}{\left(\frac{dy}{dx}\right)}$

(iii) The equation of tangent to the curve at (x, y) is

$$Y - y = \frac{dy}{dx}(X - x)$$

X, Y being the current co-ordinates

(iv) The equation of the normal to the curve at (x, y) is

$$(X - x) + \frac{dy}{dx}(Y - y) = 0$$

(v) The intercepts of the tangents on the x-axis and y-axis are $x - y\frac{dx}{dy}$ and $y - x\frac{dy}{dx}$ respectively.

(vi) The lengths of tangent and normal are respectively $y\sqrt{1+\left(\frac{dx}{dy}\right)^2}$ and $y\sqrt{1+\left(\frac{dy}{dx}\right)^2}$

(vii) The lengths of sub-tangent and sub-normal are respectively $y\frac{dx}{dy}$ and $y\frac{dy}{dx}$

***Illustration 25*:** Find the curves for which the length of normal is equal to the radius vector.

Solution : The length of normal $= y\sqrt{1+\left(\frac{dy}{dx}\right)^2}$

From the hypothesis of the problem

$y\sqrt{1+\left(\frac{dy}{dx}\right)^2}$ = radius vector = r

$$\Rightarrow \quad y^2\left\{1+\left(\frac{dy}{dx}\right)^2\right\} = r^2 = x^2 + y^2$$

$$\Rightarrow \quad y^2 + y^2\left(\frac{dy}{dx}\right)^2 = x^2 + y^2$$

$$\Rightarrow \quad y^2\left(\frac{dy}{dx}\right)^2 = x^2$$

$$\Rightarrow \quad \pm y\frac{dy}{dx} = x$$

$\Rightarrow \pm ydy = xdx$

$\Rightarrow \pm y^2 = x^2 - k^2 \quad \therefore \quad x^2 \pm y^2 = k^2$

This represents a circle or equilateral hyperbola according as we take + or – sign.

***Illustration 26*:** Show that the equation of the curve whose slope at any point is equal to $y + 2x$ and which passes through the origin is $y = 2(e^x - x - 1)$.

Solution : We have, $\frac{dy}{dx} = y + 2x$

$\Rightarrow \quad \frac{dy}{dx} - y = 2x$

$IF = e^{-\int dx} = e^{-x}$

Multiplying the equation by *IF* and integrating

$y \times e^{-x} = \int 2xe^{-x}dx + k$, k being the constant of integration.

$= 2[x \int e^{-x}\,dx - \int 1.\,(-e^{-x})dx] + k$, using integration by parts

$= -2xe^{-x} - 2e^{-x} + k \quad$ .... (i)

As curve (i) passes through (0, 0)

$0 = 0 - 2 + k \quad \therefore \quad k = 2$

Thus the curve is

$ye^{-x} = -2xe^{-x} - 2e^{-x} + 2$

$\therefore \quad y = -2x - 2 + 2e^x$

$\therefore \quad y = 2(e^x - x - 1)$

***Illustration 27*:** According to Newton's law of cooling, the rate at which a substance cools in moving are is proportional to the difference between the temperature of the substance and that of the air. If the temperature of the air is 300 K and the substance cools from 370 K to 340 K in15 minutes, find when the temperature will be 310 K.

Solution : Let T be the temperature of the substance at t minutes.

Then $\frac{dT}{dt} = -k(T - 300)$

or $\frac{dT}{T - 300} = -k\,dt$

Integrating between the limits $t = 0$, $T = 370$ and $t = 15$, $T = 340$

$$\int_{370}^{340} \frac{dT}{T - 300} = -k \int_{0}^{15} dt \quad \Rightarrow \quad \ln(T - 300)\Big|_{370}^{340} = -15k$$

$\Rightarrow \quad \ln 40 - \ln 70 = -15\,k \quad \Rightarrow \quad \ln\frac{4}{7} = -15k$

$\Rightarrow \quad \ln\frac{7}{4} = 15k$

$\Rightarrow \quad 15k = \ln\frac{7}{4} \therefore \; k = \frac{\ln\frac{7}{4}}{15}$

Integrating between the limits $t = 0$, $T = 370$ and $t = t$, $T = 310$,

$$\int_{370}^{310} \frac{dT}{T-300} = -k\int_{0}^{t} dt$$

$$\Rightarrow \quad \ln(T-300)\Big|_{370}^{310} = -kt$$

$$\Rightarrow \quad \ln 10 - \ln 70 = -kt$$

$$\Rightarrow \quad \ln\frac{1}{7} = -kt$$

$$\Rightarrow \quad \ln 7 = kt$$

$$\Rightarrow \quad \ln 7 = \frac{\ln\frac{7}{4}}{15}t$$

$$\therefore \quad t = \frac{15\ln 7}{\ln\left(\frac{7}{4}\right)} = \frac{15\ln 7}{\ln 7 - \ln 4}$$

(If value of logarithms involved be known, it can be shown that the time taken t = 52 minutes)

6. ORTHOGONAL TRAJECTORY

Let a family of curves be represented by the equation

$f(x, y, c) = 0$ ..... (i)

And we want to determine the family of curves each member of which cuts members of the family (i) at right angles. Then the faimily of curves we are seeking constitute, the **orthogonal trajectory** of family (i).

The procedure can be systematically described as thus :

(i) We first form the differential equation of the family by differentiating *i* and eliminating the arbitrary constants.

(ii) As products of slopes of tangent at a point of intersection of (i) and its orthogonal trajectory is –1, we replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$ in the differential equation of family (i).

(iii) Now we solve the resulting differential equation to obtain the family of orthogonal trajectory.

***Illustration 28*:** Find the orthogonal trajectory of the family of circles $x^2 + y^2 = 2ax$ each of which touches the y-axis at origin.

Solution : $x^2 + y^2 = 2ax$ ..... (i)

Differentiating w.r.t. x, we have

$$2x + 2y\frac{dy}{dx} = 2a$$

$$\Rightarrow \quad x + y\frac{dy}{dx} = a \qquad \text{.... (ii)}$$

Eliminating a between (i) and (ii), we get

$$x^2 + y^2 = 2\left(x + y\frac{dy}{dx}\right)x$$

$$2x^2 + 2xy\frac{dy}{dx} - x^2 = y^2$$

$$\Rightarrow \quad y^2 - 2xy\frac{dy}{dx} - x^2 = 0 \qquad \text{.... (iii)}$$

Replacing $\frac{dy}{dx}$ *by* $-\frac{dx}{dy}$ to get the differential equation corresponding to the trajectory orthogonal to (i), we have,

$$y^2 + 2xy\frac{dx}{dy} - x^2 = 0$$

$$\Rightarrow \quad x^2 - 2xy\frac{dx}{dy} - y^2 = 0 \qquad \text{.... (iv)}$$

Although (iv) can be solved by the substitution $y = vx$, the differential equation being homogeneous, $\left(viz, \frac{dx}{dy} = \frac{x^2 - y^2}{2xy} \right)$, we observe that (iv) is same as (iii) with x and y interchanged.

Hence the solution of (iv) is obtained by changing x to y in (i) i.e.,

$y^2 + x^2 = 2by$

Which is another set of circles, each of which touches the x-axis at the origin.

Student Task

Q.1 Find the curve for which area of triangle formed by x-axis, tangent drawn at any point on the curve and radius vector of point of tangency is constant, equal to a^2.

Q.2 Find orthogonal trajectory of $x^2 + y^2 = a^2$, where a is arbitrary constant.

Q.3 Find solution of differential equation $x \cos y\, dy + y \cos x\, dx + \sin y\, dx + \sin x\, dy = 0$.

Q.4 Solve following differential equation $dy = 3y\, dx + \sin 2x\, dx$.

Q.5 Solve the differential equation, $x \cos y\, dy + y \cos x\, dx + \sin y\, dx + \sin x\, dy = 0$.

Answers :

1. $x = cy + \frac{a^2}{y}$ 2. $y = cx$, 3. $x \sin y + y \sin x = \text{constant}$

4. $y.e. = -\frac{1}{13}e\{2\cos 2x + 3\sin 2x\} + c.e.^{3x}$ 5. $x \sin y + y \sin x = c$

MISCELLANEOUS EXAMPLES

Example 1 : Determine the order and degree (if it exists) for the following differential equations.

(i) $\frac{d^2y}{dx^2} = x\ln\left(\frac{dy}{dx}\right)$

(ii) $\frac{d^2y}{dx^2} = \cos\left(\frac{dy}{dx}\right)$

(iii) $y = 1 + \frac{dy}{dx} + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + ...$

Solution : (i) $\frac{d^2y}{dx^2} = x\ln\left(\frac{dy}{dx}\right)$ order is 2, $\left(\text{viz. } \frac{d^2y}{dx^2}\right)$

Since the differential equation can't written as a polynomial in the derivatives due to presence of $\ln\left(\frac{dy}{dx}\right)$, the degree is not defined.

(ii) $\frac{d^2y}{dx^2} = \cos\left(\frac{dy}{dx}\right)$, order is 2

Since the differential equation can't be written as a polynomial in derivatives due to presence of $\cos\left(\frac{dy}{dx}\right)$, the degree is not defined.

(iii) $y = 1 + \frac{dy}{dx} + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + ...$

Identifying the series on the RHS as $e^{\frac{dy}{dx}}$, we have

$y = e^{\frac{dy}{dx}}$

Taking logarithm on both sides

$\ln y = \frac{dy}{dx}$

$\Rightarrow \frac{dy}{dx} - \ln y = 0$

$\therefore$ Order is 1 and degree is also 1.

Example 2 : Determine the order and degree (if it exists) for the following differential equation.

(i) $\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} + \left(\frac{d^3y}{dx^3}\right)^{\frac{3}{2}} = 0$

(ii) $\sqrt{\frac{d^2y}{dx^2} + 4\frac{dy}{dx}} = \sqrt[3]{3\frac{dy}{dx} + 7}$

Solution : (i) Given differential equation in $\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}}+\left(\frac{d^3y}{dx^3}\right)^{\frac{3}{2}}=0$, order is obviously 3, as $\frac{d^3y}{dx^3}$ is the highest order derivative appearing in the equation. (Indeed, it's the only derivative appearing in the equation).

To get the order, we need to write it as a polynomial in derivatives. Transforming the equation as

$$\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}}=-\left(\frac{d^3y}{dx^3}\right)^{\frac{3}{2}}$$

Raising it to the 6th power (the L.C.M. of denominators of power - viz. 2 and 3)

$$\Rightarrow \left(\frac{d^3y}{dx^3}\right)^4=\left(\frac{d^3y}{dx^3}\right)^9 \Rightarrow \left(\frac{d^3y}{dx^3}\right)^9-\left(\frac{d^3y}{dx^3}\right)^4=0$$

Thus the degree is 9.

(ii) Given differential equation is $\sqrt{\frac{d^2y}{dx^2}+4\frac{dy}{dx}}=\sqrt[3]{3\frac{dy}{dx}+7}$. Raising both sides to the 6th power.

$$\Rightarrow \left(\frac{d^2y}{dx^2}+4\frac{dy}{dx}\right)^3=\left(3\frac{dy}{dx}+7\right)^2$$

The left hand side when expanded will produce a term $\left(\frac{d^2y}{dx^2}\right)^3$ giving the order of the equation as 2 and degree 3.

Example 3 : Consider the two differential equations

$$x^2\left(\frac{d^2y}{dx^2}\right)^6+y^{\frac{-2}{3}}\sqrt{1+\left(\frac{d^3y}{dx^3}\right)^5}+\frac{d^2}{dx^2}\left\{\left(\frac{d^2y}{dx^2}\right)^{\frac{-2}{3}}\right\}=0, \text{ and } \frac{dy}{dx}-6x=\left(ay+bx\frac{dy}{dx}\right)^{\frac{-3}{2}}, b\neq 0$$

Find the sum or order of first differential equation and degree of second differential equation.

Solution : Note that the order of first differential equation is not 3!

Consider the term $\frac{d^2}{dx^2}\left\{\left(\frac{d^2y}{dx^2}\right)^{\frac{-2}{3}}\right\}=\frac{d^2}{dx^2}\left(y_2^{\frac{-2}{3}}\right)=\frac{d}{dx}.\left(\frac{-2}{3}.y_2^{\frac{-5}{3}}.y_3\right)$

$$=\frac{-2}{3}\left[\frac{-5}{3}y_2^{\frac{-8}{3}}y_3.y_3+y_2^{\frac{-5}{2}}y_4\right]$$

$$=\frac{-2}{3}y_2^{\frac{-8}{3}}\left(\frac{-5}{3}y_3^2+y_2y_4\right)$$

$$=\frac{-2}{9}y^{\frac{-8}{3}}(3y_2y_4=5y_3^2)$$

First differential equation now becomes

$$x^2\left(\frac{d^2y}{dx^2}\right)^6 + y^{\frac{-2}{3}}\sqrt{1+\left(\frac{d^3y}{dx^3}\right)^5} - \frac{2}{9}y^{\frac{-8}{3}}\left[3\frac{d^2y}{dx^2}\frac{d^4y}{dx^4} - 5\left(\frac{d^3y}{dx^3}\right)^2\right] = 0$$

Clearly the order is 4, $\left(\text{viz. term } \frac{d^4y}{dx^4}\right)$.

Second differential equation is

$$\frac{dy}{dx} - 6x = \left(ay + bx\frac{dy}{dx}\right)^{\frac{-3}{2}}, b \neq 0$$

Writing this as a polynomial in derivatives,

$$\left(ay + bx\frac{dy}{dx}\right)^{\frac{3}{2}}\left(\frac{dy}{dx} - 6x\right) = 1$$

$$\Rightarrow \quad \left(ay + bx\frac{dy}{dx}\right)^3\left(\frac{dy}{dx} - 6x\right)^2 = 1$$

$$\Rightarrow \quad (bx)^3\left(\frac{dy}{dx}\right)^5 + \text{other terms} = 1$$

To find the degree, we are interested only in the highest power of $\left(\frac{dy}{dx}\right)$

$\therefore$ degree = 5

Hence sum of order of 1st differential equation and degree of 2nd differential equation = 4 + 5 = 9.

Example 4 : Solve $\frac{xdx + ydy}{xdy - ydx} = \sqrt{\frac{a^2 - x^2 - y^2}{x^2 + y^2}}$

Solution : Since $x^2 + y^2$ appears as a combination, we make the substitution

$x = r\cos\theta \Rightarrow dx = -r\sin\theta d\theta + \cos\theta dr$

$y = r\sin\theta \Rightarrow dy = r\cos\theta d\theta + \sin\theta dr$ (Note that r and θ are both variables)

$xdx + ydy = r\cos\theta(-r\sin\theta d\theta + \cos\theta dr) + r\sin\theta(r\cos\theta d\theta + \sin\theta dr)$

$= -r^2\sin\theta\cos\theta d\theta + r\cos\theta\sin\theta dr + r^2\sin^2\theta d\theta - r\sin\theta\cos\theta dr$

$= r^2 d\theta$

(A) and (B) could also be derived in another elegant way.

$x = r\cos\theta$, $y = r\sin\theta$ give $r^2 = x^2 + y^2$

$\Rightarrow$ $2rdr = 2xdx + 2ydy$ $\therefore$ $rdr = xdx + ydy$

Again,

$$\tan\theta = \frac{y}{x} \Rightarrow \sec^2\theta d\theta = \frac{xdy - ydx}{x^2}$$

$\Rightarrow$ $xdy - ydx = x^2\sec^2\theta d\theta$

$= (r^2\cos^2\theta)\sec^2\theta d\theta = r^2 d\theta$

$\therefore$ $xdy - ydx = r^2 d\theta$

The given equation now transform to

$$\frac{xdx+ydy}{xdx-ydy}=\sqrt{\frac{a^2-(x^2+y^2)}{x^2+y^2}}$$

$$\Rightarrow \quad \frac{rdr}{r^2d\theta}=\sqrt{\frac{a^2-r^2}{r^2}} \quad \Rightarrow \quad d\theta=\frac{dr}{\sqrt{a^2-r^2}}$$

Integration yields $k+\theta=\sin^{-1}\frac{r}{a}$ (k is an arbitrary constant.)

$$\Rightarrow \quad r=a\sin(k+\theta) \Rightarrow \quad x^2+y^2=a\sin\left(k+\tan^{-1}\frac{y}{x}\right)$$

Note :- To help you remember (B), we remind you of the expression for area of a curve in polar coordinates and parametric from.

$$A=\frac{1}{2}\int xdy-ydx; \text{ also } A=\frac{1}{2}\int r^2d\theta$$

Example 5 : Solve $\frac{xdx-ydy}{xdy-ydx}=\sqrt{\frac{1+x^2-y^2}{x^2-y^2}}$

Solution : The presence of the term $x^2 - y^2$ suggests the substitution $x = r\sec\theta$, $y = r\tan\theta$

$\Rightarrow \quad x^2 - y^2 = r^2$ .... (i)

Differentiating (i), we get,

$2xdx - 2ydy = 2rdr$

$xdx - ydy = rdr$ ..... (A)

and $\sin\theta=\frac{r\tan\theta}{r\sec\theta}=\frac{y}{x}$ ..... (ii)

Differentiating (ii), we get,

$$\cos\theta d\theta=\frac{xdy-ydx}{x^2} \Rightarrow xdy - ydx = x^2\cos\theta d\theta$$

$$= (r^2\sec^2\theta)\cos\theta d\theta$$

$$= r^2\sec\theta d\theta$$

$\therefore \quad xdy - ydx = r^2\sec\theta d\theta$ ..... (B)

Substituting (A) and (B) in the equation $\frac{xdx-ydy}{xdy-ydx}=\sqrt{\frac{1+x^2-y^2}{x^2-y^2}}$

We have,

$$\frac{rdr}{r^2\sec\theta d\theta}=\sqrt{\frac{1+r^2}{r^2}}=\frac{\sqrt{1+r^2}}{r}$$

$$\Rightarrow \quad \frac{dr}{\sec\theta d\theta}=\sqrt{1+r^2}$$

$$\Rightarrow \quad \frac{dr}{\sqrt{1+r^2}}=\sec\theta d\theta$$

Integrating we obtain,

$\ln\left|r+\sqrt{1+r^2}\right| = \ln|\sec\theta + \tan\theta| + \ln k$; k being the constant of integration

$\Rightarrow \quad r+\sqrt{1+r^2} = (\sec\theta+\tan\theta)k$

$\Rightarrow \quad \sqrt{x^2-y^2}+\sqrt{1+x^2-y^2} = k\left(\dfrac{x+y}{\sqrt{x^2-y^2}}\right)$ is the general solution

Example 6 : Solve $xdy - ydx - \sqrt{x^2+y^2}dx = 0$

Solution : **1st method**

We have,

$$x\frac{dy}{dx} = y+\sqrt{x^2+y^2}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y+\sqrt{x^2+y^2}}{x} \qquad \text{.... (i)}$$

Let $y = vx \Rightarrow \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

$$\Rightarrow \quad x\frac{dv}{dx} = \sqrt{1+v^2}$$

$$\Rightarrow \quad \frac{dv}{\sqrt{1+v^2}} = \frac{dx}{x}$$

Integrating, we get

$$\ln\left|v+\sqrt{1+v^2}\right| = \ln x + \ln k \qquad (k \text{ being the constant of integration})$$

$$\Rightarrow \quad v+\sqrt{1+v^2} = kx$$

$$\Rightarrow \quad \frac{y}{x}+\sqrt{1+\frac{y^2}{x^2}} = kx$$

$\Rightarrow \quad y+\sqrt{x^2+y^2} = kx^2$ is the general solution.

2nd method (by inspection)

The equation can be rewritten as

$$\frac{xdy-ydx}{\sqrt{x^2+y^2}} - dx = 0 \Rightarrow \frac{xdy-ydx}{x\sqrt{1+\frac{y^2}{x^2}}} - dx = 0$$

$$\Rightarrow \quad \frac{\frac{xdy-ydx}{x^2}}{\sqrt{1+\frac{y^2}{x^2}}} - \frac{dx}{x} = 0 \Rightarrow \frac{d\left(\frac{y}{x}\right)}{\sqrt{1+\frac{y^2}{x^2}}} - \frac{dx}{x} = 0$$

Integrating,

$$\ln\left|\frac{y}{x}+\sqrt{1+\frac{y^2}{x^2}}\right|-\ln x-\ln k \text{ ; } k \text{ being the constant of integration}$$

$$\Rightarrow \quad \ln\left|\frac{y+\sqrt{x^2+y^2}}{x^2}\right|=\ln k$$

$$\Rightarrow \quad y+\sqrt{x^2+y^2}=kx^2 \text{ is the general solution}$$

Example 7 : **Statement-1 :** The degree of the differential equation, $e^{y'''}-xy''+y=0$ is not defined.
Statement-1 : The differential equation mentioned in statement-1 can't be written as a polynomial in derivatives.

(1) Statement-1 is True, Statement-2 is True, Statement-2 is a correct explanation for Statement-1
(2) Statement-1 is True, Statement-2 is True, Statement-2 is not a correct explanation for Statement-1
(3) Statement-1 is True, Statement-2 is False
(4) Statement-1 is False, Statement-2 is True

Solution : Recall that the degree of a differential equation which can be expressed as a polynomial in derivatives is the degree of the highest order derivative which then occurs. Both Statement-1 and Statement-2 are true and Staement-2 explains Statement-1 also. Hence answer is (1).

Example 8 : **Statement-1 :** The differential equation $\frac{dy}{dx}=\frac{2xy}{x^2+y^2}$ can't be solved by the substitution $x = vy$.
Statement-1 : When the differential equation is homogeneous of first order and first degree, then the substitution that solves the equation is $y = vx$.

(1) Statement-1 is True, Statement-2 is True, Statement-2 is a correct explanation for Statement-1
(2) Statement-1 is True, Statement-2 is True, Statement-2 is not a correct explanation for Statement-1
(3) Statement-1 is True, Statement-2 is False
(4) Statement-1 is False, Statement-2 is True

Solution : The differential equation can be put in the form $\frac{dx}{dy}=\frac{x^2+y^2}{2xy}$ and it being homogeneous of first order and first degree with roles of x and y interchanged, can be solved by the substitution $x = vy$. Thus Statement-1 is false. Again Statement-2 is true. The answer then is S(4).

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Q.1 The degree of the differential equation $\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} + 4 - 3\frac{d^2y}{dx^2} + 5\frac{d^3y}{dx^3} = 0$ is

(a) 1 (b) 2
(c) 3 (d) Not defined

Q.2 The degree of the differential equation corresponding to the family of curves $y = a(x + a)^2$, where a is an arbitrary constant is

(a) 1 (b) 2
(c) 3 (d) 4

Q.3 The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where c is a positive parameter, is of

(a) Order 1, degree 3
(b) Order 1, degree 2
(c) Order 2, degree 3
(d) Order 2, degree 2

Q.4 The differential equation of the family of curves $y = P(x + Q)^2$ is

(a) $yy'' = (y')^2$ (b) $2yy'' = (y')^2$
(c) $2yy'' = y' + y$ (d) $2yy'' = y' - y$

Q.5 The differential equation of all circles passing through the origin and having their centres on the x-axis is

(a) $y^2 = x^2 - 2xy\frac{dy}{dx}$

(b) $x^2 = y^2 + xy\frac{dy}{dx}$

(c) $x^2 = y^2 + 3xy\frac{dy}{dx}$

(d) $y^2 = x^2 + 2xy\frac{dy}{dx}$

Q.6 The solution of $\frac{dy}{dx} = \frac{ax + h}{by + k}$ represents a parabola when

(a) $a = 0, b = 0$ (b) $a = 1, b = 2$
(c) $a = 0, b \neq 0$ (d) $a = 2, b = 1$

Q.7 The differential equation of all ellipse centred at the origin having major and minor axes along coordinates axes is

(a) $xyy_2 - xy_1^2 + yy_1 = 0$
(b) $xyy_2 + xy_1^2 - yy_1 = 0$
(c) $xyy_2 + xy_1^2 - yy_1 = 0$
(d) $xyy_2 + xy_1^2 = 0$

Q.8 The differential equation of all parabolas with axis parallel to the axis of y is

(a) $y_2 = 2y_1 + x$ (b) $y_3 = 2y_1$
(c) $y_2^3 = y_1$ (d) $y_3 = 0$

Q.9 The solution of the equation $2xy - y = 3$ represents a family of

(a) Circle (b) Straight line
(c) Ellipse (d) Parabola

Q.10 If $\frac{dp}{dy} = 3^{\cos y}\sin y$, then P is equal to

(a) $\sin y + C$ (b) $3^{\cos y} + C$
(c) $\frac{-3^{\cos y}}{\ln 3} + C$ (d) $3^{\sin y} + C$

Q.11 The solution of $y' - y = 1, y(0) = 1$, is given by $y(x) =$

(a) $-exp(x)$ (b) $-exp(-x)$
(c) 1 (d) $2exp(x) - 1$

Q.12 The general solution of the differential equation $(1 + y^2)dx + (1 + x^2)dy = 0$ is

(a) $(x - y) = c(1 - xy)$
(b) $(x - y) = c(1 + xy)$
(c) $(x + y) = c(1 - xy)$
(d) $(x + y) = c(1 + xy)$

Q.13 The solution of the equation $y\frac{dy}{dx} = x - 1, y(1) = 1$, is

(a) $y^2 = x^2 + 2x + 2$
(b) $y^2 = 2x^2 + x + 1$
(c) $y^2 = 2x^2 - x - 1$
(d) $y^2 = x^2 - 2x + 2$

Q.14 Solution of the differential equation $\sin x \cdot \cos y\, dy + \cos x \cdot \sin y\, dx = 0$ is

(a) $\sin x + \sin y = C$ (b) $\cos x + \cos y = C$

(c) $\sin x \cdot \sin y = C$ (d) $\frac{\sin x}{\sin y} = C$

Q.15 Solution of differential equation $\frac{dy}{dx} = \frac{2}{x+y}$ is

(a) $x + y + 2 = ke^{y/2}$
(b) $x - y + 2 = ke^{y/2}$
(c) $x + y + 2 = ke^{-y/2}$
(d) $x - y + 2 = ke^{-y/2}$

Q.16 The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with

(a) Variable radii and a fixed centre at (0, 1)
(b) Variable radii and a fixed centre at (0, –1)
(c) Fixed radius 1 and variable centre along the x-axis
(d) Fixed radius 1 and variable centre along the y-axis

Q.17 If $y = y(x)$ and $\left(\frac{2+\sin x}{y+1}\right)\frac{dy}{dx} = -\cos x, y(0) = 1$, then $y\left(\frac{\pi}{2}\right) =$

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$

(c) $-\frac{1}{3}$ (d) 1

Q.18 The slope of tangent at (x, y) to a curve passing through (2, 1) is $\frac{x^2+y^2}{2xy}$, then the equation of the curve is

(a) $x(x^2 + y^2) = 10$ (b) $x(x^2 - y^2) = 6$
(c) $2(x^2 - y^2) = 6y$ (d) $2(x^2 - y^2) = 3x$

Q.19 The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{Q\left(\frac{y}{x}\right)}{Q'\left(\frac{y}{x}\right)}$ is

(a) $Q\left(\frac{y}{x}\right) = kx$ (b) $xQ\left(\frac{y}{x}\right) = k$

(c) $Q\left(\frac{y}{x}\right) = ky$ (d) $y \cdot Q\left(\frac{y}{x}\right) = k$

Q.20 The solution of differential equation $x^2y^2dy = (1 - xy^3)dx$ is

(a) $x^3y^3 = x^2 + C$ (b) $2x^3y^3 = 3x^2 + C$
(c) $x^3y^3 = x^2 + x + C$ (d) $x^3y^3 = 3x^2 + C$

Q.21 The solution of differential equation $(1+x^2)\frac{dy}{dx} + y = e^{\tan^{-1}x}$

(a) $ye^{\tan^{-1}x} = \frac{1}{2}e^{2\tan^{-1}x} + C$

(b) $y = \frac{1}{2}e^{2\tan^{-1}x} + C$

(c) $ye^{\tan^{-1}x} = 2e^{2\tan^{-1}x} + C$

(d) $y \cdot \tan^{-1}x = \frac{1}{2}e^{2\tan^{-1}x} + C$

Q.22 The solution of the differential equation $ydx + (x + x^2y)dy = 0$ is

(a) $\log y = Cx$ (b) $-\frac{1}{xy} + \log y = C$

(c) $\frac{1}{xy} + \log y = C$ (d) $-\frac{1}{xy} = C$

Q.23 The family whose x and y intercepts of a tangent at any point are respectively double of the x and y co-ordinates of that point is

(a) $x^2 + y^2 = C$ (b) $x^2 - y^2 = C$
(c) $xy = C$ (d) $x^2y = C$

Q.24 The solution of the equation $y' - \cos(x - y)$ is

(a) $y + \cot\left(\frac{x-y}{2}\right) = c$

(b) $x + \cot\left(\frac{x-y}{2}\right) = c$

(c) $x + \tan\left(\frac{x-y}{2}\right) = c$

(d) $y + \tan\left(\frac{x-y}{2}\right) = c$

Q.25 Solution of $y\,dx - x\,dy = x^2 y\,dx$ is

(a) $y^2 e^{x^2} = cx^2$

(b) $ye^{-x^2} = cx^2$

(c) $y''' + y'' + y = 0$

(d) $y' = y$

Section-B
(Previous Years Question)

Q.1 The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$ in $(-1,1)$ satisfying $f(0) = 0$. Then $\int_{-\sqrt{3}/2}^{\sqrt{3}/2} f(x)dx$ is

(a) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (b) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

(c) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (d) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Q.2 A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x,y) be $\frac{y}{x} + \sec\left(\frac{y}{x}\right), x > 0$. Then the equation fo the curve is

(a) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$

(b) $\operatorname{cosec}\left(\frac{y}{x}\right) = \log x + 2$

(c) $\sec\left(\frac{2y}{x}\right) = \log x + 2$

(d) $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$

Q.3 The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with

(a) variable radii and a fixed centre at (0,1)

(b) variable radii and fixed centre at (0, -1)

(c) fixed radius 1 and variable centres along the x-axis

(d) fixed radius 1 and variable centres along the y-axis

Q.4 For the primitive integral equation $ydx + y^2\,dy = x\,dy; x \in R, y > 0,$ $y = y(x), y(1) = 1,$ then $y(-3)$ is

(a) 3 (b) 5

(c) 1 (d) 2

Q.5 The solution of primitive equation $(x^2 + y^2)dx = xy,$ is $y = y(x)$. If $y(1) = 1$ and $y(x_0) = e,$ then x_0 is

(a) $\sqrt{\frac{e^2+1}{2}}$ (b) $\sqrt{2(e^2+1)}$

(c) $\sqrt{2(e^2-1)}$ (d) $\sqrt{3}e$

Q.6 If $y = y(x)$ and $\frac{2+\sin x}{y+1}\left(\frac{dy}{dx}\right) = -\cos x, y(0) = 1,$ then $y\left(\frac{\pi}{2}\right)$ equals

(a) 1/ 3 (b) 2/ 3

(c) $-1/3$ (d) 1

Q.7 If $y(t)$ is a solution of $(1+t)\frac{dy}{dt} - ty = 1$ and $y(0) = -1,$ then $y(1)$ is equal to

(a) $\frac{-1}{2}$ (b) $\frac{1}{2}$

(c) 1 (d) -1

Q.8 If $x^2 + y^2 = 1,$ then

(a) $yy'' - 2(y')^2 + 1 = 0$

(b) $yy'' + 2(y')^2 + 1 = 0$

(c) $yy'' + (y')^2 - 1 = 0$

(d) $yy'' + 2(y')^2 + 1 = 0$

Q.9 A solution of the differential equation $\left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + y = 0$ is

(a) $y = 2$

(b) $y = 2x$

(c) $y = 2x - 4$

(d) $y = 2x^2 - 4$

Q.10 The order of the differential equatrion whose general solution is given by $y = (c_1 + c_2)\cos(x + c_3) - c_4 e^{x+c_5}$, where $c_1, c_2, c_3 c_4, c_5$ are arbitrary constants, is

(a) 5 (b) 4

(c) 3 (d) 2

Q.11 If $y^2 = P(x)$, a polynomial of degree 3, then $2\frac{d}{dx}\left(y^3\frac{d^2y}{dx^2}\right)$ equals

(a) P'''(x) + P'(x)

(b) P''(x) P'''(x)

(c) P(x) P'''(x)

(d) a constant

Q.12 Let the population of rabbits surving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$.

If $p(0) = 100$ then $p(t)$ equaals

(a) 300 – 200 $e^{-t/2}$

(b) 600 – 500 $e^{t/2}$

(c) 400 – 300 $e^{-t/2}$

(d) 400 – 300 $e^{t/2}$

Q.13 The population p(t) at time t of a certain mouse s pecies satisfies the differential equation $\frac{dp(t)}{dt} = 0.5p(t) - 450$.

If p(0) = 850, then the time at which the population becomes zero is

(a) $\frac{1}{2}$ ln 18 (b) ln 18

(c) 2 ln 18 (d) ln 9

Q.14 Let I be the purchase value of an equipment and V(t) be the value after it has been used for t years. The value V(t) depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T - t)$; where k > 0 is a constant and T is the total life in years of the equipment. Then the scrap value V(T) of the equipment is

(a) $I - \frac{k(T-t)^2}{2}$ (b) e^{-kT}

(c) $T^2 - \frac{I}{k}$ (d) $I - \frac{kT^2}{2}$

Q.15 If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(ln2)$ is equal to

(a) 13 (b) -2

(c) 7 (d) 5

Q.16 Solution of the differential equation $\cos x dy = y(\sin x - y)dx, 0 < x < \pi/2$ is

(a) $\sec x = (\tan x + c)y$

(b) $y\sec x = \tan x + c$

(c) $y\tan x = \sec x + c$

(d) $\tan x = (\sec x + c)y$

Q.17 The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where c_1 and c_2 are arbitrary constants, is

(a) $y'' = y'y$ (b) $yy'' = y'$

(c) $yy'' = (y')^2$ (d) $y' = y^2$

Q.18 The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition y(1)=1 is

(a) $y = x ln x + x$ (b) $y = ln x + x$

(c) $y = x ln x + x^2$ (d) $y = xe^{(x-1)}$

Q.19 The The differential equation of the family of circles with fixed radius 5 units and centre on the line y = 2 is

(a) $(x-2)^2 y'^2 = 25 - (y-2)^2$

(b) $(x-2)y'^2 = 25-(y-2)^2$

(c) $(y-2)y'^2 = 25-(y-2)^2$

(d) $(y-2)^2 y'^2 = 25-(y-2)^2$

Q.20 The differential equation of all circles passing through the origin and having their centres on the x-axis is

(a) $y^2 = x^2 + 2xy\frac{dy}{dx}$ (b) $y^2 = x^2 - 2xy\frac{dy}{dx}$

(c) $x^2 = y^2 + xy\frac{dy}{dx}$ (d) $x^2 = y^2 + 3xy\frac{dy}{dx}$

Q.21 The differential equation whose solution is $Ax^2 + By^2 = 1$, where A and B are arbitary constants is of

(a) second order and second degree
(b) frist order and second degree
(c) first order and first first degree
(d) second order and first degree

Q.22 If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is

(a) $x\log\left(\frac{y}{x}\right) = cy$ (b) $y\log\left(\frac{x}{y}\right) = cx$

(c) $\log\left(\frac{x}{y}\right) = cy$ (d) $\log\left(\frac{y}{x}\right) = cx$

Q.23 The differential equation representing the family of curves $y^2 = 2c\left(x + \sqrt{c}\right)$, where c > 0, is a parameter, is a parameter, is of order and degree as follows

(a) order 1, degree 1 (b) order 1, degree 2
(c) order 2, degree 2 (d) order 1, degree 3

Q.24 The solution of the differential equation $ydx + \left(x + x^2y\right)dy = 0$ is

(a) $\frac{1}{xy} + \log y = C$ (b) $-\frac{1}{xy} + \log y = C$

(c) $-\frac{1}{xy} = C$ (d) $\log y = Cx$

Q.25 The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant is

(a) $\left(x^2 - y^2\right)y' = 2xy$

(b) $2\left(x^2 + y^2\right)y' = xy$

(c) $2\left(x^2 - y^2\right)y' = xy$

(d) $\left(x^2 + y^2\right)y' = 2xy$

Q.26 If $x = e^{y+e^{y+\ldots to\ \infty}}, x > 0$ then $\frac{dy}{dx}$ is

(a) $\frac{1-x}{x}$ (b) $\frac{1}{x}$

(c) $\frac{x}{1+x}$ (d) $\frac{1+x}{x}$

Q.27 The solution of the differential equation $\left(1+y^2\right) + \left(x - e^{\tan^{-1}y}\right)\frac{dy}{dx} = 0$ is

(a) $2xe^{\tan^{-1}y} = e^{2\tan^{-1}}y + k$

(b) $xe^{\tan^{-1}y} = \tan^{-1}y + k$

(c) $xe^{2\tan^{-1}y} = e^{\tan^{-1}y} + k$

(d) $(x-2) = ke^{-\tan^{-1}y}$

Q.28 The degree and order of the differentia equation of the family of all parabolas whoas axis is x-axis, are respectively

(a) 1, 2 (b) 3,2
(c) 2, 3 (d) 2, 1

Q.29 The order and degree of the differential equation $\left(1+3\frac{dy}{dx}\right)^{\frac{2}{3}} = 4\frac{d^3y}{dx^3}$ are

(a) $1, \frac{2}{3}$ (b) 3, 1
(c) 3, 3 (d) 1 , 2

Q.30 The solution of the equation $\frac{d^2y}{dx^2} = e^{-2x}$

(a) $\frac{1}{4}e^{-2x}$ (b) $\frac{1}{4}e^{-2x} + cx + d$

(c) $\frac{1}{4}e^{-2x} + cx^2 + d$ (d) $\frac{1}{4}e^{-2x} + c + d$

Section-C
(Board Pattern)

Q.1 Find order and degree of the differential equation that represents all the parabola each of which has latus recturm '4*a*' and whose axes are parallel to x-axis.

Q.2 From one differential equation representing the family of curves given by $(x-a)^2 + 2y^2 = a^2$ where *a* is an arbitrary constant.

Q.3 A saving account pays 8% interest per year, compounded continuously. In addition, income from another investment is creditted to the second continuously at the rate of Rs. 400 per year. Form a differential equation to model this account.

Q.4 Find the differential equation of all circles in first quadrant touching the coordinate axis.

Q.5 Show that $y=\left(x+\sqrt{x^2+1}\right)^m$ is satisfies the differential equation $(x^2+1).\frac{d^2y}{dx^2}+x.\frac{dy}{dx}-m^2y=0$.

Q.6 Shwo that $y=\frac{c-x}{1+cx}$ is a solution of differential equation $\left(1+x^2\right).\frac{dy}{dx}+(1+y^2)=0$.

Q.7 Verified that $y=\log_e\left(x+\sqrt{a^2+x^2}\right)$ satisfies the differential equation $(a^2+x^2)\frac{d^2y}{dx^2}+x\frac{dy}{dx}=0$.

Q.8 Solve the differential equation $y.\sqrt{1+x^2}+(x\sqrt{1+y^2})\frac{dy}{dx}=0$.

Q.9 Find the particular solution of the differential equation $\ln\left(\frac{dy}{dx}\right)=3x+4y$, given that $y = 0$ when $x = 0$.

Q.10 Solve the differential equation $\frac{dy}{dx}=\cos(x+y)+\sin(x+y)$.

Q.11 Solve the differential equation $y.dx+x.\log\left(\frac{y}{x}\right)dy-2x.dy=0$.

Q.12 Find the particular solution of the differential equation $x.e^{y/x}-y.\sin\frac{y}{x}+x.\frac{dy}{dx}.\sin\frac{y}{x}=0$.

Q.13 Solve the differential equation $\frac{dy}{dx}=\frac{x+y-5}{x-y-1}$.

Q.14 Find the equation of a curve passing through the origin given that the slope of the tangent to the curve at any point (x, y) is equal to the sum of the co-ordinates of the points.

Q.15 Find the equation of the curve at any point (x, y) on it is $\left(\frac{y}{x}-\cot\frac{y}{x}.\cos\frac{y}{x}\right), x>0, y>0$ and the curve passes through $\left(1,\frac{\pi}{4}\right)$ then find the equation of the curve.

ANSWER KEY

CHAPTER-01 (DEFINITE INTEGRATION AND THEIR APPLICATION)

SECTION-A (Straight objective type questions)

1. (b)	2. (a)	3. (d)	4. (d)	5. (a)	6. (a)	7. (a)
8. (c)	9. (c)	10. (d)	11. (b)	12. (c)	13. (d)	14. (d)
15. (b)	16. (a)	17. (b)	18. (b)	19. (d)	20. (b)	21. (d)
22. (b)	23. (b)	24. (c)	25. (c)	26. (c)	27. (b)	28. (c)
29. (b)	30. (b)	31. (c)	32. (c)	33. (d)	34. (a)	35. (c)
36. (c)	37. (b)	38. (d)	39. (b)	40. (a)	41. (a)	42. (a)
43. (b)	44. (c)	45. (b)	46. (c)	47. (c)	48. (b)	49. (d)
50. (a)	51. (a)	52. (a)	53. (a)	54. (b)	55. (b)	56. (c)
57. (d)	58. (b)	59. (c)	60. (c)	61. (b)	62. (d)	63. (d)
64. (a)	65. (b)	66. (d)				

SECTION-B (Previous Year Question)

1. (a)	2. (b)	3. (b)	4. (c)	5. (b)	6. (a)	7. (c)
8. (c)	9. (a)	10. (d)	11. (a)	12. (d)	13. (a)	14. (a)
15. (c)	16. (a)	17. (c)	18. (c)	19. (b)	20. (c)	21. (c)
22. (a)	23. (b)	24. (b)	25. (c)	26. (a)	27. (b)	28. (d)
29. (a, c,d)	30. (a)	31. (b,c)	32. (a,b,c)	33. (b,c,d)	34. (a,d)	35. (a,b,c,d)
36. (b)	37. (0)	38. (c)	39. (c)	40. (c)	41. (b)	42. (c)
43. (c)	44. (a)	45. (a)	46. (c)	47. (a,c)	48. (a)	49. (b)
50. (c)						

SECTION-C (School/Board level type questions)

1. $\sqrt{2}-1$ 2. 9/2 or 1/2 3. $1+\frac{4}{\pi}-\frac{2\sqrt{2}}{\pi}$ 4. $\frac{\pi}{2ab(a+b)}$ 5. $\frac{1}{4}\log 2-\frac{\pi}{8}+\frac{1}{4}$

6. $\frac{e^2}{2}-e$ 7. (i) $\frac{\pi}{4}-\frac{1}{2}\log 2$, (ii) $\frac{\pi}{2}-\log 2$ 8. $\sqrt{2}\pi$ 9. $\pi/2$

11. (i) $a\left(\frac{\pi}{2}-1\right)$, (ii) $\frac{\pi}{2ab}$ 12. (i) $\frac{19}{2}$, (ii) $\frac{11}{4}$ 13. $\pi/12$ 14. (i) 0, (ii) $\frac{\pi}{8}\log 2$

15. $-\frac{\pi}{2}\log 2$ 16. $\frac{\pi}{2}-\log 2$ 17. $\pi^2/4$ 18. $\frac{\pi}{2}(\pi-2)$ 20. 0

21. $a\pi$ 22. $\frac{3\pi+1}{\pi^2}$ 23. 0 24. $-\pi\log\frac{1}{2}$ 25. $200\sqrt{2}$

29. $\frac{\pi}{4}-\frac{1}{2}$ 30. $\frac{\pi}{2}$ 31. $e-\frac{2}{\log 2}$ 32. $\frac{\pi}{8}\log 2$

33. $\frac{1}{4}\log 2-\frac{\pi}{8}+\frac{1}{4}$ 34. $\frac{93}{2}$ 35. 0 36. e^2-1 37. 93/2

38. $\frac{16}{3}a^2$ 39. 1/6 40. 1/3 41. $\frac{1}{2}\left(\frac{\pi}{2}-1\right)$

42. $\frac{\sqrt{2}}{6}+\frac{9\pi}{8}-\frac{9}{4}\sin^{-1}\left(\frac{1}{3}\right)$ 44. $\left(\frac{2\pi}{3}-\frac{\sqrt{3}}{2}\right)$ 45. 7 46. $\frac{4\sqrt{3}}{3}+\frac{16\pi}{3}$

47. $a = \frac{1}{\sqrt{3}}$ 48. 8 50. $ab\left[e\sqrt{1-e^2} + \sin^{-1} e\right]$ 51. 11

52. $\pi/8$ 53. $\left\{\frac{5}{2}\left(\sin^{-1}\frac{2}{\sqrt{5}} + \sin^{-1}\frac{1}{\sqrt{5}}\right) - \frac{3}{2}\right\}$

CHAPTER-02 (Differential Equation)

SECTION-A (Straight objective type questions)

1. (c)	2. (c)	3. (a)	4. (b)	5. (d)	6. (c)	7. (b)
8. (d)	9. (d)	10. (c)	11. (d)	12. (c)	13. (d)	14. (c)
15. (a)	16. (c)	17. (a)	18. (d)	19. (a)	20. (b)	21. (a)
22. (b)	23. (c)	24. (b)	25. (a)			

SECTION-B (Previous Year Question)

1. (b)	2. (a)	3. (c)	4. (a)	5. (d)	6. (a)	7. (a)
8. (b)	9. (c)	10. (c)	11. (c)	12. (d)	13. (c)	14. (d)
15. (c)	16. (a)	17. (c)	18. (a)	19. (d)	20. (a)	21. (d)
22. (d)	23. (d)	24. (b)	25. (a)	26. (a)	27. (a)	28. (a)
29. (c)	30. (b)					